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FortiSOAR Administrator Lab Guide

for FortiSOAR 7.3

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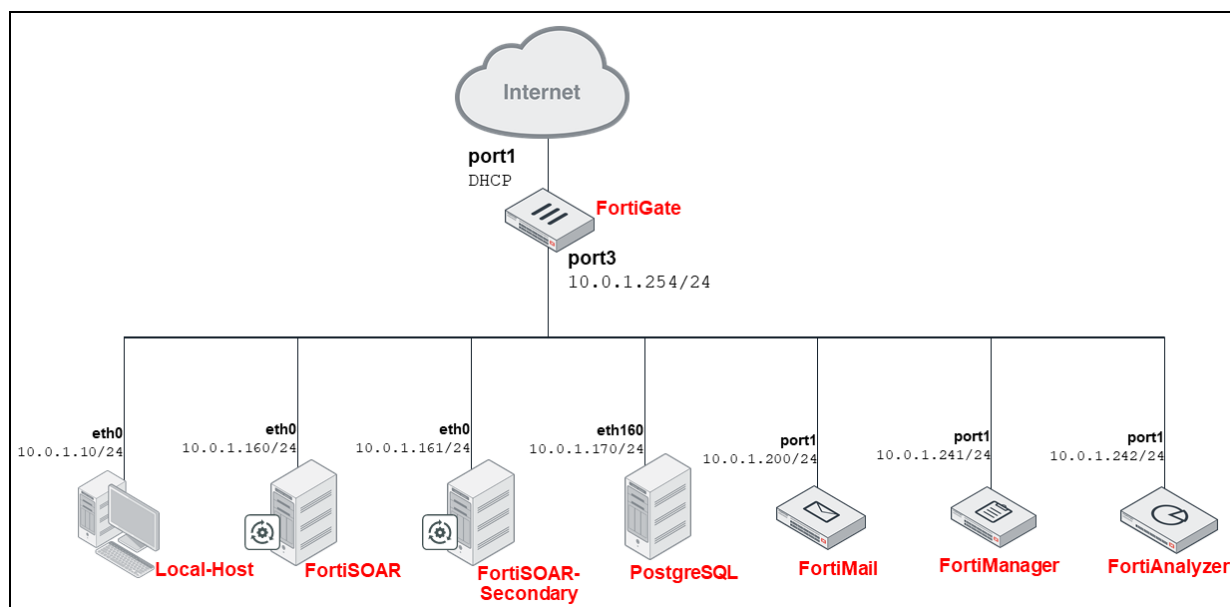
5/10/2023

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Network Topology



Lab 1: FortiSOAR Overview

In this lab, you will review multiple FortiSOAR features and become familiar with the interface. You will learn more about the features in future labs.

Objectives

- Review the FortiSOAR dashboards
- Review the **Incident Response** modules
- Review the **Queue & Shift Management** module
- Review the **Indicators** module
- Review the **Playbooks** module
- Configure data ingestion

Time to Complete

Estimated: 50 minutes

Exercise 1: Examining the FortiSOAR Dashboards

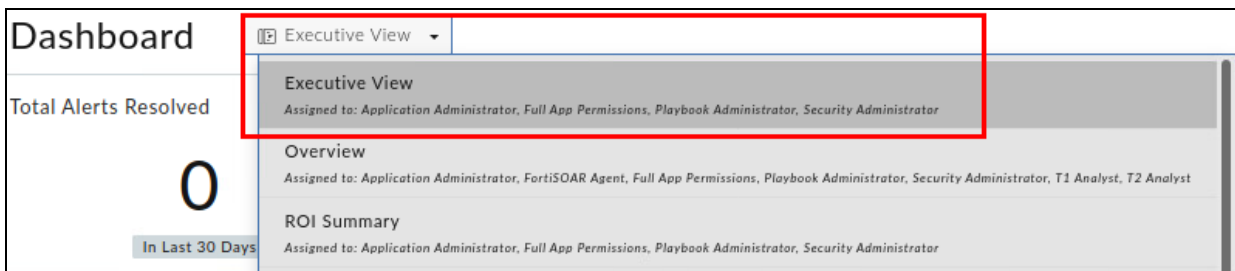
In this exercise, you will review the FortiSOAR dashboards. The main dashboard is the landing page that appears after you log in to FortiSOAR. You can then navigate to other FortiSOAR dashboards from this page.

Examine the Executive View Dashboard

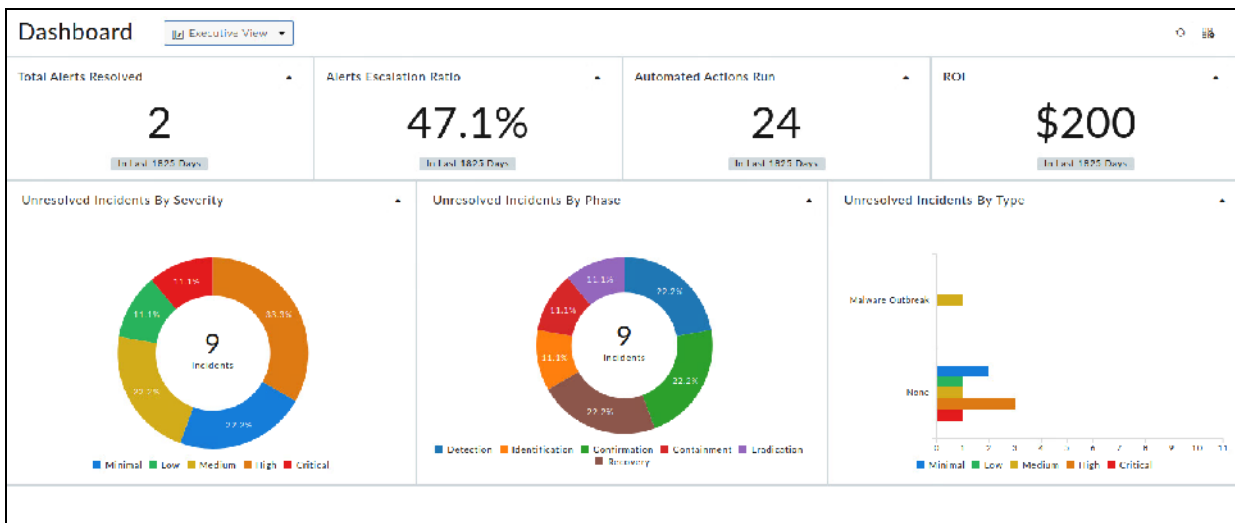
You will examine the **Executive View** dashboard. This dashboard provides an executive summary of the incident response on FortiSOAR. It provides information, such as various time widgets related to resolved incidents and the number of unresolved incidents by severity and type.

To examine the Executive View dashboard

1. Log in to the FortiSOAR-Primary GUI with the username `csadmin` and password `Fortinet1!`.
2. In the **Dashboard** field, select **Executive View**.



The **Executive View** dashboard appears.



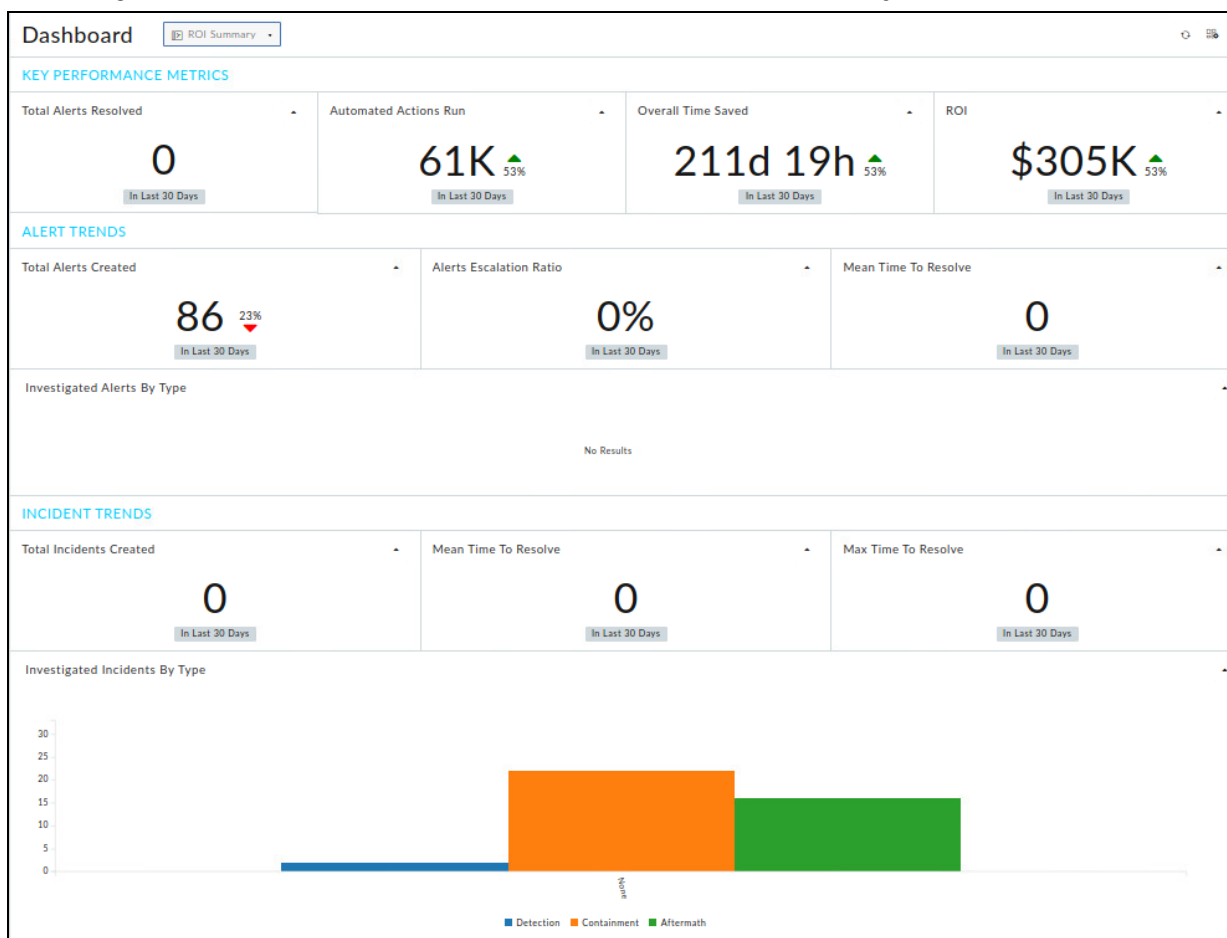
The data that you see on your FortiSOAR instance is different from what appears in this lab guide.

Examine the ROI Summary Dashboard

You will examine the **ROI Summary** dashboard. This is another built-in dashboard that displays the return on investment (ROI) that you gain by using FortiSOAR.

To examine the ROI Summary dashboard

1. Continuing on the FortiSOAR GUI, in the **Dashboard** field, select **ROI Summary**.



The data that you see on your FortiSOAR instance is different from what appears in this lab guide.

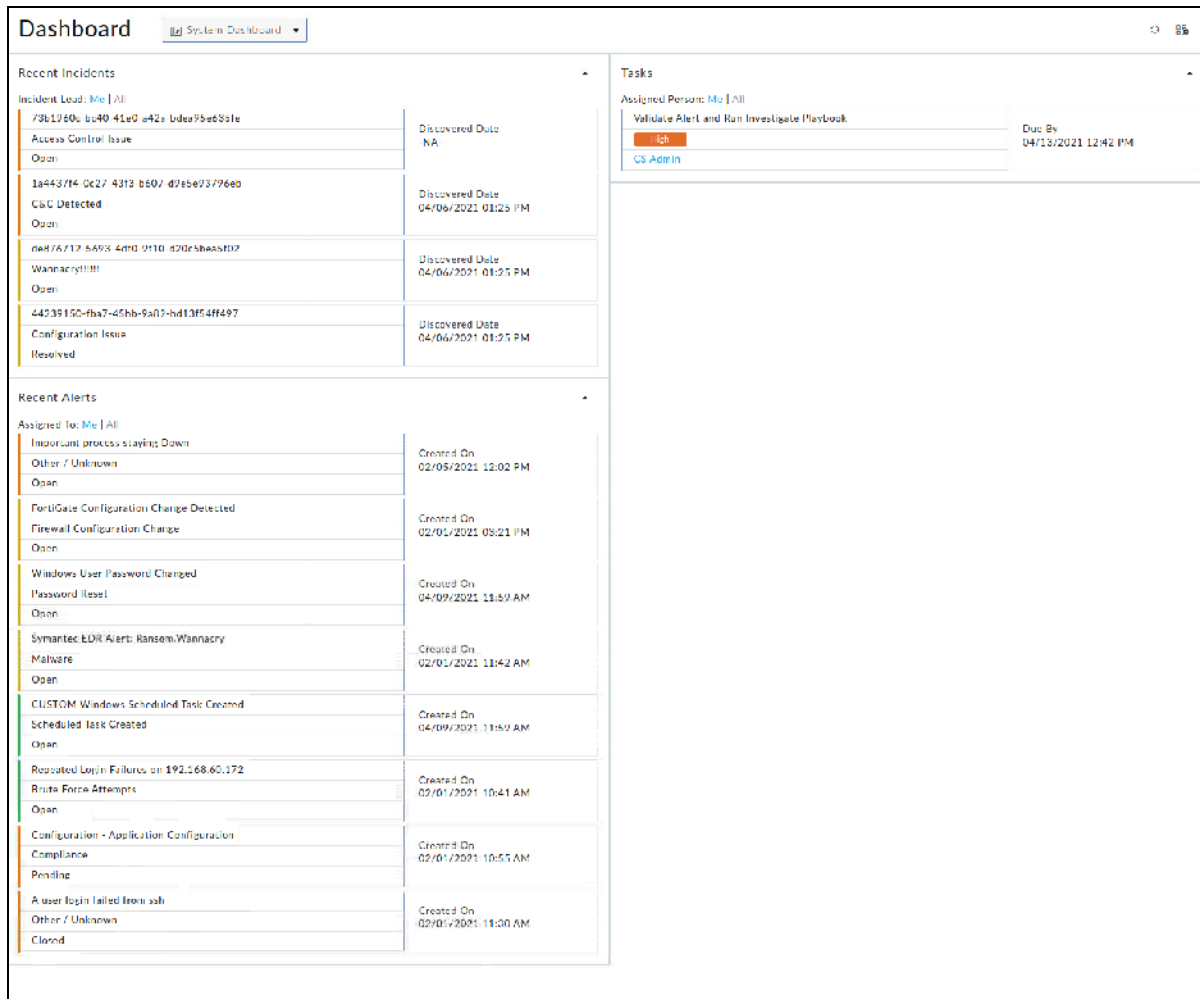
Examine the System Dashboard

You will examine the **System Dashboard**. The **System Dashboard** is included by default on FortiSOAR to monitor various FortiSOAR incidents, alerts, tasks, and their respective statuses. The advantage of having the

System Dashboard is that you do not necessarily have to log in as an administrator to check the various usage levels.

To examine the System Dashboard

1. Continuing on the FortiSOAR GUI, in the **Dashboard** field, select **System Dashboard**.



Dashboard System Dashboard

Recent Incidents

Incident Lead: Me | All

Incident ID	Title	Discovered Date
7361760c-bc40-41e0-e42e-bde95e658e	Access Control Issue	NA
1a4437f4-0c27-4313-b507-d7e6e73796e0	C&C Detected	04/09/2021 08:25 PM
6a87a717-5a93-4d9d-9110-d20f3baa5d02	Wannacry!!!!	04/06/2021 01:25 PM
44239150-fba7-45bb-9a07-nd13f54ff497	Configuration Issue	04/06/2021 01:25 PM

Recent Alerts

Assigned To: Me | All

Alert Title	Created On
Important process staying Down	02/05/2021 12:02 PM
Other / Unknown	
FortiGate Configuration Change Detected	02/01/2021 09:21 PM
Firewall Configuration Change	02/01/2021 09:21 PM
Windows User Password Changed	04/09/2021 11:59 AM
Malware	02/01/2021 11:42 AM
Custom Windows Scheduled Task Created	04/09/2021 11:59 AM
Scheduled Task Created	04/09/2021 11:59 AM
Repeated Login Failures on 192.168.60.172	02/01/2021 10:41 AM
Configuration - Application Configuration	02/01/2021 10:55 AM
Compliance	02/01/2021 11:00 AM
A user login failed from ash	02/01/2021 11:00 AM

Tasks

Assigned Person: Me | All

Task	Due By
Validate Alert and Run Investigate Playbook	04/13/2021 12:42 PM



There are various other built-in dashboards on FortiSOAR that you can explore.

2. Log out of the FortiSOAR GUI.

Exercise 2: Examining the Incident Response Modules

In this exercise, you will review some of the **Incident Response** modules. The **Incident Response** modules include the **Alerts**, **Incidents**, **Hunts**, **MITRE ATT&CK Techniques**, **Tasks**, and **Communications** modules.

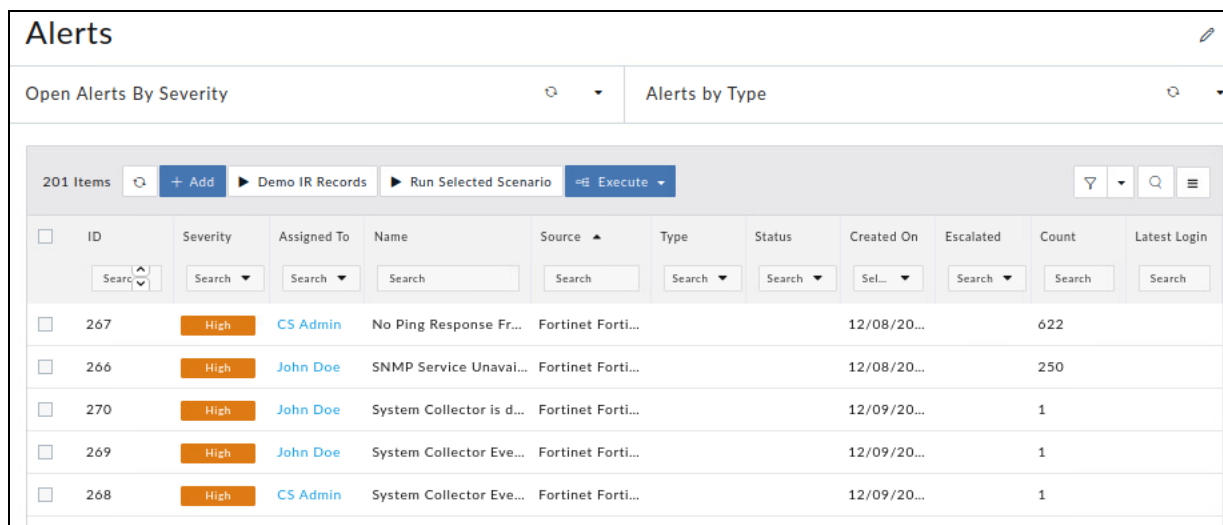
Examine the Alerts Module

You will examine the **Alerts** module. Alerts in FortiSOAR are notifications indicating that a potential attack has been directed at an organization's systems and further investigation may be warranted. Alerts are related to events and often contain essential information for addressing the attack by including vulnerabilities and exploits being leveraged by the potential attack.

To examine the Alerts module

1. Log in to the FortiSOAR-Primary GUI with the username `csadmin` and password `Fortinet1!`.
2. Click **Incident Response > Alerts**.

The alerts list view shows all the alerts that the current user's team owns, and provides high-level details about each alert.



ID	Severity	Assigned To	Name	Source	Type	Status	Created On	Escalated	Count	Latest Login
267	High	CS Admin	No Ping Response Fr...	Fortinet Forti...			12/08/20...		622	
266	High	John Doe	SNMP Service Unavai...	Fortinet Forti...			12/08/20...		250	
270	High	John Doe	System Collector is d...	Fortinet Forti...			12/09/20...		1	
269	High	John Doe	System Collector Eve...	Fortinet Forti...			12/09/20...		1	
268	High	CS Admin	System Collector Eve...	Fortinet Forti...			12/09/20...		1	



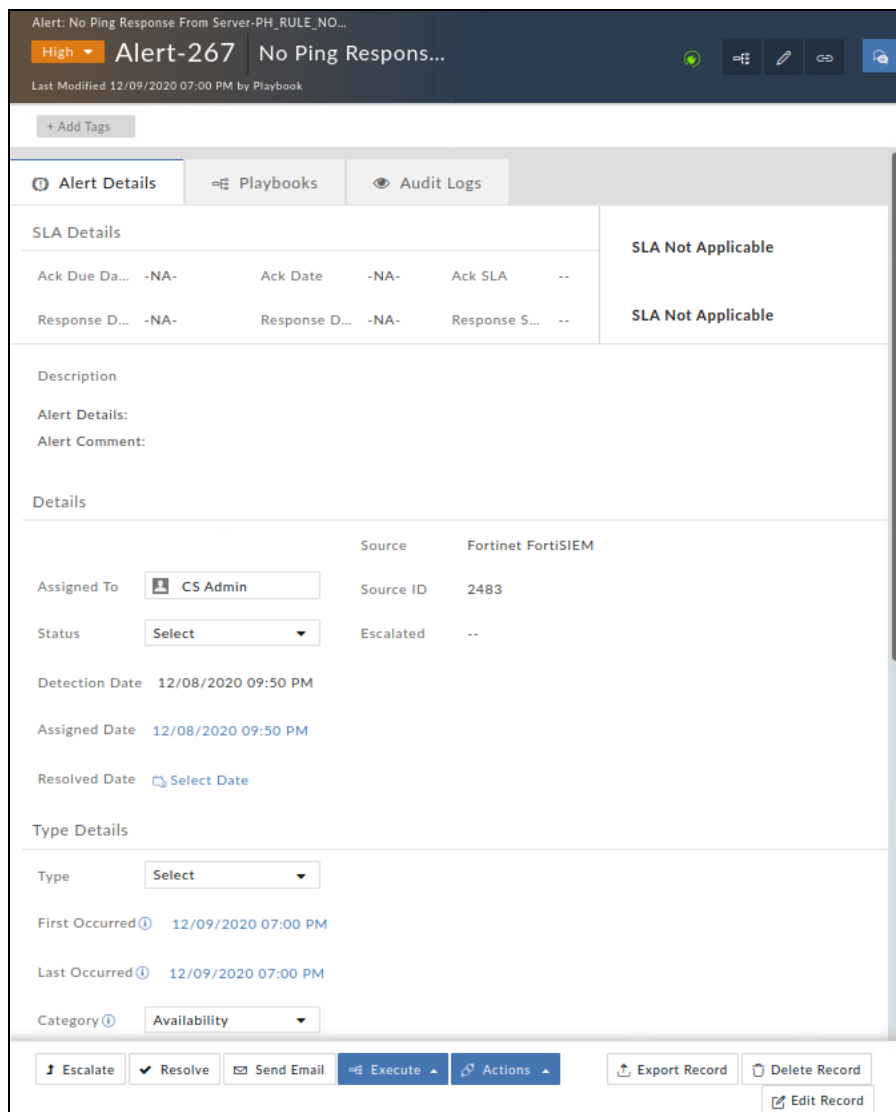
The alerts you see were created for you to review. For a new deployment where FortiSOAR is deployed for the very first time, you will not see any alerts until you create some or configure data ingestion.

The alerts that you see on your FortiSOAR instance are different from what appears in this lab guide.

To review an alert record

1. Continuing on the FortiSOAR GUI, click one of the alerts.
A new window opens.

The alert record contains details about the incident, such as acknowledge date, due date, status, assigned to, severity, and so on.



Alert: No Ping Response From Server-PH_RULE_NO...

High Alert-267 No Ping Respons...

Last Modified 12/09/2020 07:00 PM by Playbook

+ Add Tags

Alert Details Playbooks Audit Logs

SLA Details

Ack Due Da...	Ack Date	Ack SLA	
-NA-	-NA-	--	SLA Not Applicable
Response D...	Response D...	Response S...	SLA Not Applicable

Description

Alert Details:

Alert Comment:

Details

Source Fortinet FortiSIEM

Assigned To CS Admin Source ID 2483

Status Select Escalated --

Detection Date 12/08/2020 09:50 PM

Assigned Date 12/08/2020 09:50 PM

Resolved Date Select Date

Type Details

Type Select

First Occurred 12/09/2020 07:00 PM

Last Occurred 12/09/2020 07:00 PM

Category Availability

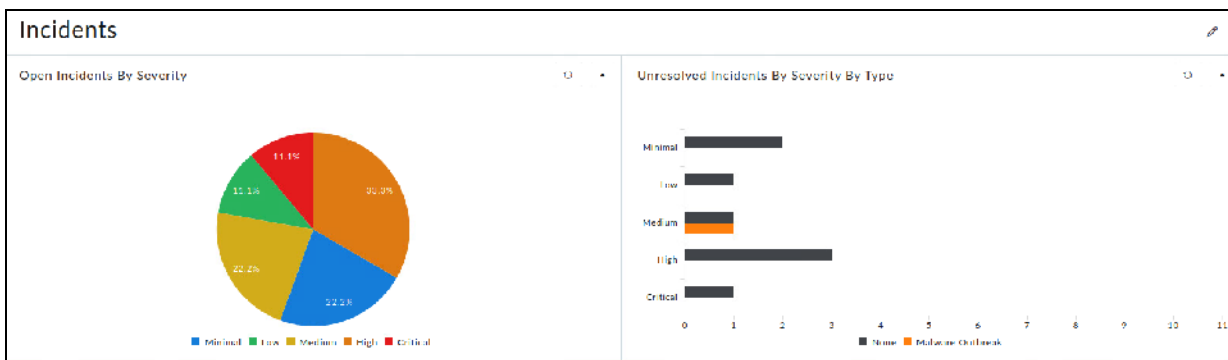
Escalate Resolve Send Email Execute Actions Export Record Delete Record Edit Record

Examine the Incidents Module

You will examine the **Incidents** module. The **Incidents** module records incidents that represent a collection of information discovered during an incident response investigation. Incidents are triggered based on the suspicion or confirmation of a security breach. Incidents may be related to a cyberattack or a physical security breach.

To examine the Incidents module

- Continuing on the FortiSOAR GUI, click **Incident Response > Incidents**.
Review the two charts at the top, which are included by default. The pie chart shows open incidents by severity, and the bar chart shows unresolved incidents by severity and type.



The incident records are located under the charts.

ID	Severity	Name	Source	Type	Phase	Status	Incident Lead	Created On	Ack SLA	Response SLA
7	Minimal	FGT config issue			Indication	Open		02/05/2021 1...		
10	Low	CUSTOM Windows Scheduled Task Created			Recovery	Awaiting		04/12/2021 1...		
4	Medium	WannaCry!!!!			Identification	Open	CS Admin	02/01/2021 1...		
2	Medium	Configuration Issue		Malware Outbr...	Attempted	Resolved	CS Admin	02/01/2021 1...	Met	Met
6	High	C&C Detected			Containment	Open	CS Admin	02/05/2021 1...		
3	Medium	Outbound Unencrypted PII Detected	Symantec DLP	Malware Outbr...	Confirmation	Open		02/01/2021 1...	Awaiting Action	Awaiting Action
5	Critical	Failed Login			Confirmation	Open		02/01/2021 0...		
11	Minimal	Windows User Added to Groups			Detection	In Progress		04/12/2021 1...		
9	High	FGT Configuration change			Detection	Open		04/12/2021 1...		
1	Low	Access Control Issue			Recovery	Open	CS Admin	02/01/2021 1...		

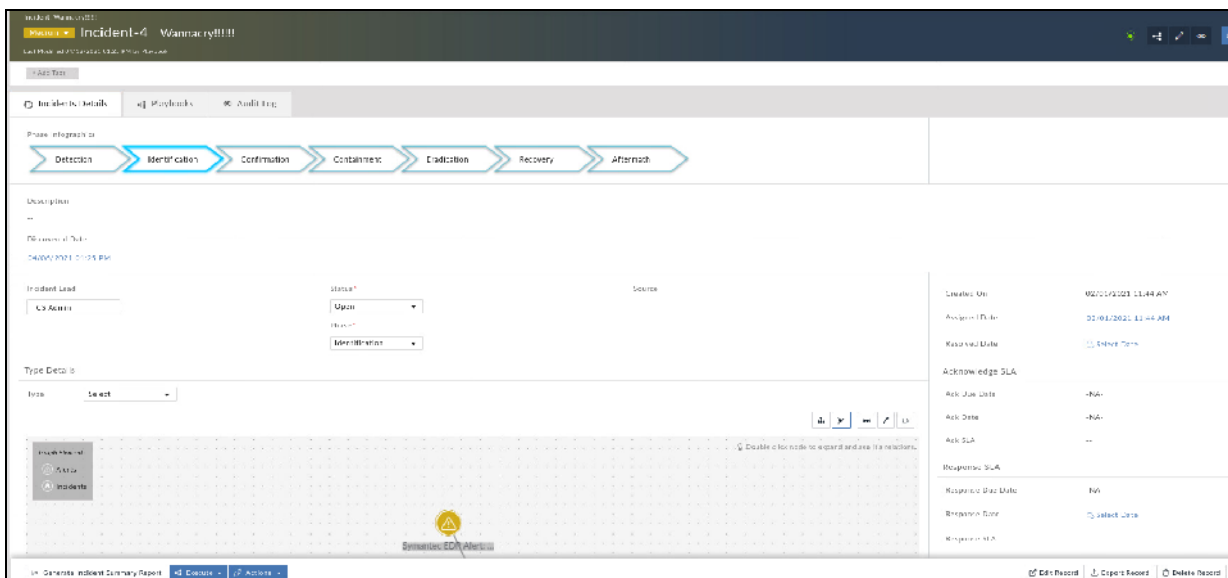


The incidents that you see on your FortiSOAR instance are different from what appears in this lab guide.

To review an incident record

1. Click any of the incident records.
2. Compare the incident record to an alert record.

There are some very notable differences between them. The incident record contains more detailed information about the phase of the attack. It contains a graph element that maps and links various records and attributes the incident is associated with.



The data that you see on your FortiSOAR instance is different from what appears in this lab guide.

3. Log out of the FortiSOAR GUI.

Exercise 3: Examining the Queue & Shift Management Module

In this exercise, you will review the **Queue & Shift Management** module by creating a queue and generating a shift.

This module allows you to automatically assign records to users in a queue and checks whether a user is working or off-shift. It also allows you to define queue rules so that the assignments are based on matching conditions.

Examine the Queue & Shift Management module

You will examine the **Queue & Shift Management** module.

To create a queue and configure assignment parameters

1. Log in to the FortiSOAR-Primary GUI with the username `csadmin` and password `Fortinet1!`.
2. Click **Queue & Shift Management**.
3. Click **Create New Queue**.
4. Configure the following settings:

Field	Value
Queue Status	Active
Queue Name	Demo Queue
Description	Automatic assignment of critical alerts
Module Types	Alerts (click Add)

5. Click **Next: Define Rules**.
6. Click **Add Rule**.
7. In the **Rule Name** field, type `Demo Rule`.
8. In the **When Record Type** fields, select **Alert** and **Created**.
9. In the logic field, select **ALL OF THE BELOW ARE TRUE (AND)**.
10. Click **Add Condition**.
11. In the condition fields, select **Severity**, **Equals**, and **Critical**.

Rule Name

Demo Rule

When Record Type **Alert** Is **Created**

ALL OF THE BELOW ARE TRUE (AND)

Severity Equals Critical

+ ADD CONDITION + ADD CONDITIONS GROUP

12. Click **Next: Define Members**.
13. In the **Add records by** field, select **Teams**.
14. Select **SOC Team**, and then click **Add**.
15. In the **Queue Leader** field, select **CS Admin**.
16. In the **Queue Owners** field, select **SOC Team**, and then click **Add**.
17. Ensure the **Update Record Ownership To Match Queue Ownership** checkbox is selected.

Add records by Queue Leader

Teams Select teams... Add CS Admin (me)

1 Items

	First Name	Last Name	Title	Work Phone	Company
☐	Search	Search	Search	Search	Search
☐	CS	Admin	Admin	116462759691	

« < 1 of 1 > » Items Per Page 30

Queue Owners

Select teams + Add

SOC Team X

☒ Update Record Ownership To Match Queue Ownership

18. Click **Next: Configure User Assignment**.
19. In the **Select Assignment Method** field, select **Assign to Queue Lead**.
20. Ensure that the **Enable Shift-Based Assignment** checkbox is cleared.



In a real-world environment with more analysts, you can enable shift-based assignment. This allows you to automatically assign all records either to a lead or in a round-robin fashion to analysts currently working.

21. Click **Save and Close**.

To generate a shift for a calendar year

1. In the **Queue & Shift Management** section, click the **Shift Management** tab.
2. Click **Generate Shifts**.
3. Configure the following settings:

Field	Value
Name	Demo Shift
Start Time	08 00
Timezone	America/Toronto
Duration (Hours)	8 hours 0 minutes
Days Of The Week	Monday, Tuesday, Wednesday, Thursday, Friday
Starting On	Current Date
# Of Weeks To Generate	52
Add Shift Members By	Team
Select teams...	SOC Team (click Add)
Shift leader	CS Admin

4. Click **Generate**.
5. Review the shifts that the wizard generated.

To create alerts and validate auto assignment

1. Click **Incident Response > Alerts**.
2. Click **Add**, and then configure the following settings:

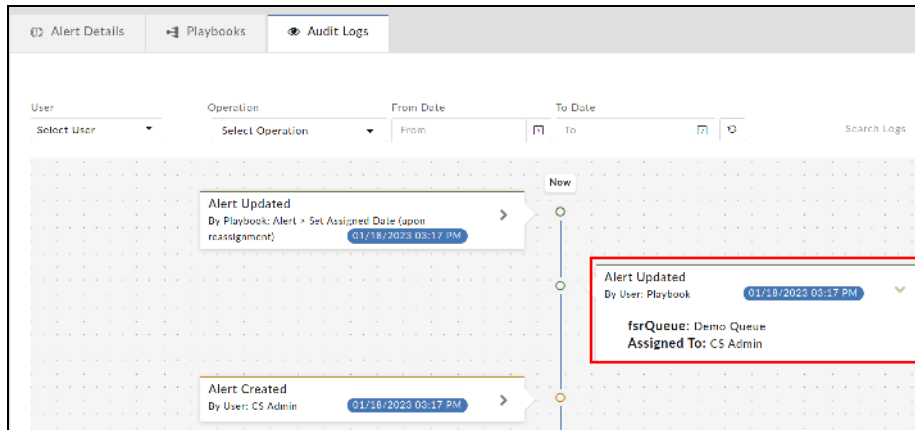
Field	Value
Name	Demo minimal severity alert
Description	Pings coming from 10.77.77.77
Severity	Minimal

3. Click **Save**.
4. Click **X** to close the record.
5. Click **Add**, and then configure the following settings:

Field	Value
Name	Demo critical severity alert
Description	Suspicious host with IP 10.254.254.254

Field	Value
Severity	Critical

- Click **Save**.
- Click the **Audit Logs** tab.
- Click > to expand the **Alert Updated** entries.



You previously configured a queue rule to assign new critical alerts to the **Demo Queue**. The minimal severity alert, on the other hand, does not have the same action in its audit logs, and remains unassigned.

- Log out of the FortiSOAR GUI.

Exercise 4: Examining the Indicators Module

In this exercise, you will review the **Indicators** module.

The **Indicators** module contains details for all data that FortiSOAR collects from system log entries or files that identify potentially malicious activity on a system or network. It contains records of identifiable information regarding a threat, such as an IP address or URL.

After FortiSOAR creates an alert, it extracts the metadata from the raw alert data, and then creates indicators with details. These details can include:

- The type of indicator, such as the IP address, URL, attachment, or domain
- The value of the indicator, such as the IP address or domain name
- Whether the indicator has been sighted in any other alerts
- The indicator of compromise (IOC) status of the indicator

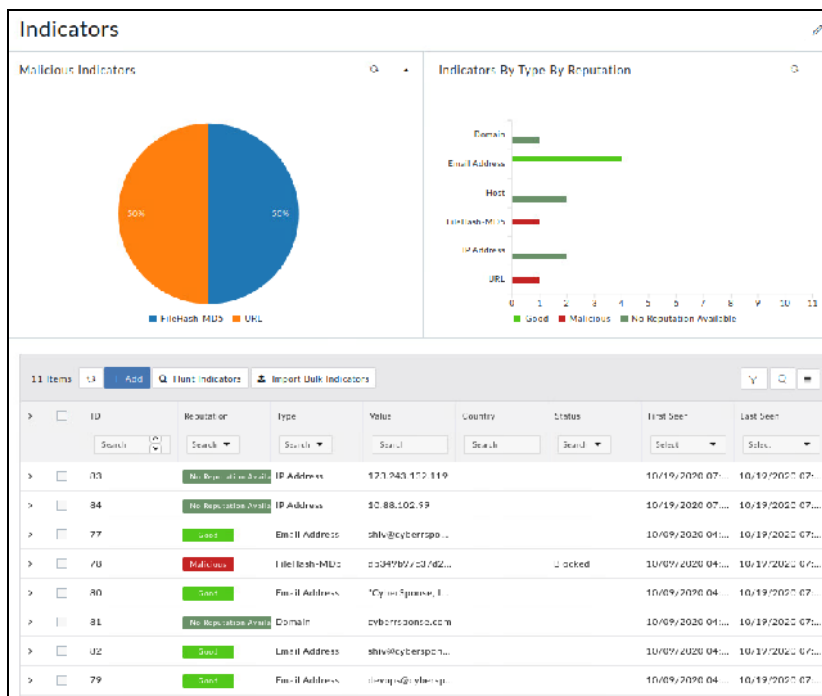
Examine the Indicators Module

You will examine the **Indicators** module.

To examine the Indicators module

1. Log in to the FortiSOAR-Primary GUI with the username `csadmin` and password `Fortinet1!`.
2. Click **Threat Intelligence > Indicators**.

By default, the module contains two charts at the top, followed by a grid column that displays the various types of indicators.





If the **Malicious Indicators** and **Indicators By Type By Reputation** chart windows are collapsed, click the arrows to expand them. Note that the data you see on your FortiSOAR instance is different from what appears in this lab guide.

- Click the **IP Address 10.254.254.254** indicator record, which was automatically created when you created the critical severity alert in the previous exercise.

>	☐	ID	Reputation	Type	Value	First Seen	Last Seen	Country	Expires
		Search	Search	Search	Search	Select	Select	Search	Select
>	☐	21	TBD	IP Address	10.254.254.254	02/28/2023...	02/28/2023...		03/20
>	☐	20	TBD	IP Address	10.77.77.77	02/28/2023...	02/28/2023...		03/20

- Review the fields in the record.
- In the **Sightings** section, click the **Alerts** tab.
- Confirm that the **Demo critical severity** alert has been automatically linked to this indicator.

Sightings

Source Data

Access Control

Audit Log

Alerts 1

Campaigns 0

Hunts 0

Events 0

Assets 0

Incidents 0

Indicators 0

Tasks 0

War Rooms 0

1 Items

+ Add

Link

Execute

☐	Name	Type	Severity	Status	Assigned To
	Search	Search	Search	Search	Search
☐	Demo critical alert	Other / Unknown	critical	Open	CS Admin

<

1

>

of 1

Items Per Page 30

- Click the **Audit Log** tab.
- Expand the **Indicator Created** and **Alert Linked** events, and then review the messages.

Indicator Updated
By Playbook: Enrich Indicators (Type All)
04/19/2023 01:22 PM

Indicator Updated
By Playbook: Indicator - Set First Seen Date
04/19/2023 01:22 PM

Alert Linked
By Playbook: Extract Indicators
04/19/2023 01:22 PM
Alert: Demo critical severity alert

Indicator Created
By Playbook: Extract Indicators
04/19/2023 01:22 PM

TLP: "Amber"
Type: "IP Address"
UUID: "062dceb-98b5-4d58-9283-311a41f5bc90"
Value: "10.254.254.254"
Status: "TBD"
Expired: "No"
Last Seen: "04-19-2023 12:22 PM UTC"
Expires On: "05-09-2023 12:22 PM UTC"
Reputation: "TBD"
recordTags: []
Enrichment Status: "None"

- Log out of the FortiSOAR GUI.

Exercise 5: Examining the Playbooks Module

The **Automation** modules include the **Connectors**, **Playbooks**, **Rule Engine**, **Schedules**, and **SLA Templates** modules. In this exercise, you will review the **Playbooks** module.

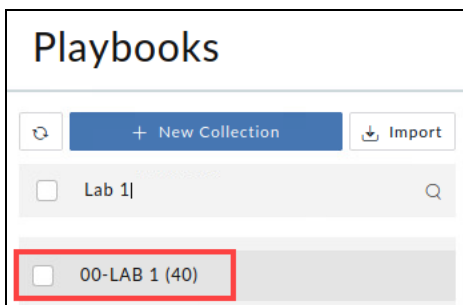
The **Playbooks** module is one of the most important modules on FortiSOAR. This is where all the playbook collections are located, and where you design playbooks.

Examine the Playbooks Module

You will review the **Playbooks** module and familiarize yourself with the playbook designer.

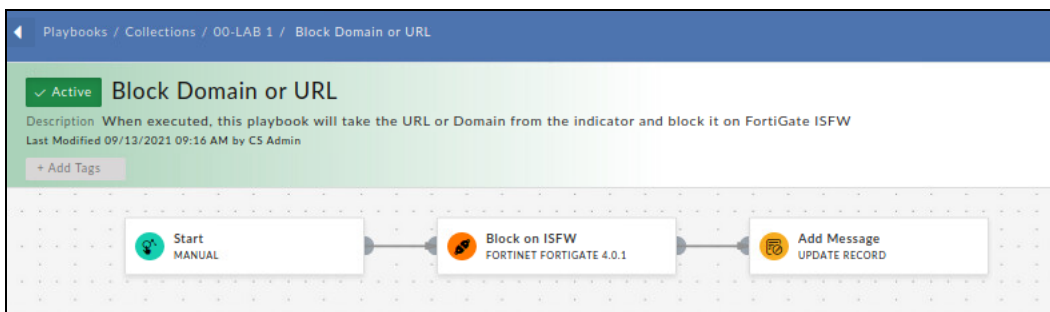
To review a playbook

1. Log in to the FortiSOAR-Primary GUI with the username `csadmin` and password `Fortinet1!`.
2. Click **Automation > Playbooks**.
3. In the **Search Collection** box, type `Lab 1`.



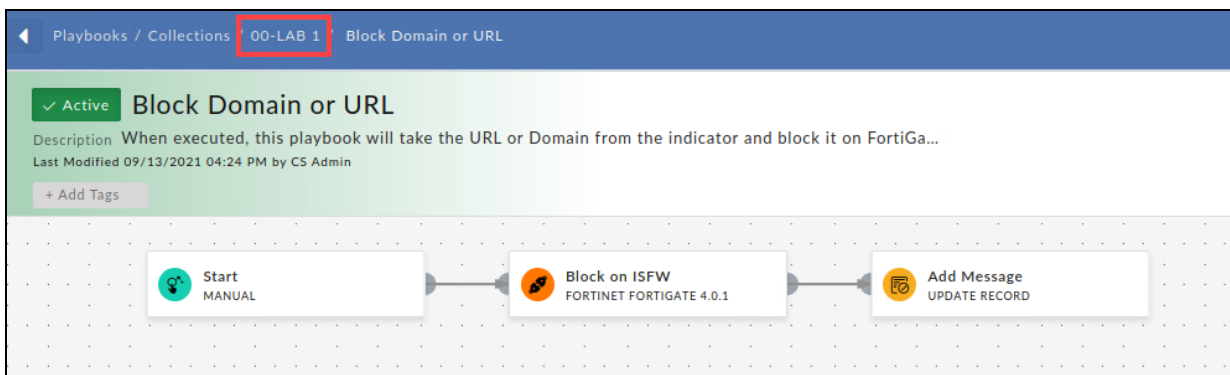
4. Click **Block Domain or URL**.

This is a three-step playbook that, when executed, will take the URL or domain from the indicator, and block it on the FortiGate.



To review the playbook designer

1. Continuing in the **Playbooks** module, click the `00-LAB 1` folder to return to the playbook collection.

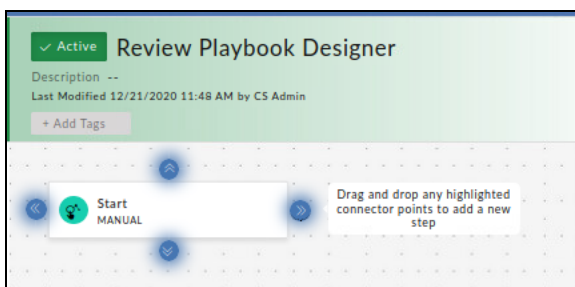


2. Click **Add Playbook**.
3. In the **Name** field, type `Review Playbook Designer`.
4. Click **Create**.

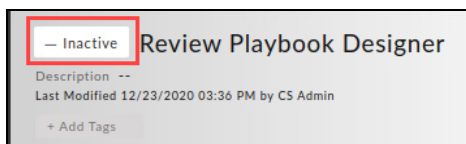
The playbook designer opens, and displays the option to add a trigger step for the playbook.

5. Review the trigger types.
6. Click **Manual**.
7. In the **Step Name** field, type `Start`.
8. Select **Does not require a record input to run**.
9. In the **Choose record modules on which the playbook would be available** drop-down list, select **Alerts**.
10. Click **Save** to add the trigger step to the playbook designer.

Now, you can add other steps to the playbook.

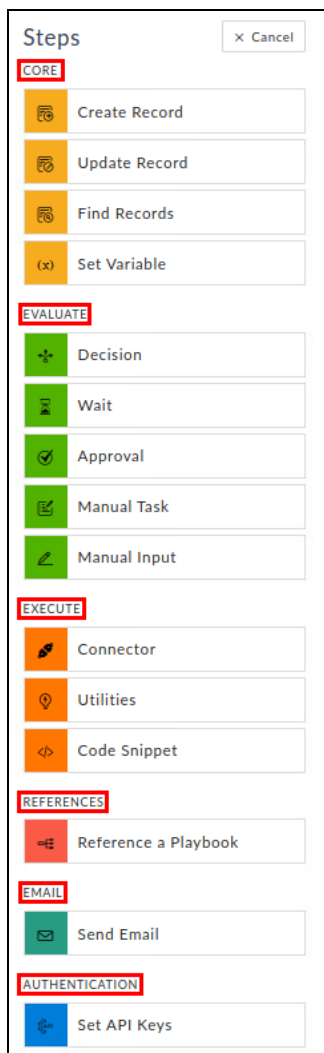


If the playbook is not **Active**, toggle it from **Inactive** to **Active**.



11. Drag and drop any of the highlighted blue connectors to add a new playbook step.
This automatically opens a new pane with the option to select a playbook step.

The steps are categorized as **CORE**, **EVALUATE**, **EXECUTE**, **REFERENCES**, **EMAIL**, and **AUTHENTICATION**.



12. Click **Create Record**.

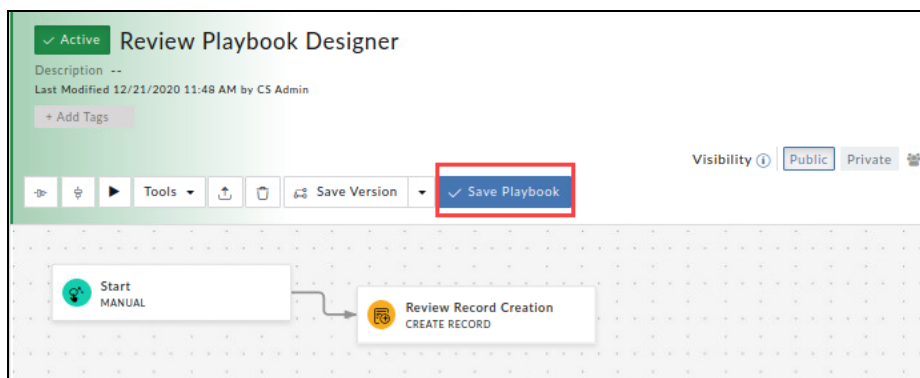
This step creates a new record in the database of a module that you select.

13. Configure the following settings (if you do not see some fields, click **Load More Fields**):

Field	Value
Step Name	Review Record Creation
Model	Alerts
Name	Review Record Creation
Severity	Low
Status	Open
Type	Other/Unknown

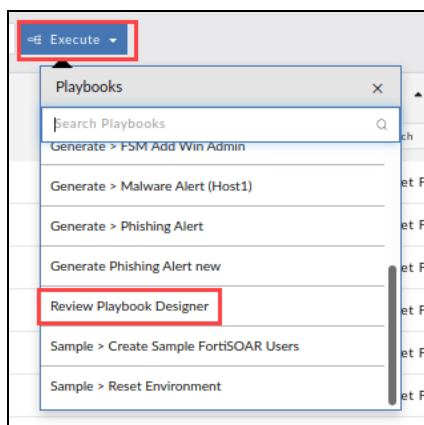
Field	Value
Destination Port	53
Domain	google.com
Source	Windows 10
Source IP	10.10.10.10
Source Port	9876
Tags	Sample

14. Click **Save**.
15. Click **Save Playbook**.



To execute a playbook

1. Continuing on the FortiSOAR GUI, click **Incident Response > Alerts**.
2. Click **Execute**.
3. Select **Review Playbook Designer**.



FortiSOAR executes the playbook, and creates a sample record in the **Alerts** module.

4. Click the **Review Record Creation** alert that was just created.
The alert opens in a new window.

- Review the record and the information in the record, including the extracted indicators from this alert.

The screenshot displays the FortiSOAR interface. At the top, a 'Sample' tab is selected. Below it, a 'Details' section shows a playbook record with the following fields: 'Assigned To' (Select), 'Status' (Open), 'Source' (Windows 10), 'Source ID' (), 'Assigned Date' (), 'Resolved Date' (), and 'Escalated' (No). The 'Type Details' section shows 'Type' (Other / Unknown), 'Source IP' (10.10.10.10), 'Destination IP' (8.8.8.8), 'Source Port' (8076), and 'Destination Port' (80). Below this, a 'Brokers' tab is selected, showing a table of alerts. The table has columns for 'Alert ID', 'Alert Name', 'Alert Type', 'Alert Status', 'Alert Severity', 'Alert Source', 'Alert Destination', 'Alert Action', 'Alert Result', and 'Alert Count'. The table contains three rows of alerts, all with a status of 'No Expansion Available'.

Alert ID	Alert Name	Alert Type	Alert Status	Alert Severity	Alert Source	Alert Destination	Alert Action	Alert Result	Alert Count
01/17/2023 04:32 PM	01/17/2023 04:32 PM	IPID	Alert	High	10.10.10.10	8.8.8.8	Port	No Expansion Available	10
01/17/2023 05:00 PM	01/17/2023 05:00 PM	TCP	Alert	High	10.10.10.10	8.8.8.8	Port	No Expansion Available	10
01/17/2023 05:00 PM	01/17/2023 05:00 PM	TCP	Alert	High	10.10.10.10	8.8.8.8	Port	No Expansion Available	10

- Log out of the FortiSOAR GUI.

Exercise 6: Configuring Data Ingestion

In this exercise, you will configure data ingestion to automatically pull security events from FortiAnalyzer into FortiSOAR.

Ingest Data from FortiAnalyzer

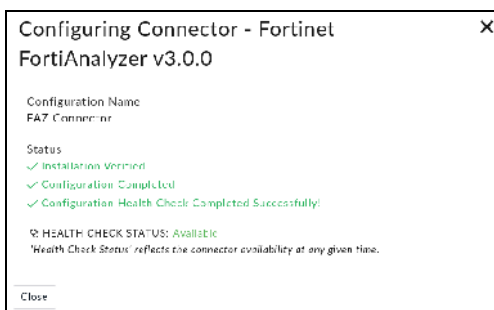
You will configure the FortiAnalyzer connector on FortiSOAR, generate a web filter violation on FortiGate, which will create an event on FortiAnalyzer, and then ingest data into FortiSOAR.

To configure the FortiAnalyzer connector on FortiSOAR

1. Log in to the FortiSOAR-Primary GUI with the username `csadmin` and password `Fortinet1!`.
2. Click **Automation > Data Ingestion**.
3. In the **Fortinet FortiAnalyzer** connector section, click **Add Configuration**.
4. Configure the following settings:

Field	Value
Configuration Name	FAZ Connector
Mark As Default Configuration	Enabled
Server URL	10.0.1.242
Username	admin
Password	Fortinet1!
ADOM Name	root
Port	443
Verify SSL	Disabled

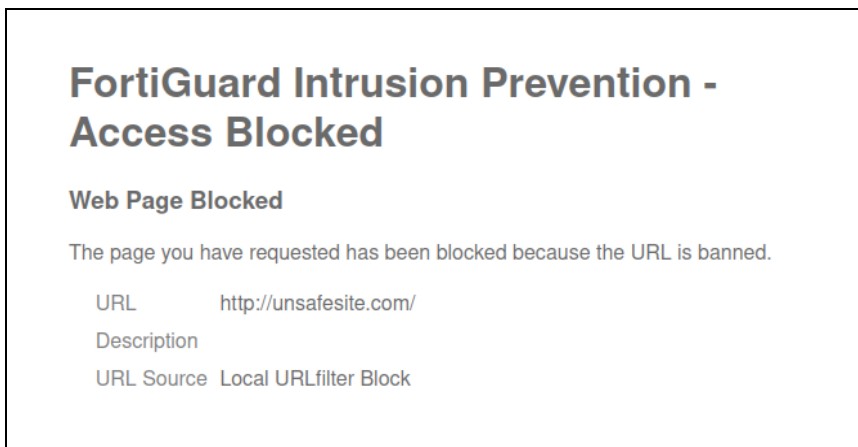
5. Click **Save**.
6. Ensure that the **HEALTH CHECK STATUS** is **Available**.



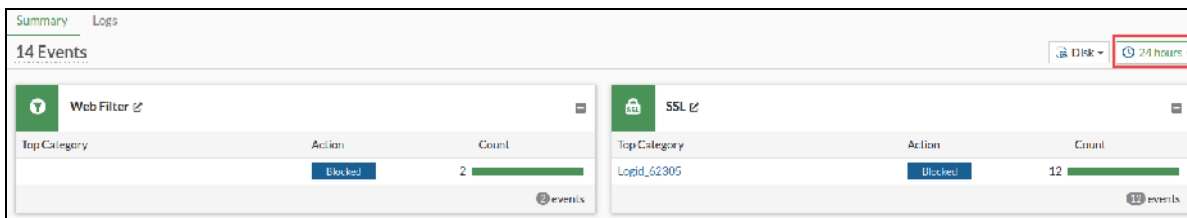
7. Close the connector configuration window.

To generate an event on FortiAnalyzer

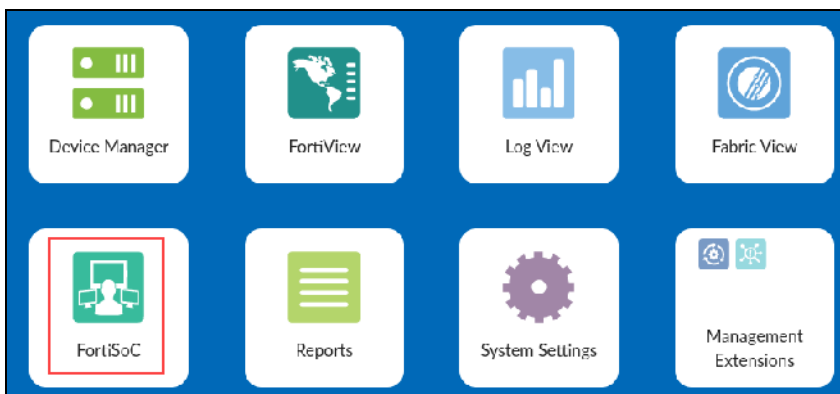
1. In Firefox, open a new browser tab, and then browse to the URL `www.unsafesite.com`.
2. Confirm that you see a web filter violation.



3. Log in to the FortiGate GUI with the username `admin` and password `Fortinet1!`.
4. Click **Log & Report > Security Events > Web Filter**.
5. In the time frame field, select **24 hours**.



6. In the **Summary** section, click **Web Filter** to view the logs.
7. Double-click any entry to see more information.
8. Log in to the FortiAnalyzer GUI with the username `admin` and password `Fortinet1!`.
9. Click **FortiSoC**.



10. Click **Event Monitor > All Events**.
11. In the **Event Type** field, enter `"webfilter"`.

#	Event	Event Status	Event Type	Count	Severity	First Occurrence	Last Update	Additional Info	Handler
1	URL Filter Created	Success	Web Filter	2	Medium	2023-01-16 08:37...	2023-01-16 08:37...		FortiGate Web Filter
	URL Filter Created	Success	Web Filter	1	Medium	2022-12-01 13:43...	2022-12-01 13:43...		FortiGate Web Filter
	URL Filter Created	Success	Web Filter	3	Medium	2022-12-01 08:33...	2022-12-01 08:35...		FortiGate Web Filter

12. Click the down arrow and review the **URL Filter Created** event in more detail.
13. Click **Handlers > Event Handler List**, and then double-click **FortiGate Web Filter**.
14. Review the event handler configuration.

Status	Name	Filters
✓	Default-Malicious-Code-Detection-By-Endpoint	> 8 Filters
✓	Default-Compromised Host-Detection-IOC-By-Endpoint	> 3 Filters
✓	Default-Botnet-Communication-Detection-By-Endpoint	> 9 Filters
✓	FortiGate Web Filter	> 1 Filter

To ingest data into FortiSOAR

1. Return to the FortiSOAR GUI—if the session has expired, log in again with the username `csadmin` and password `Fortinet1!`.
2. Click **Automation > Data Ingestion**, and then click the down arrow for **Fortinet FortiAnalyzer**.

Fortinet FortiAnalyzer Connector Version 1.2.0	1 Configuration Available	Self	Agent	Set Playbook Logging Levels
---	---------------------------	------	-------	-----------------------------

3. Click **Configure Ingestion**.
The data ingestion wizard opens.
4. Click **Let's start by fetching some data**.
5. Configure the following settings:

Field	Value
Search Query	eventtype='webfilter'
Limit	10
Maximum Logs To Pull Per Event	5
Pull Sample Events in Past X Minutes	2880

6. Click **FETCH DATA**.
7. Accept the default field mappings, and then click **Save Mapping & Continue**.

- On the **Scheduling Data Ingestion** page, configure the following settings:

Field	Value
Do you want to schedule the ingestion?	Yes
Configure Schedule Settings	Every X minutes - Set the value to every 2 minutes (* / 2)

Schedule Settings

Do you want to schedule the ingestion?

Yes

Configure Schedule Settings

By : **Every X minutes** | Hourly | Daily | Weekly | Monthly | Yearly

* / 2	*	*	*	*
minute	hour	day of month	month	day of week

" Every 2 minutes "

☐ Limit execution to one active instance at a time ⓘ

- Click **Save Settings & Continue**.
- Click **Done**.

To review the ingested alerts

- Wait 5 minutes to allow data ingestion to occur, and then click **Incident Response > Alerts**.
- Double-click the new **URL Filter Created** alert.

21 Items												Add	Execute	Deploy Scenario					
☐	ID	Priority	Weight	Severity	Assigned To	Name	Source	Type	Status	Created On	Escalated								
	Search	Search	Search	Search	Search	Search	Search	Search	Search	Search	Search								
☐	31	1	Medium			URL Filter Created	Fortinet FortiAnalyzer	Other / Unknown	Open	04/19/2023 02:18 ...	No								
☐	30	1	Low			Review Record Creation	Windows 10	Other / Unknown	Open	04/19/2023 01:35 ...	No								
☐	29	1	Critical	CS Admin		Demo critical severity alert		Other / Unknown	Open	04/19/2023 01:22 ...	No								

- Review the **Description** and **Details** sections.

4. Click the **Audit Logs** tab.
5. Review the **Alert Created** entry.

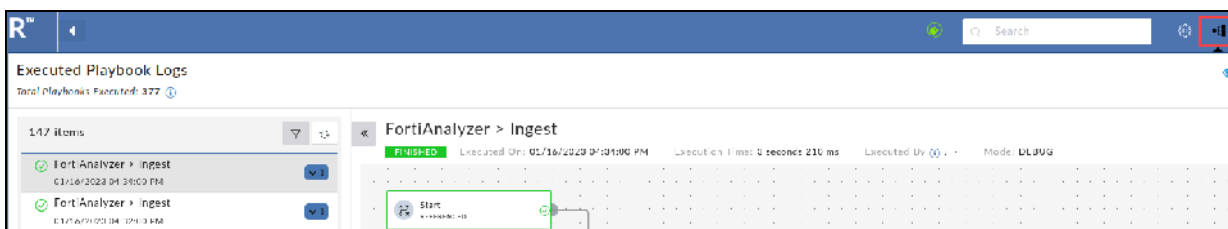
FortiSOAR Administrator 7.3 Lab Guide
Fortinet Technologies Inc.



The FortiGate and FortiAnalyzer were preconfigured for this lab exercise. The following items were configured:

- Web filter profile
- Firewall policy
- Sending logs to FortiAnalyzer
- Event handler to generate incidents on FortiAnalyzer

6. Click the **Executed Playbook Logs** icon, and then review the scheduled ingestion you configured earlier in this exercise.



This lab exercise shows only one possible configuration to automatically ingest data from other sources. FortiSOAR offers a vast number of connectors in the **Content Hub** for you to explore.

7. Log out of the FortiSOAR GUI.

Lab 2: Teams and Roles

In this lab, you will configure the SMTP connector on FortiSOAR. You will also create additional FortiSOAR administrator users, establish a team hierarchy, assign roles to users, and review record ownership.

Objectives

- Configure the SMTP connector
- Create an additional FortiSOAR administrator account
- Create teams, alerts, and review the team hierarchy and team relationships
- Create roles and assign roles to users

Time to Complete

Estimated: 45 minutes

Exercise 1: Configuring the SMTP Connector

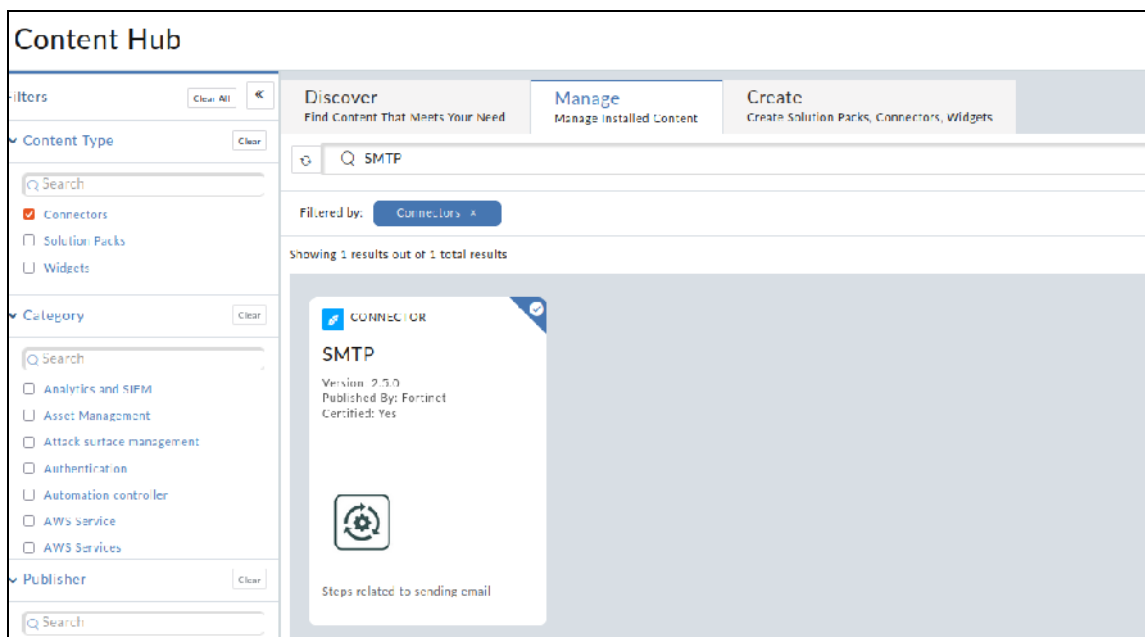
In this exercise, you will configure the SMTP connector. It is an important configuration so that FortiSOAR users can receive email alerts and password notifications, and monitor system health.

Configure the SMTP Connector

You will configure the SMTP connector to integrate with FortiMail.

To configure the SMTP connector

1. Log in to the FortiSOAR-Primary GUI with the username `csadmin` and password `Fortinet1!`.
2. Click **Automation > Connectors**.
3. On the **Manage** tab, search for **SMTP**, and then click the **SMTP** connector to edit it.



4. In the **Select Configuration** field, select **Add New Configuration**.
5. In the **Configuration Name** field, type `FortiMail`.
6. Select the **Mark As Default Configuration** checkbox.
7. In the **HOST** field, type `10.0.1.200`.
8. In the **Default From** field, type `soar@internal.lab`.
9. In the **Username** field, type `soar@internal.lab`.
10. In the **Password** field, type `Fortinet1!`.
11. Select the **Use TLS** checkbox.

Your configuration should match the following example:

The screenshot shows the 'SMTP' configuration page in FortiSOAR. At the top, it displays the connector icon, name 'SMTP', version '2.4.2', and status 'Certified: Yes'. Below this, it indicates 'Steps related to sending email' and shows the connector is 'Active' with a 'Deactivate Connector' button. A green checkmark confirms it is 'Compatible with your integration'. The 'Configurations' tab is selected, showing a list of configurations with 'FortiMail' as the default. The configuration details for 'FortiMail' include: 'Mark As Default Configuration' (checked), 'Visibility' set to 'Public', 'Host' (HOST) as '10.0.1.200', 'Port' as '25', 'Default From' as 'soar@internal.lab', 'Username' as 'soar@internal.lab', and a masked 'Password'. A note states: 'Password fields are write only. If you do not change this field, your Password will not be overwritten.' The 'Use TLS' checkbox is checked. The 'Timeout' is set to '10' seconds.

12. Click **Save**.
13. Verify that the **HEALTH CHECK STATUS** is **Available**.

The screenshot shows a dialog box titled 'Configuring Connector - SMTP v2.5.0'. It displays the configuration name 'FortiMail' and a 'Status' section with three green checkmarks: 'Installation Verified', 'Configuration Completed', and 'Configuration Health Check Completed Successfully'. Below this, it shows 'HEALTH CHECK STATUS: Available' with a green checkmark. A note at the bottom states: 'Health Check Status reflects the connector availability at any given time.' A 'Close' button is located at the bottom left of the dialog.

14. Close the SMTP connector.
15. Log out of the FortiSOAR GUI.

Exercise 2: Creating FortiSOAR Administrator Accounts

In this exercise, you will create a FortiSOAR administrator account, and then reset the password using the SMTP connector you configured in the previous exercise.

Create an Administrator Account

All new users, including the **csadmin** user, must change their passwords when they first log in to FortiSOAR, regardless of the complexity of the passwords assigned to users.

To create an administrator account

1. Log in to the FortiSOAR-Primary GUI with the username **csadmin** and password **Fortinet1!**.
2. Click **Settings**.
3. In the **Security Management** section, click **Users**.

Security

System

- System Configuration
- Audit Log
- License Manager
- Notifications
- Data Archival

Agent Configurations

- Secure Message Exchange
- Agents
- Security Management**
- Team Hierarchy
- Teams
- Roles
- Users**
- Appliances
- Authentication
- Password Vault

Users

This section is for managing FortiSOAR users, allowing system admins to create users, map roles & teams, set access type, authorization and other important user details.

Designate Super Admin ⓘ
Select Appropriate Super Admin User:

Select User
CS Admin ☒ Set As Super Admin

Manage Users

6 Items

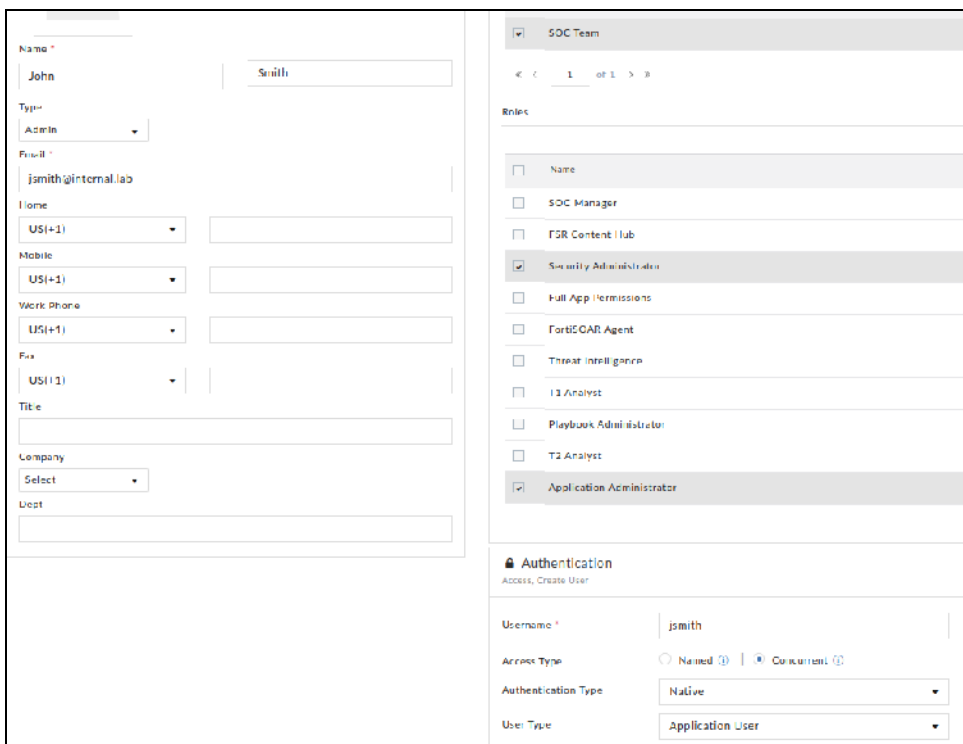
☐	First Name	Last Name	Locked	Title	Work Phone	Company	Type	Login Status
☐	CS	Admin	No	Admin	+16162759691			
☐	France Analyst	Manager	No				Employee	
☐	New SOC	Manager	No				Employee	
☐	US Analyst	Manager	No				Employee	
☐	US L1	Analyst	No				Employee	
☐	US L2	Analyst	No				Employee	

<< 1 of 1 >>

Items Per Page: 30

4. Click **Add** to add a new user.
5. In the **Name** fields, type **John** and **Smith**.
6. In the **Type** field, select **Admin**.
7. In the **Email** field, type **jsmith@internal.lab**.
8. In the **Teams** section, select the **SOC Team** checkbox.
9. In the **Roles** section, select the **Security Administrator** and **Application Administrator** checkboxes.

10. In the **Authentication** section, confirm that the **User Type** is **Application User**.
11. In the **Username** field, type `jsmith`.



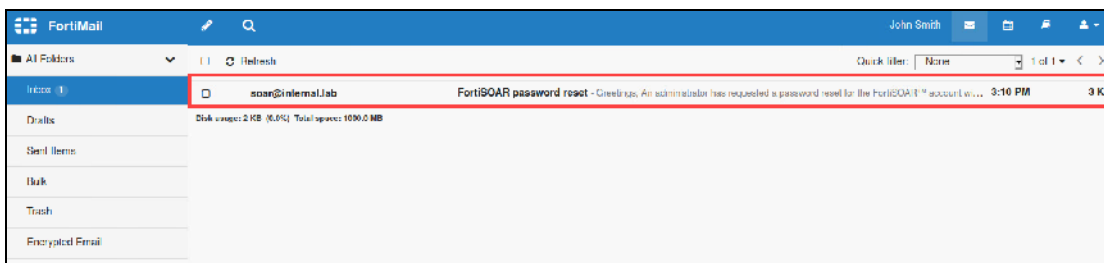
12. Click **Save**.



Notice that we did not specify a password for the user John Smith. This is because all new users, including the **csadmin** user, must change their passwords when they first log in to FortiSOAR, regardless of the complexity of the passwords assigned to users.

To reset a password

1. In the Firefox browser, open a new tab, and then go to the FortiMail webmail GUI.
2. Log in with the username `jsmith` and password `Fortinet1!`.
3. Open the **FortiSOAR password reset** email from `soar@internal.lab`.



4. Click the link in the email body.
The link takes you to the FortiSOAR password reset page.
5. In the **New Password** and **Confirm Password** fields, type `Fortinet1!`.

FortiSOAR™ v7.3.0-2034

RESET PASSWORD

Password must conform the following restrictions:

- Your password must have at least 8 characters
- Password must contain one lower-case alphabet, one upper-case alphabet, one digit, and any one of these special characters ~!@#%*^&*!)?_

New Password

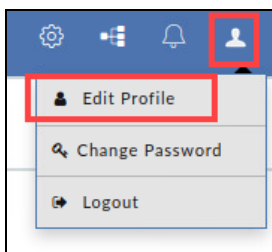
Confirm Password

CHANGE PASSWORD

6. Click **Reset Password**.

To review user profile options

1. Log in to the FortiSOAR GUI with the username `jsmith` and password `Fortinet1!`.
2. In the upper-right corner, click the user profile icon, and then click **Edit Profile**.



More user-specific tabs, including **2-Factor**, **Notifications**, and **Theme Settings** are displayed.

Authentication

Access, Reset Password

Username *

Access Type ☐ Named ⓘ | ☒ Concurrent ⓘ

Authentication Type

User Type

2-Factor

Edit 2-Factor Authentication settings

Notifications

Edit Notification Settings

Theme Settings

Edit Your Theme

3. Log out of the FortiSOAR GUI.

Exercise 3: Configuring a Team Hierarchy and Examining Record Ownership

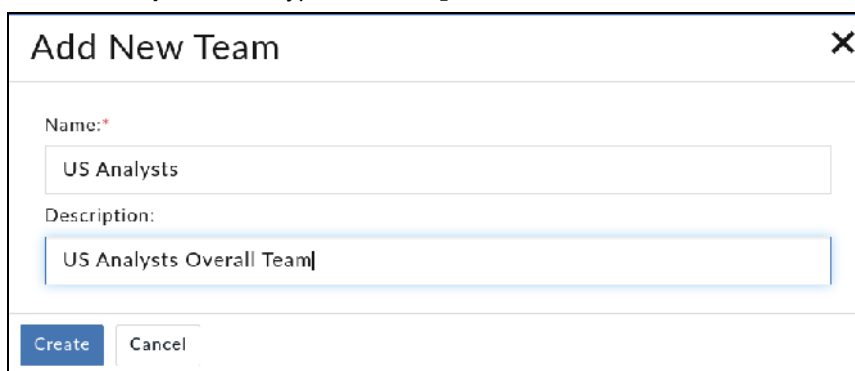
In this exercise, you will configure a team hierarchy and examine record ownership based on it. In FortiSOAR, it is important to carefully design a team hierarchy so that users can only view data that they are supposed to. You will establish a new team hierarchy, create relationships, and then log in as different users to validate your configuration.

Create Teams, Configure a Hierarchy, and Examine Record Ownership

You will create teams, establish a hierarchy, create alerts, and then examine which records are viewable based on configured team relationships.

To create teams

1. Log in to the FortiSOAR GUI with the username `csadmin` and password `Fortinet1!`.
2. Click **Settings**.
3. In the **Security Management** section, click **Teams**.
4. Click **Add**.
5. In the **Name** field, type `US Analysts`.
6. In the **Description** field, type `US Analysts Overall Team`.



7. Click **Create**.
8. In the **Assign Team Members** section, click **Link**.
9. Select **US Analyst Manager**, and then click **Save Relationship**.
10. In the **Security Management** section, click **Teams**.
11. Repeat steps 4–9 to create the following additional teams:

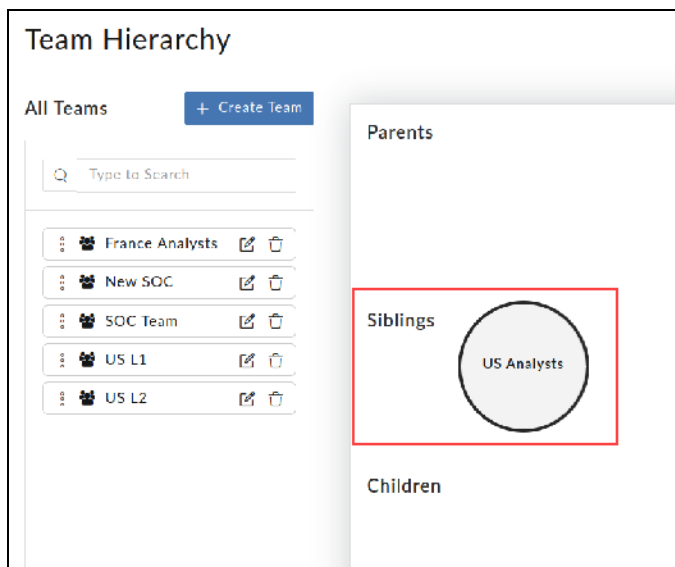
Name	Description	Team member
US L1	US L1 Team	US L1 Analyst

Name	Description	Team member
US L2	US L2 Team	US L2 Analyst
France Analysts	France Analysts Overall Team	France Analyst Manager
New SOC	New SOC Parent Team	New SOC Manager

6 Items	+ Add
Name	Description
France Analysts	France Analysts Overall Team
New SOC	New SOC Parent Team
SOC Team	Management team
US Analysts	US Analysts Overall Team
US L1	US L1 Team
US L2	US L2 team

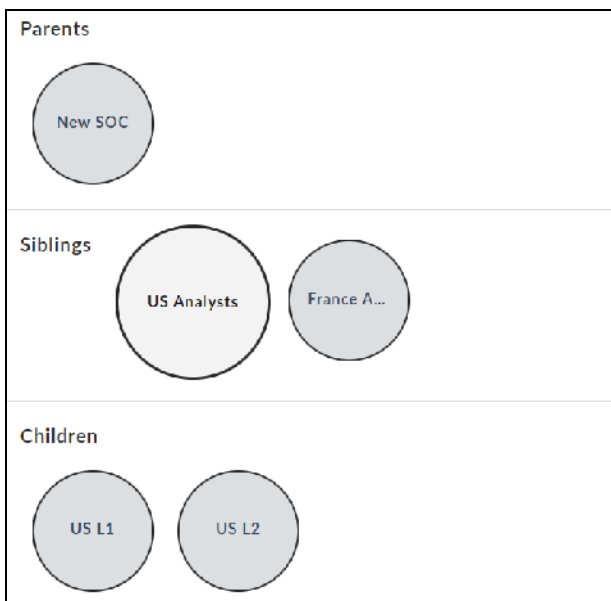
To configure the team hierarchy

1. Continuing on the FortiSOAR GUI, in the **Security Management** section, click **Team Hierarchy**.
2. Select the **US Analysts** team, and then drag it to the **Siblings** section.



3. Add the **France Analysts** team to the **Siblings** section.
4. Add the **New SOC** team to the **Parents** section.
5. Add the **US L1** and **US L2** teams to the **Children** section.

Your configuration should match the following example:



6. Click **Save**.
7. Click **Revert** to reset the team in focus.



The **Revert** button does not make any changes to existing hierarchies. It is used to reset the interface so you can create new hierarchies or change the team in focus. Before reverting, ensure that you have saved your configured hierarchies.

8. In the **All Teams** section, double-click the **France Analysts** team, and then review its relationships.
9. Click **Revert** to reset the team in focus.
10. In the **All Teams** section, double-click the **US L1** team, and then review its relationships.
11. Click **Revert** to reset the team in focus.

Stop and think!

Why does the **France Analysts** team not have a parent, while its sibling **US Analysts** does?

Why does the **US L1** team not have a sibling, even though both **US L1** and **US L2** are children of **US Analysts**?

In FortiSOAR, relationships must be explicitly defined. In other words, siblings can have different parents, and children of the same parent may not necessarily be siblings.

Additionally, keep in mind that grandparent teams can access records of their children, their grandchildren, their great grandchildren, and so on. You will see this demonstrated later in the exercise when you review the accessible records from the perspective of the **New SOC** team.

To create alerts

1. Continuing on the FortiSOAR GUI, click **Incident Response > Alerts**, and then click **Add**.
2. In the **Name** field, type `US Analysts Record`.
3. In the **Description** field, type `Owned by US Analysts team`.

4. Click **Save**.
5. Repeat steps 1–4 to create the following additional records:

Name	Description
US L1 Record	Owned by US L1 child team
US L2 Record	Owned by US L2 child team
France Analysts Record	Owned by France Analysts team
New SOC Record	Owned by New SOC parent team

To assign ownership to records

1. Continuing on the FortiSOAR GUI, click **Incident Response > Alerts**.
2. Sort alerts by the **Created On** column in descending order so that newest entries are listed first.

ID	Incident Weight	Severity	Assigned To	Name	Source	Type	Status	Created On
23	1	Low		New SOC Record	Either / Unknown	Open		01/19/2023 10:44 AM
25	1	Low		France Analysts Record	Either / Unknown	Open		01/19/2023 10:44 AM
26	1	Low		US L2 Record	Either / Unknown	Open		01/19/2023 10:44 AM
24	1	Low		US L1 Record	Other / Unknown	Open		01/19/2023 04:47 PM
33	1	Low		US Analysts Record	Other / Unknown	Open		01/19/2023 04:47 PM

3. Click the **US Analysts Record**.
4. In the **Access Control** section, click **+**, and then click **Assign Team**.

Severity: Alert-29 Us Analysts Record

Last Modified 01/17/2023 04:40 PM by CS Admin

Add Tags

Details

Assigned To: Select
Source: --
Assigned Date: Select Date

Status: Open
Source ID: --
Resolved Date: Select Date

Escalated: No

Type Details

Type: Select

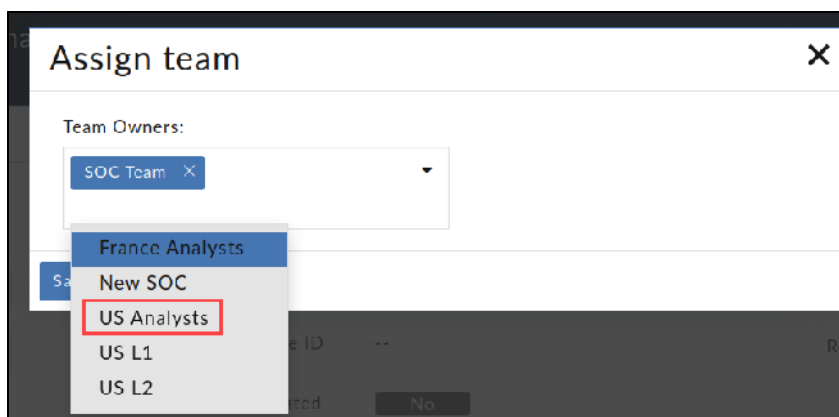
Indicators
Correlations
Tasks
Source Data
External Communications

Access Control

+

Assign Team

5. In the white box under **SOC Team**, click an open area, and then select **US Analysts**.
6. Leave **SOC Team** assigned.



7. Click **Save**.
8. Repeat steps 1–7 for the records in the following table:

Alert	Team Owners
US L1 Record	SOC Team, US L1
US L2 Record	SOC Team, US L2
France Analysts Record	SOC Team, France Analysts
New SOC Record	SOC Team, New SOC

9. Log out of the FortiSOAR GUI.

To validate access to records



As you perform this procedure, make a note of how the viewable records change for each user you log in as. Think of why only certain records are viewable when you are logged in as that user. Review the team hierarchy you set up previously, if required.

1. Log in to the FortiSOAR GUI with the username `usanalystmanager` and password `Fortinet1!`.
2. Click **Incident Response > Alerts**.
3. Log out of the FortiSOAR GUI.
4. Log in to the FortiSOAR GUI with the username `usl1analyst` and password `Fortinet1!`.
5. Click **Incident Response > Alerts**.
6. Log out of the FortiSOAR GUI.
7. Log in to the FortiSOAR GUI with the username `usl2analyst` and password `Fortinet1!`.
8. Click **Incident Response > Alerts**.
9. Log out of the FortiSOAR GUI.
10. Log in to the FortiSOAR GUI with the username `franceanalystmanager` and password `Fortinet1!`.
11. Click **Incident Response > Alerts**.
12. Log out of the FortiSOAR GUI.
13. Log in to the FortiSOAR GUI with the username `newsocmanager` and password `Fortinet1!`.

14. Click **Incident Response > Alerts**.
15. Log out of the FortiSOAR GUI.

The following list is a summary of viewable records per user:



- **US Analyst Manager:** All records except **New SOC**, which is its parent team
 - **US L1 Analyst:** Only its own record
 - **US L2 Analyst:** Only its own record
 - **France Analyst Manager:** Its own record and **US Analyst Manager's** record, which is explicitly defined as a sibling team
 - **New SOC Manager:** All records except **France Analyst Manager's** record, with which it has no relationship defined
-

Exercise 4: Creating Roles

In this exercise, you will create a role, assign the role to a user, and review access for a read-only user.

Create a Role and Assign a Role to a User

You will create a role and assign it to a user.

To create a role

1. Log in to the FortiSOAR-Primary GUI with the username `csadmin` and password `Fortinet1!`.
2. Click **Settings**.
3. In the **Security Management** section, click **Roles**.
4. Click **Add**.
5. In the **Role Name** field, type `Show Everything`.
6. In the **Description** field, type `Read permissions to all modules`.
7. In the **Set Role Permissions** section, in the **Read** column, click **+**.

New Role

Role Details

Role Name: `Show Everything`

Description: `Read permissions to all modules`

Set Role Permissions

Module	Create	Read	Update	Delete	Fields Permissions
Agents (1)	+ -	+ -	+ -	+ -	Set Field Permissions
Alerts (1)	+ -	✓	+ -	+ -	Set Field Permissions
Announcements (1)	+ -	✓	+ -	+ -	Set Field Permissions
Appliances (1)	+ -	✓	+ -	+ -	Set Field Permissions
Application (1)	+ -	✓	+ -	+ -	Set Field Permissions

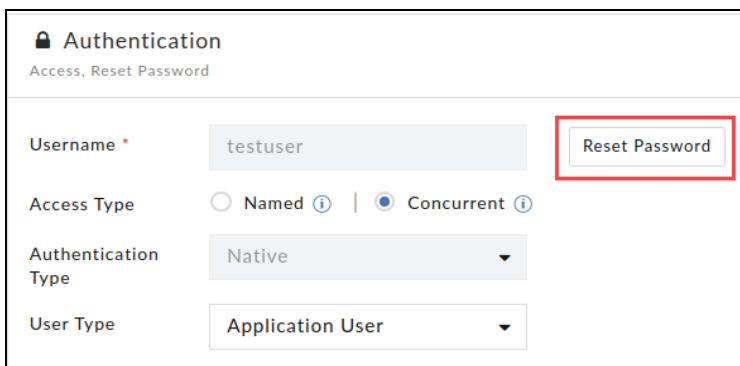
8. Click **Save**.

To create additional users

1. Continuing on the FortiSOAR GUI, click **Users**.
2. Click **Add**.
3. Configure the following settings:

Field	Description
Name	Test User
Type	Admin
Email	testuser@internal.lab
Roles	Show Everything
Teams	SOC Team
Username	testuser

- Click **Save**.
- Edit the **Test User** account.
- In the **Authentication** section, click **Reset Password**.



The screenshot shows the 'Authentication' section of the FortiSOAR GUI. It includes fields for 'Username' (testuser), 'Access Type' (radio buttons for 'Named' and 'Concurrent', with 'Concurrent' selected), 'Authentication Type' (Native), and 'User Type' (Application User). A red box highlights the 'Reset Password' button.

- In the **New Password** field, type `Fortinet1!`.
- In the **Confirm Password** field, type `Fortinet1!`.
- Select the **Send Email to User** checkbox.
- Click **Submit**.

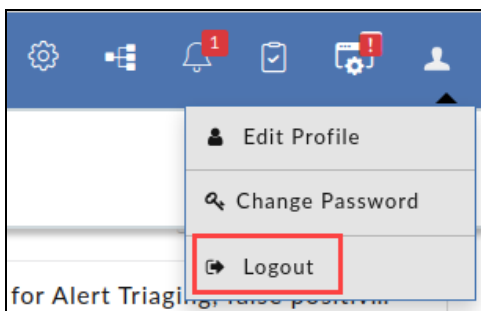
To check user role permissions

- On the FortiMail webmail GUI, log in with the username `testuser` and password `Fortinet1!`.
You should see the password reset email.

Stop and think!

Why does this email not contain a password reset link?

- Log out of the FortiMail webmail GUI.
- Return to the FortiSOAR GUI, and then log out of the **csadmin** account.
Ensure that you are logging out of the account and not only closing the browser tab.



4. Log in with the username `testuser` and password `Fortinet1!`.
5. Explore the FortiSOAR GUI.



Note that while you can view new tabs and configurations, you do not have the option to modify any content or settings since the **Show Everything** profile is read-only.

6. Log out of the FortiSOAR GUI, and then log back in with the username `csadmin` and password `Fortinet1!`.
7. Click **Settings**.
8. Click **Users**.
9. Edit the **Test User** account.
10. In the **Team and Role** section, add the **US Analysts** team.
11. Click **Save**.
12. In the **Security Management** section, click **Teams**.
13. Double-click **France Analysts**.
14. In the **Assign Team Members** section, click **Link**.
15. Select **Test User**.
16. Click **Save Relationship**.
17. In the **Security Management** section, click **Users**.
18. Edit the **Test User** account, and confirm that this user is part of the **SOC**, **US Analysts**, and **France Analysts** teams.



You can change team membership either through a user's profile under **Users** or link different users to a team through the **Teams** interface.

19. Log out of the FortiSOAR GUI.

Lab 3: Reports and Audit Logs

In this lab, you will create, schedule, and view reports. You will also view and purge audit logs, and manage the recycle bin.

Objectives

- Create, schedule, and view reports
- Manage audit logs
- Manage the recycle bin

Time to Complete

Estimated: 40 minutes

Exercise 1: Creating Reports

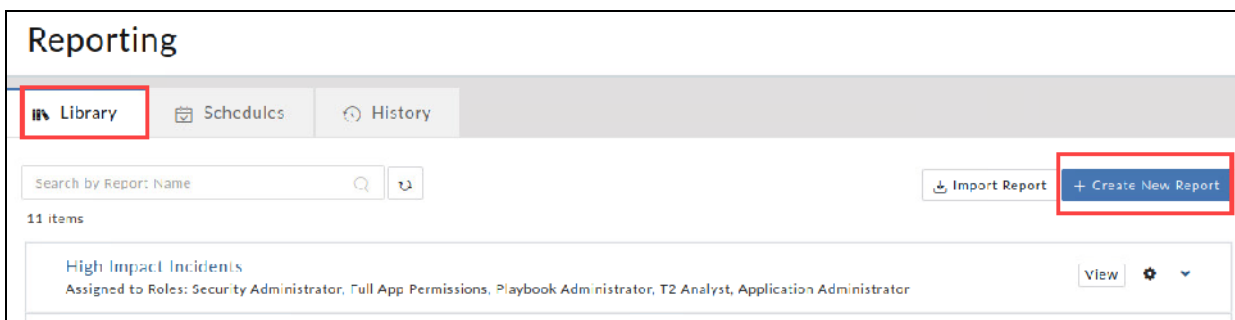
In this exercise, you will create, schedule, and run a simple report. Next, you will view the report.

Create and Schedule a Report

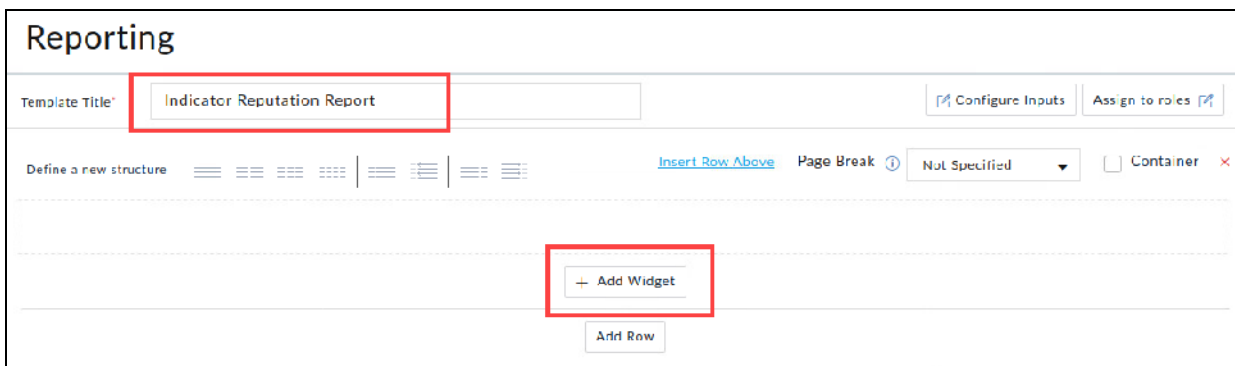
You will create a report template on indicators based on their reputations, and schedule the report to run every 5 minutes.

To create a report

1. Log in to the FortiSOAR-Primary GUI with the username `csadmin` and password `Fortinet1!`.
2. Click **Reports**.
3. Click the **Library** tab.
4. Click **Create New Report**.



5. In the **Template Title** field, type `Indicator Reputation Report`.
6. Click **Add Row**.
7. Click **Add Widget**.



8. In the **Charts and Metrics** section, select **Chart - Choose from multiple chart widgets like Bar, Pie, Donut etc.**
9. In the **Title** field, type `Indicator Reputations`.
10. In the **Data Source** field, select **Indicators**.
11. In the **Chart Type** field, select **Pie**.

12. In the **Filter Criteria** section, make sure **ALL OF THE BELOW ARE TRUE (AND)** is selected.
13. Click **ADD CONDITION**.
14. In the **Select a field** field, select **Reputation**.
15. In the **Equals** field, select **Is Null**.
16. In the **True** field, select **False**.

Filter Criteria

ALL OF THE BELOW ARE TRUE (AND)

Reputation Is Null False

+ ADD CONDITION + ADD CONDITIONS GROUP

17. In the **Slices** field, select **Reputation**.
- Your configuration should match the following example:

Title*

Indicator Reputations

Data Source*

Indicators

Chart Type*

Pie

Section Show/Hide

☒ Always Show ☐ Hide widget if its output has no records

Only Me | All (field) :

Filter Criteria

ALL OF THE BELOW ARE TRUE (AND)

Reputation Is Null False

+ ADD CONDITION + ADD CONDITIONS GROUP

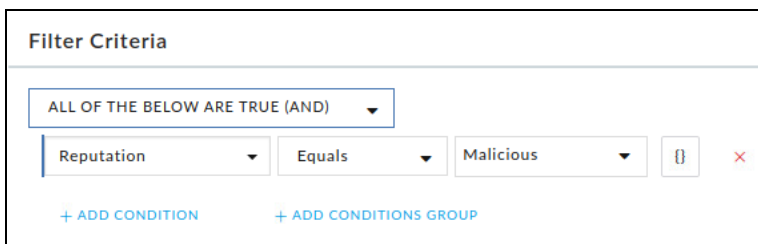
Slices*

Reputation

18. Click **Save**.
19. Click **Add Widget**.
20. In the **Record - Listing** section, select **Single Line Item - List record fields in a vertically stacked format**.
21. In the **Title** field, type `Malicious Indicators`.
22. In the **Data Source** field, select **Indicators**.
23. In the **Maximum Record Limit** field, select **50**.
24. In the **Single Line Item Template** section, in the **Select a field** field, select **Value**.
25. Click **Add Column**.

A screenshot of the 'Single Line Item Template' section. It features a dropdown menu with 'Value' selected, which is highlighted by a red rectangle. To the right of the dropdown is a blue button labeled 'Add Column'.

26. In the **Single Line Item Template** section, in the **Select a field** field, select **Type**.
27. Click **Add Column**.
28. In the **Single Line Item Template** section, in the **Select a field** field, select **Reputation**.
29. Click **Add Column**.
30. In the **Filter Criteria** section, confirm that **ALL OF THE BELOW ARE TRUE (AND)** is selected.
31. Click **ADD CONDITION**.
32. In the **Select a field** field, select **Reputation**.
33. Make sure the second field is set to **Equals**.
34. In the third field, select **Malicious**.

A screenshot of the 'Filter Criteria' section. At the top, a dropdown menu shows 'ALL OF THE BELOW ARE TRUE (AND)'. Below this, there are three input fields: 'Reputation', 'Equals', and 'Malicious', each with a dropdown arrow. To the right of these fields is a button with a plus sign and a red 'X' button. At the bottom, there are two blue links: '+ ADD CONDITION' and '+ ADD CONDITIONS GROUP'.

35. In the **Default Sort** section, click **ADD SORTING PARAMETER**.
36. In the **Select** field, select **Value**.
37. Confirm that **Ascending** is selected.

Your configuration should match the following example:

Edit Single Line Widget

Title*
Malicious Indicators

Data Source*
Indicators

Max Record Limit
50

Single Line Item Template

Select a field Add Column

Malicious Indicators

Only Me | All (field) :

Value × Type × Reputation ×

Filter Criteria

ALL OF THE BELOW ARE TRUE (AND) ▼

Reputation ▼ Equals ▼ Malicious ▼ {} ×

+ ADD CONDITION + ADD CONDITIONS GROUP

Default Sort

Value ▼ ● Ascending ○ Descending ×

+ ADD SORTING PARAMETER

38. Click **Save**.
39. Click **Configure Inputs**.

✍ Configure Inputs Assign to roles ✍

40. Click **Add New Input**.
41. In the **Input Type** field, select **Date Range**.
42. In the **Label** field, type *Dates*.
43. Confirm that the **Identifier** field is set to *dates*.
44. In the **Default Value** section, select **Relative**.
45. In the second field, select **Last Year**.
46. Select the **Required** checkbox.

Your configuration should match the following example:

Configure Inputs

☐ Enable Auto-refresh

DATES

Input Type *
Date Range

Label *
Dates

Identifier *
dates

Default Value
Relative Last Year

☒ Required

47. Click **Save**.

48. Click **Assign to roles**.

☒ Configure Inputs **Assign to roles** ☒

49. Select the **Application Administrator**, **Full App Permissions**, **Playbook Administrator**, **Security Administrator**, **T1 Analyst**, and **T2 Analyst** checkboxes.

Assign to Role(s)

Roles: Search

☒ **Application Administrator**
Full access to general application-wide features for system configuration

☐ **FortiSOAR Agent**
Agent appliances will be auto-assigned this role. Defaults to access to files and attachments

☒ **Full App Permissions**
Essentially the root user, use carefully

☒ **Playbook Administrator**
Permitted across all major modules as well as the Security role

☒ **Security Administrator**
Manages the Roles and Teams area of the administration menu

☐ **Show Everything**
Read permissions to all modules

☒ **T1 Analyst**
Responsible for Alert Triage, false positive filtering and escalating potentially malicious alerts to Incidents for review by T2 Analysts.

☒ **T2 Analyst**
Responsible for Incident Investigation and other remediation and containment related tasks.

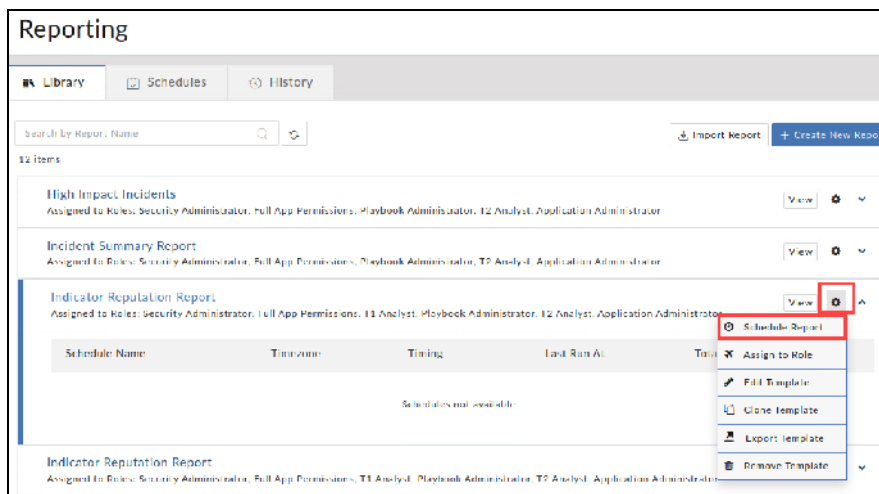
☐ **Threat Intelligence**
Responsible for researching Indicators, and updating/overriding the reputation

50. Click **OK**.

51. Click **Apply Changes**.

To schedule a report

1. Continuing on the FortiSOAR GUI, click **Reports**.
2. Click the **Library** tab.
3. In the **Indicator Reputation Report** row, click the **Settings** icon, and then select **Schedule Report**.



4. In the **Name** field, type `Email Report`.
5. Select the **Start Schedule** checkbox.
6. In the **Schedule Frequency** section, click **Every X minute**.
This changes the frequency to every 5 minutes, which is the default.
7. In the **Email Address** field, type `csadmin@internal.lab`.
Your configuration should match the following example:

Schedule Details

Report Inputs

Dates*

Relative ▼ Last Year ▼

Schedule Properties

Name: * ⓘ

Email Report

☒ Start Schedule

Schedule Frequency

By : Every X minute | Hourly | Daily | Weekly | Monthly | Yearly

* / 5 * * * *

minute hour day of month month day of week

" Every 5 minutes "

Timezone: ⓘ

Select ▼

☐ Limit execution to one active instance at a time ⓘ

Start Time: ⓘ -- End Time: ⓘ --

Notification

Email Address ⓘ

csadmin@internal.lab

✓ Save × Close

- Click **Save**.
- Click the **Schedules** tab.
- Confirm that the schedule is running.
The status of the schedule should be **Active**.

Reporting

Library Schedules History

By Schedule Name By Status By Report

Search by Schedule Name 🔍 All ALL ↻

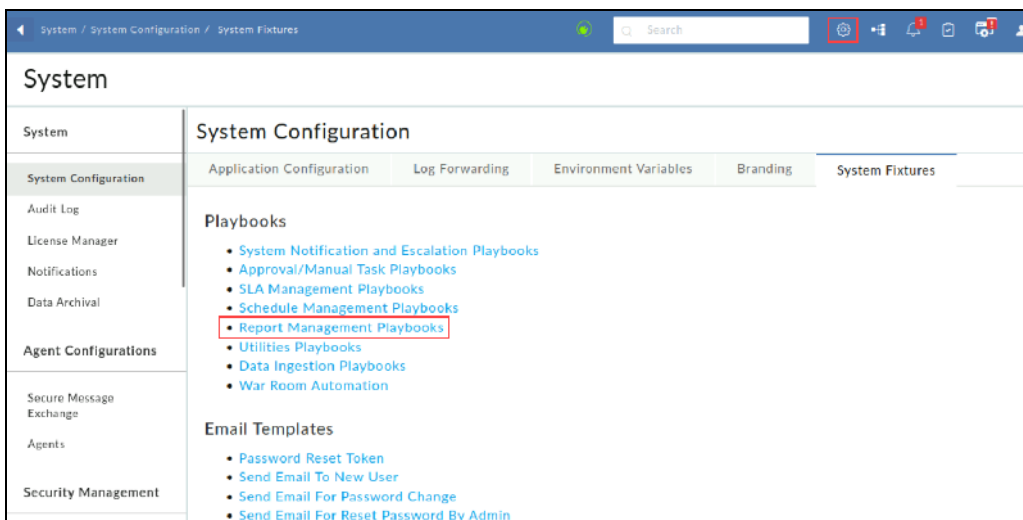
1 items

Active	Email Report	
Last Run At: Scheduled For: Indicator Reputation Report		
Total Run Count: 0 Timing: Every 5 minutes Timezone:		

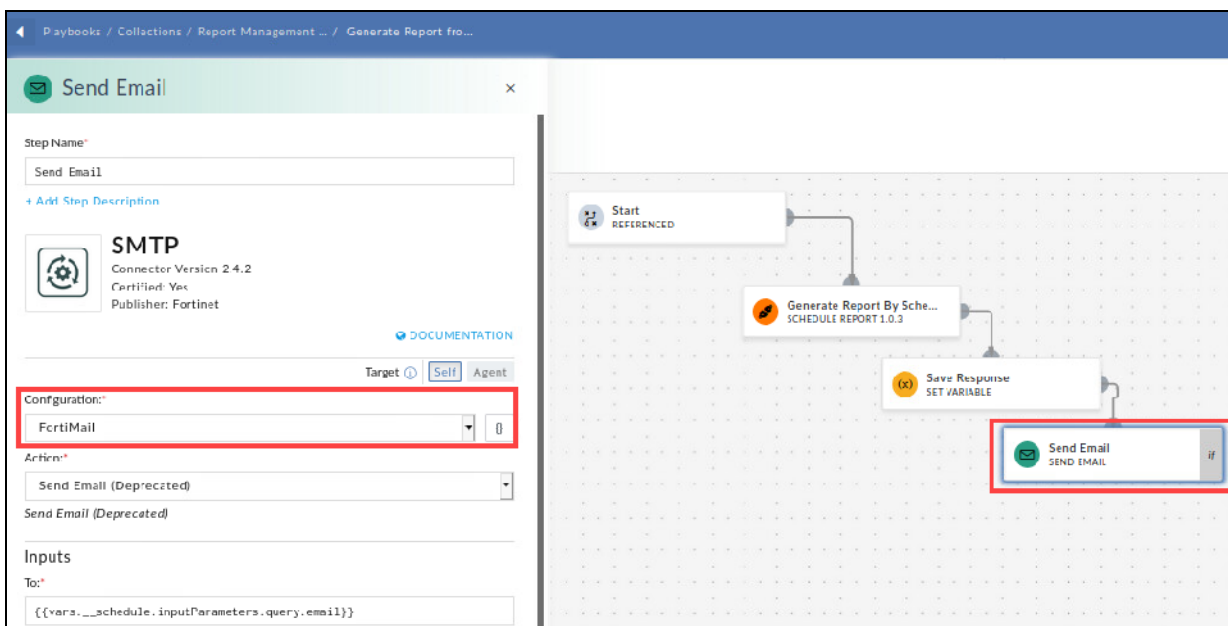
✎ 🛑 🗑️ ▼

To configure the email settings

1. Continuing on the FortiSOAR GUI, click **Settings**.
2. Click **System Configuration**.
3. Click **System Fixtures**.
4. Click **Report Management Playbooks**.



5. Click the **Generate Report from Schedule** playbook.
6. Double-click the **Send Email** step to edit it.
7. In the **Configuration** field, select **FortiMail**.

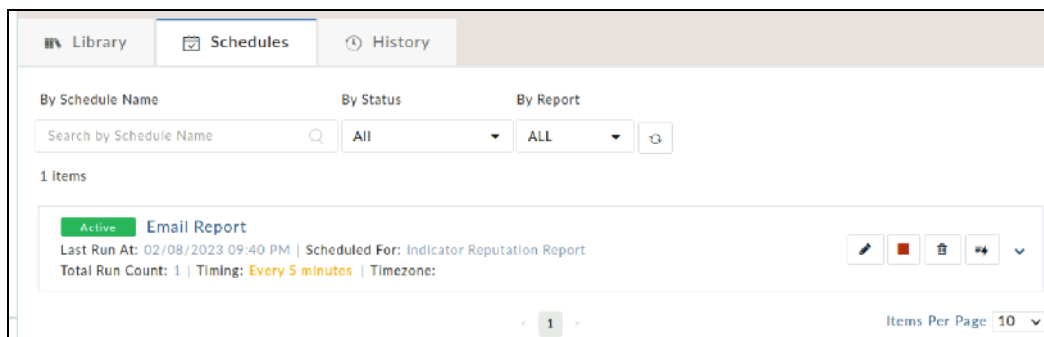


8. Click **Save**.
9. Click **Save Playbook**.

To view the report

1. Continuing on the FortiSOAR GUI, click **Reports**.
2. Click the **Schedules** tab.
3. Confirm that the report generated.

The **Total Run Count** column should display a value greater than **0**. If not, wait a few more minutes, and then refresh the page.



4. Click the **History** tab.
5. In the **File** field, click the **Indicator Reputation Report** link.
6. Click **Open with Firefox** to view the report.
You also have the option to save the file in PDF format.
7. Log out of the FortiSOAR GUI.
8. On the FortiMail webmail GUI, log in with the username `csadmin` and password `Fortinet1!`.
9. Open the email with the subject **CyOps:Email Report**.
10. Open the attachment and view the report.
11. Log out of the FortiMail webmail GUI.

Exercise 2: Managing Logs and the Recycle Bin

In this exercise, you will view and purge logs. You will also enable the recycle bin and delete a record. Data management is important in a SOAR environment. As the amount of data increases, managing logs and deleting unused entries can help maintain performance.

Manage Audit Logs

You will manage user and system account logs, and then schedule purging of the audit logs.

To access user-specific audit logs

1. Log in to the FortiSOAR GUI with the username `csadmin` and password `Fortinet1!`.
2. Click **Settings**.
3. Click **Users**.
4. Edit the **csadmin** user.
5. In the **Audit Logs - User Specific Audit Logs** section, review the entries for the **csadmin** user.
6. Log out of the FortiSOAR GUI.
7. Log in to the FortiSOAR GUI with the username `jsmith` and password `Fortinet1!`.
8. Click the user icon, and then click **Edit Profile**.
9. Scroll to the bottom.
The **Audit Logs - User Specific Audit Logs** section is not available.
10. Under the **Roles** section, click **View Effective Role Permissions**.
11. Review the aggregated list of permissions.
This user does not have permissions to the **Audit Logs Activities** module.
12. Click **Settings**.
Note that under **System**, you also cannot see the audit log feature.
13. Click **Roles**.
14. Edit the **Security Administrator** role.
15. In the **Set Role Permissions** section, scroll down to the **Audit Log Activities** module, and then select the **Create**, **Read**, and **Update** checkboxes.

Set Role Permissions						
Module		Create	Read	Update	Delete	Fields Permissions
Agents ⓘ		☐	☐	☐	☐	Set Field Permissions
Alerts ⓘ		☐	☐	☐	☐	Set Field Permissions
Announcements ⓘ		☐	☐	☐	☐	Set Field Permissions
Appliances ⓘ		☐	☐	☐	☐	Set Field Permissions
Application ⓘ		☒	☒	☒	☒	
Approvals ⓘ		☐	☐	☐	☐	Set Field Permissions
Assets ⓘ		☐	☐	☐	☐	Set Field Permissions
Attachments ⓘ		☐	☐	☐	☐	Set Field Permissions
Audit Log Activities ⓘ		☒	☒	☒	☐	

- Click **Save**.
- Navigate to the user profile for **jsmith** again.
You can now view the user's audit logs.

John Smith						
Audit Logs						
21 Items						
Operation From Date To Date Select Operation From To						
Title	Record Type	Operation	User	Status	Transaction Date	
> DB Login Success for user [jsmith]	Person	Login Success	John Smith	10.0.1.10	01/19/2023 06:20 PM	
> User [jsmith] logged out	Person	Log Out	John Smith	10.0.1.10	01/19/2023 06:18 PM	
> DB Login Success for user [jsmith]	Person	Login Success	John Smith	10.0.1.10	01/19/2023 06:16 PM	
> User [jsmith] logged out	Person	Log Out	John Smith	10.0.1.10	01/19/2023 06:11 PM	
> DB Login Success for user [jsmith]	Person	Login Success	John Smith	10.0.1.10	01/19/2023 06:07 PM	

- Log out of the FortiSOAR GUI.

To manage system audit logs

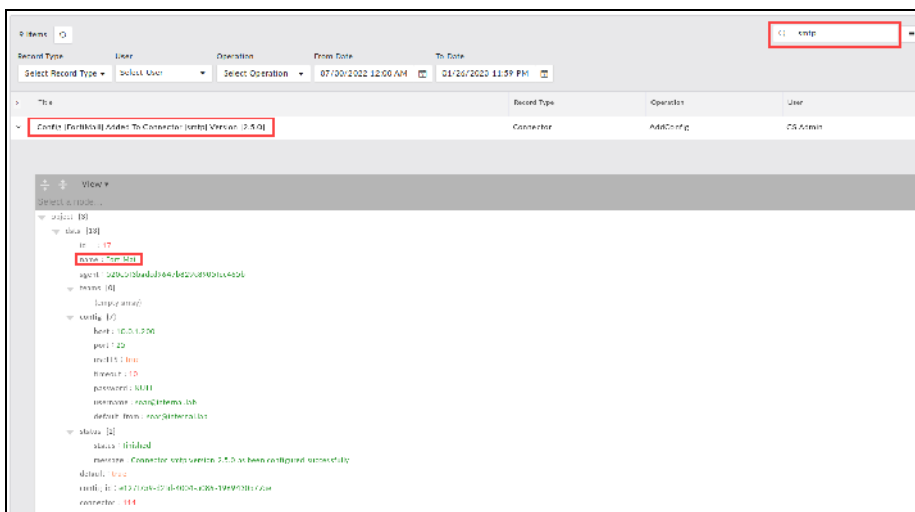
- Log in to the FortiSOAR GUI with the username **csadmin** and password **Fortinet1!**.
- Click **Settings**.
- Under **System**, click **Audit Log**.
- In the **Operation** field, select **Log Out**.

You can see a filtered list of results.

23 Items						
Record Type User Operation From Date To Date Select Record Type Select User Log Out 07/30/2022 12:00 AM 01/26/2023 11:59 PM						
Title	Record Type	Operation	User	Status	Transaction Date	
> User [csadmin] logged out	Person	Log Out	CS Admin	10.0.1.10	01/25/2023 08:27 PM	
> User [csadmin] logged out	Person	Log Out	CS Admin	10.0.1.10	01/25/2023 05:52 PM	
> User [csadmin] logged out	Person	Log Out	CS Admin	10.0.1.10	01/25/2023 01:26 PM	

- Review the entries.
- In the **Operation** field, select **Select Operation** to clear the filter.

7. Click the magnifying glass icon to search, and then enter SMTP.
8. Review the **Config [FortiMail] Added To Connector** log entry from a previous exercise.



9. Clear the search query for SMTP.
10. Refresh the page.
11. In the **User** field, review the list of users available for selection.



The **csadmin** user has been given permissions in this lab environment to view all audit log entries in the system, including other users' activities.

12. Select **Playbook**.
13. Review log entries related to the user.
14. In the **User** field, select **John Smith**.
15. Review log entries related to the user.

To purge audit logs

1. Click **Roles**.
2. Edit the **Security Administrator** role.
3. In the **Set Role Permissions** section, scroll to the **Audit Log Activities** role.
4. Select the **Delete** checkbox.

Set Role Permissions						
Module		Create		Read		Fields Permissions
		+	-	+	-	
Agents ⓘ	☐	☐	☐	☐	☐	Set Field Permissions
Alerts ⓘ	☐	☐	☐	☐	☐	Set Field Permissions
Announcements ⓘ	☐	☐	☐	☐	☐	Set Field Permissions
Appliances ⓘ	☐	☐	☐	☐	☐	Set Field Permissions
Application ⓘ	☒	☒	☒	☒	☒	
Approvals ⓘ	☐	☐	☐	☐	☐	Set Field Permissions
Assets ⓘ	☐	☐	☐	☐	☐	Set Field Permissions
Attachments ⓘ	☐	☐	☐	☐	☐	Set Field Permissions
Audit Log Activities ⓘ	☒	☒	☒	☒	☒	

5. Click **Save**.

6. Click **Audit Log**.

Now, you can see the **Purge Logs** button.

Audit Log					
					Purge Logs
494 Items					
Record Type	User	Operations	From Date	To Date	
Select Record Type	CS Admin (me)	Select Operation	07/30/2022 12:00 AM	01/24/2023 11:59 PM	
Title	Record Type	Operation	User		
IDB Login Success for user [codmin]	Person	Login Success	CS Admin		

7. Click **Purge Logs**.

8. In the **Purge all logs before** section, click the calendar icon.

9. Select a day that is one month before today.

10. Click **Close**.

11. In the **PURGE LOGS OF SPECIFIC EVENT TYPE** section, clear the checkboxes for all event types except the following: **Login Failed**, **Login Success**, and **Log Out**.

12. Click **Purge Logs**.

13. Click **I Have Read The Warning - Purge Logs**.

Purge Audit Logs

Provide the date and time until which the audit logs are to be purged. Logs before the specified time frame will be permanently deleted.

Purge all logs before, 12/26/2022 12:00 AM

PURGE LOGS OF SPECIFIC EVENT TYPE

Various event types are recorded within audit logs. You can select the event types for which you want to purge the logs. By default, logs for all event types are purged.

☐ Create
 ☐ Update
 ☐ Delete
 ☐ Bulk Delete
 ☐ Bulk Insert
 ☐ Soft Delete
 ☐ Archival Start
 ☐ Archival Failure
 ☐ Restore
 ☐ Link
 ☐ Unlink
 ☐ Import
 ☐ Update During Import
 ☐ Overwrite During Import
 ☐ Comment
 ☐ Event
 ☐ Trigger
 ☐ Resume
 ☐ Retry
 ☐ Terminate
 ☐ Version Created
 ☐ Version Deleted
 ☒ Login Success
 ☒ Login Failed
 ☒ Log Out
 ☐ Install
 ☐ Uninstall
 ☐ Upgrade
 ☐ AddConfig
 ☐ DeleteConfig
 ☐ UpdateConfig
 ☐ Executed Action
 ☐ Activate
 ☐ Deactivate
 ☐ Validate
 ☐ Publish
 ☐ Clone
 ☐ Revert
 ☐ Collect
 ☐ Start
 ☐ Stop

Purge Logs Cancel

To schedule the purging of audit logs

1. Click **Settings**.
2. In the **System Configuration** section, click **Application Configuration**.
3. In the **Purge Logs** section, select the **Enable Purging** checkbox.
4. In the **Keep logs of** field, select **Last 3 Months**.

Purge Logs

Scheduled Purging for Audit Logs and Playbook Execution Logs

☒ Enable Purging

Purge Criteria

A system schedule to periodically clear audit logs will be created under Automation > Schedules

Schedule Status: Schedule Not Found

Keep logs of: Last 3 Months

5. Click **Save**.
6. Under the main navigation pane, click **Automation > Schedules**.
7. Select **Purge Audit Logs**.
8. In the **Purge Audit Logs** row, click the edit icon (✎).

Schedules

By Schedule Name

By Status

Search by Schedule Name

All

+ Create New Schedule

5 Items

☐	Name	Scheduled	Timezone	Last Run At	Total Run Count	Status	Expired	Actions
☐	Ingestion_fortinet: fortisim (c36e89...	Every minute			4299	Inactive		 
☐	Purge Executed Playbook Logs	At 12:00 AM		06/23/2021 12:00 AM	74	Active		 
☐	Email Report	Every 5 minutes			1	Inactive		 
☐	Purge Integration Action Logs	Every hour		06/23/2021 04:00 PM	229	Active		 
☐	Purge Audit Logs	At 12:00 AM			0	Active		 

< < 1 of 1 > >

Items Per Page 30



The scheduler runs the **AuditLog Cleanup** playbook every day at 12 AM, which deletes all audit logs from three months before.

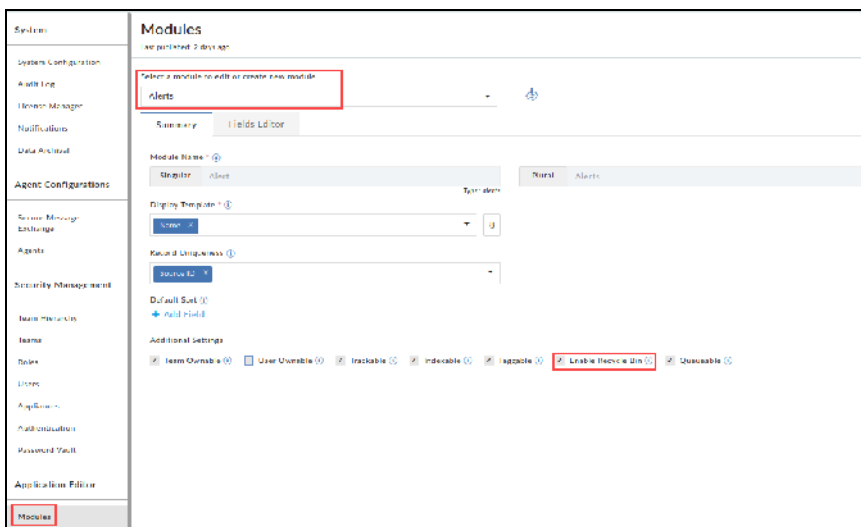
- Click **Close**.
- Log out of the FortiSOAR GUI.

Manage the Recycle Bin

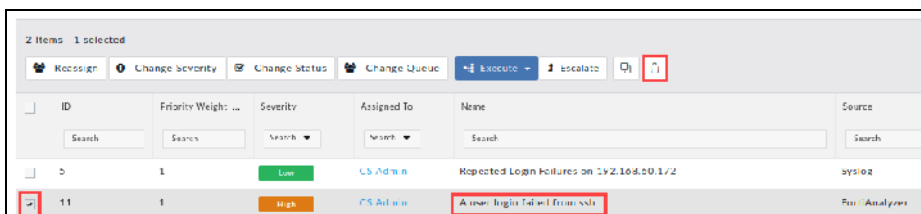
You will enable the recycle bin for the alerts module, move an alert into it, and permanently delete that record. You will also schedule purging of soft-deleted records.

To enable the recycle bin for alerts

- Log in to the FortiSOAR GUI with the username `csadmin` and password `Fortinet1!`.
- Click **Settings**.
- In the **Application Editor** section, click **Modules**.
- In the **Select a module to edit or create new module** field, select **Alerts**.
- In the **Additional Settings** section, select the **Enable Recycle Bin** checkbox.



6. Click **Save**.
7. Click **Publish All Modules**.
8. Click **OK**.
9. Under the main navigation pane, click **Incident Response > Alerts**.
10. Click the magnifying glass icon, and then enter `ssh`.
11. Check the first column for the **A user login failed from ssh** alert.
12. Click the delete icon.



13. Click **Move to Recycle Bin**.
14. In the navigation pane, click **Recycle Bin**.
15. In the **Select a module** field, select **Alerts**.
16. Select the **A user login failed from ssh** alert checkbox.
17. Click **Empty Recycle Bin**.



You also have the option to delete all records at once, or to restore records by taking them out of the recycle bin. Consider enabling the recycle bin for modules which may require soft deletion in case of mistakes.

18. Click **OK**.
19. Under the main navigation pane, click **Automation > Schedules**.
20. Select the **Recycle Bin Cleanup** checkbox.
21. Under the **Actions** field, click the edit icon (pencil).
22. Select the **Start Schedule** checkbox.
23. Click **Save**.

24. Close the **Schedule Details** window.
25. Log out of the FortiSOAR GUI.

Lab 4: High Availability

In this lab, you will configure high availability (HA), discover the HA monitoring and takeover processes, perform a fire drill on the secondary node, and externalize the FortiSOAR database to an external PostgreSQL server.

Objectives

- Configure FortiSOAR HA with an internal PostgreSQL database
- Configure FortiSOAR HA monitoring and perform a takeover
- Perform a fire drill on the secondary node
- Externalize the database to an external PostgreSQL server

Time to Complete

Estimated: 1 hour 15 minutes

Exercise 1: Configuring HA

In this exercise, you will configure FortiSOAR instances to form an HA cluster. When you have FortiSOAR instances configured in an HA cluster, you can achieve load balancing in an active-active setup, or redundancy in an active-passive setup.

Configure an HA Cluster

You will configure two FortiSOAR VMs in an active-passive HA cluster.

To configure the primary node

1. Open an SSH connection to the FortiSOAR-Primary VM at IP 10.0.1.160.
2. Log in with username `csadmin` and password `changeme`.
3. Enter the following command to enable root privileges:
`sudo -i`
4. Enter the following command to change the host name:
`csadm hostname --set primary.internal.lab`



Changing the host name on FortiSOAR causes FortiSOAR to regenerate the certificates and restart all services.

5. Type `hostname`, and then press `Enter` to confirm that the host name is `primary.internal.lab`.

```
[root@fortisoar ~]# hostname  
primary.internal.lab
```

6. Enter the following command to open and edit the `pg_hba.conf` file:
`vi /var/lib/pgsql/14/data/pg_hba.conf`
7. Scroll to the `Trusted hosts` section, and then press `i` to enable insert mode in VI.
8. Add the following statement to the config file:

```
host all all 10.0.1.161/32 trust  
  
# local          DATABASE  USER  METHOD  [OPTIONS]  
# host           DATABASE  USER  ADDRESS METHOD  [OPTIONS]  
# hostssl        DATABASE  USER  ADDRESS METHOD  [OPTIONS]  
# hostnossl      DATABASE  USER  ADDRESS METHOD  [OPTIONS]  
#  
# (The uppercase items must be replaced by actual values.)  
#  
# TYPE  DATABASE  USER  ADDRESS  METHOD  
# "local" is for Unix domain socket connections only  
local  all  all  md5  
# IPv4 local connections:  
host  all  all  127.0.0.1/32  md5  
# IPv6 local connections:  
host  all  all  ::1/128  md5  
# Password hosts  
# Trusted hosts  
host all all 10.0.1.161/32 trust
```

9. Press `Esc` to exit insert mode in VI.
10. Type `:wq`, and then press `Enter` to save and exit the file.

To configure the secondary node

1. Open an SSH session to the FortiSOAR-Secondary VM at IP `10.0.1.161`.
2. Log in with username `csadmin` and password `changeme`.
3. Enter the following command to enable root privileges:
`sudo -i`
4. Enter the following command to change the host name:
`csadm hostname --set secondary.internal.lab`
5. Type `hostname`, and then press `Enter` to confirm that the host name is `secondary.internal.lab`.
6. Enter the following command to open and edit the `pg_hba.conf` file:
`vi /var/lib/pgsql/14/data/pg_hba.conf`
7. Scroll to the `Trusted hosts` section, and then press `i` to enable insert mode in VI.
8. Add the following statement to the config file:
`host all all 10.0.1.160/32 trust`
9. Press `Esc` to exit insert mode in VI.
10. Type `:wq`, and then press `Enter` to save and exit the file.
11. Enter the following command to open and edit the `resolv.conf` file:
`vi /etc/resolv.conf`
12. Press `i` to enable insert mode in VI, and then add the following statement to the top of the list of DNS servers:
`nameserver 10.0.1.254`

```
# Generated by NetworkManager
search internal.lab
nameserver 10.0.1.254
nameserver 8.8.8.8
nameserver 10.0.1.10
```

13. Press `Esc` to exit insert mode in VI.
14. Type `:wq`, and then press `Enter` to save and exit the file.
15. Enter the following command to ping the primary node and verify DNS resolution:

```
ping primary.internal.lab
[root@secondary ~]# ping primary.internal.lab
PING primary.internal.lab (10.0.1.160) 56(84) bytes of data.
64 bytes from primary.internal.lab (10.0.1.160): icmp_seq=1 ttl=64 time=0.364 ms
64 bytes from primary.internal.lab (10.0.1.160): icmp_seq=2 ttl=64 time=0.397 ms
^C
--- primary.internal.lab ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1012ms
rtt min/avg/max/mdev = 0.364/0.380/0.397/0.025 ms
```

16. Return to the SSH session connected to the primary FortiSOAR VM, and then enter the following command to ping the secondary node and verify DNS resolution:

```
ping secondary.internal.lab
[root@fortisoar ~]# ping secondary.internal.lab
PING secondary.internal.lab (10.0.1.161) 56(84) bytes of data.
64 bytes from secondary.internal.lab (10.0.1.161): icmp_seq=1 ttl=64 time=0.454 ms
64 bytes from secondary.internal.lab (10.0.1.161): icmp_seq=2 ttl=64 time=0.455 ms
^C
--- secondary.internal.lab ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 0.454/0.454/0.455/0.021 ms
```

To configure the HA cluster

1. Return to the SSH session connected to the FortiSOAR-Secondary VM, and then enter the following command to start the clustering process:

```
csadm ha join-cluster --status passive --role secondary --primary-node
primary.internal.lab
```

2. Enter the username `csadmin` and password `changeme`.

You should see the HA clustering process start.

```
[root@secondary ~]# csadm ha join-cluster --status passive --role secondary --primary-node
primary.internal.lab
Enter username for SSH connection to 'primary.internal.lab' (or press enter if the username
is 'csadmin'):
Enter SSH password for user 'csadmin' on 'primary.internal.lab':
Validating SSH credentials
Fetching HA configuration details of primary.internal.lab
Performing allowlist on primary.internal.lab
Configuring as passive server
Comparing version with primary.internal.lab
Validating rabbitmq credentials
Successfully validated rabbitmq credentials
Validating elasticsearch credentials
Successfully validated elasticsearch credentials
Validating postgresql credentials
Successfully validated postgresql credentials
Comparing license
Successfully compared license
Stopping service cyops-ha
Successfully stopped service cyops-ha
Checking whether postgresql server is running
Postgresql server is running
Stopping postgresql server
Successfully stopped postgresql server
```

The clustering process takes 2–5 minutes, and then you should see the message `Successfully configured as passive server`.

```
Adding primary server details in /var/lib/pgsql/14/data/postgresql.conf
Creating a standby signal file at [/var/lib/pgsql/14/data/standby.signal]
Checking whether postgresql server is running
Postgresql server is not running
Starting postgresql server
Successfully started postgresql server
Taking backup of configuration file
Synchronizing license
Synchronizing jwt keys with primary server
Synchronizing parameters prod keys with primary server
Clearing API cache
Stopping all services
Disabling all services except HA, auth and nginx, for passive server
Enabling HA service
Enabling nginx service
Enabling auth service
Enabling postgresql service
Starting all services
Setting up API files as per the new model definition
Starting service cyops-ha
Successfully started service cyops-ha
Removing backup directory
Successfully configured as passive server
```

3. Close the FortiSOAR-Secondary SSH session.
4. Return to the SSH session connected to the primary FortiSOAR VM, and then enter the following command:

```
csadm ha list-nodes
```

```
[root@fortisoar ~]# csadm ha list-nodes
```

nodeId	nodeName	status	role	comment
* 520e5f3badcd9647b829c8905fec465b927ee24c7eabd15e684c996bd04ea968	primary.internal.lab	active	primary	primary server
	secondary.internal.lab	passive	secondary	Joined cluster with primary.internal.lab

The FortiSOAR-Secondary VM (`secondary.internal.lab`) has joined the cluster as a passive secondary node.

5. Close the primary FortiSOAR SSH session.
6. On the Local-Host VM, open Firefox, and then log in to the primary FortiSOAR GUI with the username `csadmin` and password `Fortinet1!`.
7. Click **Settings**.
8. Click **License Manager**.

License Manager

Type	Perpetual
Edition	Enterprise
Allowed User Seats	2 Users
User Seats Consumed	1 Named, 2 Concurrent ⓘ
Threat Intel Management Service Subscription	Disabled ⓘ

Nodes

Node Name	Status	Role	License Details
primary.internal.lab	Active	Primary	Serial Number: PSRVMP1M20000484 Total Users: 2 Device UUID: 520e5f3badcd9647b029c0905fec465b ⓘ Update License
secondary.internal.lab	Passive	Secondary	Serial Number: PSRVMP1M21000105 Total Users: 2 Device UUID: 927ee24c7eabd15e684c996bd04ea968 ⓘ Update License

You can see the HA status and roles of the two FortiSOAR instances.

9. Log out of the FortiSOAR GUI.

Exercise 2: Configuring HA Heartbeat Monitoring and Triggering a Takeover

In this exercise, you will learn how to configure HA heartbeat monitoring and trigger a failover.

Configure HA Heartbeat Monitoring

You will configure HA heartbeat monitoring on the primary node.

To configure HA heartbeat monitoring

1. Log in to the FortiSOAR-Primary GUI with the username `csadmin` and password `Fortinet1!`.
2. Click **Settings**.
3. Click **System Configuration**.
4. Click the **Application Configuration** tab.
5. In the **System & Cluster Health Monitoring** section, click **Enable Notification**.
6. In the **Service** field, make sure the **SMTP** service is selected.
7. In the **Email** field, type `csadmin@internal.lab`.
8. In the **Monitoring Interval (Minutes)** field, type `2`, and in the **Replication Lag (GB)** field, type `1`.

System & Cluster Health Monitoring
Receive email notifications for system & cluster health parameters that need attention.

☒ **Enable Notification**
A notification will be sent if any of the FortiSOAR services fail to run, or if one or more of the monitored components exceed their threshold values. Define the email addresses to be notified below:

Service: SMTP
Please ensure your SMTP service is configured.

Email: csadmin@internal.lab

Monitoring Interval (Minutes): 2
The monitoring job will run at this schedule:

System Health Thresholds
A notification will be generated when the resource consumption on the server is high. Define the respective thresholds below:

Memory Utilization (%)	CPU Utilization (%)	Disk Utilization (%)
60	80	60
Swap Memory Utilization (%)	Workflow Queue Threshold	WAL Files Size (GB)
50	100	20

Cluster Health
An email notification will also be generated if there are missed heartbeats from any of the cluster nodes, or if the replication lag is high. Define the respective thresholds below:

Missed Heartbeat Count	Replication Lag (GB)
3	1

9. Click **Save**.

Configure a Cluster Takeover

You will use the secondary FortiSOAR node to take over the cluster and make it the primary node.

To configure a takeover

1. Open an SSH session to the FortiSOAR-Secondary VM.
2. Log in with username `csadmin` and password `changeme`.
3. Enter the following command to enable root privileges:
`sudo -i`
4. Enter the following command to check the current status and role of this FortiSOAR instance:
`csadm ha list-nodes`
5. Enter the following command to initiate the takeover process:
`csadm ha takeover`

```
[root@secondary ~]# csadm ha takeover
A takeover operation makes the current node as the primary node
of the cluster and all cluster nodes must now join cluster again
to this new primary node. [OK to proceed? [yes/no]:yes]
Starting takeover operation
Creating physical replication slot slot_phy_trs_b20e373badcd964/b829c8903ec465b on localho
st
Promoting node as primary
Marking checkpoint
Resetting the password for 'cyberpsql' user
Enabling all services
Stopping all services
Starting all services
Clearing API cache
Awaiting 5 seconds for services to start up
Reindexing elasticsearch
Removing default database configuration file
Starting all services
Successfully performed takeover
```

6. At the prompt to confirm the takeover operation, enter `yes`.
7. At the prompt to confirm the invoking of the join-cluster operation, enter `yes`.

```
Do you want to invoke 'join-cluster' on other cluster nodes? [yes/no]:yes
Checking whether postgresql server is running
Postgresql server is running
Replication user 'replicator' already exists
Checking whether postgresql server is running
Postgresql server is running
Stopping postgresql server
Successfully stopped postgresql server
Modifying settings in /var/lib/pgsql/14/data/postgresql.conf
Modifying settings in /var/lib/pgsql/14/data/pg_hba.conf
Checking whether postgresql server is running
Postgresql server is not running
Starting postgresql server
Successfully started postgresql server
Opening port in firewall
```

8. To have the former primary node join the cluster as a passive secondary node, type `1` (one), and then press Enter.

```
Successfully registered
Awaiting 5 seconds for services to start up
-----
Confirm:
Choose one of the following option for node primary.internal.lab to
join cluster with current primary node secondary.internal.lab:
1. Join as a passive secondary node
2. Join as an active secondary node
3. Do not join back to the cluster
-----
Enter your choice (Default (1)):1
Invoking 'join-cluster' on node 'primary.internal.lab'
'join-cluster' started for node 'primary.internal.lab'. Logs are at /var/log/cyops/cyops-au
th/join-cluster-primary.internal.lab
'join-cluster' completed for node 'primary.internal.lab'
Waiting for other nodes join-cluster to finish....[OK]

'join-cluster' logs for node 'primary.internal.lab' from file /var/log/cyops/cyops-auth/joi
n-cluster-primary.internal.lab
```

9. The process should take 2–5 minutes, and then you should see the message Successfully configured as passive server.

```
Adding primary server details in /var/lib/pgsql/14/data/postgresql.conf
Creating a standby signal file at [/var/lib/pgsql/14/data/standby.signal]
Checking whether postgresql server is running
Postgresql server is not running
Starting postgresql server
Successfully started postgresql server
Taking backup of configuration file
Synchronizing license
Synchronizing jwt keys with primary server
Synchronizing parameters prod keys with primary server
Clearing API cache
Stopping all services
Disabling all services except HA, auth and nginx, for passive server
Enabling HA service
Enabling nginx service
Enabling auth service
Enabling postgresql service
Starting all services
Setting up API files as per the new model definition
Starting service cyops-ha
Successfully started service cyops-ha
Removing backup directory
Successfully configured as passive server
```

10. Enter the following command to confirm that the secondary.internal.lab FortiSOAR is the new active primary node in the cluster:

```
csadm ha list-nodes
```

```
[root@secondary ~]# csadm ha list-nodes
```

nodeId	nodeName	status	role	comment
* 927ee24c7eabd15e684c996bd04ea968	secondary.internal.lab	active	primary	Performed Takeover at Feb -09-2023 17:27:14PM UTC
520e5f3badcd9647b829c8905fec465b	primary.internal.lab	passive	secondary	Joined cluster with secon dary.internal.lab

11. Close the FortiSOAR-Secondary SSH session.



If the FortiSOAR instances are stuck with duplicate licenses, you must revert both FortiSOAR VMs. See [Revert FortiSOAR on page 110](#) for the revert instructions

Exercise 3: Performing a Fire Drill

In this exercise, you will trigger a fire drill on the HA cluster, which will test your system's disaster recovery configuration. Running the fire drill suspends replication to other nodes, and configures the node you run the command on as a standalone node pointing to its local database.

Initiate a Fire Drill

You will initiate a fire drill on the secondary node before restoring it to the HA cluster.

To initiate a fire drill on the secondary node

1. Open an SSH session to the FortiSOAR-Primary VM.
2. Log in with username `csadmin` and password `changeme`.
3. Enter the following command to enable root privileges:
`sudo -i`
4. Enter the following command to confirm that you are on the `primary.internal.lab` FortiSOAR that has the passive status and secondary role:

```
csadm ha list-nodes
```

nodeId	nodeName	status	role	comment
927ee24c7eabd15e684c996bd04ea968	secondary.internal.lab	active	primary	Performed Takeover at Feb-09-2023 17:27:14PM UTC
* 520e5f3badcd9647b829c8905fec465b	primary.internal.lab	passive	secondary	Joined cluster with secondary.internal.lab

5. Enter the following command to start the fire drill process:

```
csadm ha firedrill
```

```
[root@fortisoar ~]# csadm ha firedrill
Taking backup of postgresql configuration files
Taking backup of database configuration file
Creating physical replication slot slot_phy_fsr_firedrill on localhost
Starting firedrill operation
Promoting node as primary
Marking checkpoint
Resetting the password for 'cyberpgsql' user
Enabling all services
Stopping all services
Disabling celerybeatd service
Disabling cyops-postman service
Starting all services
Clearing API cache
Awaiting 5 seconds for services to start up
Reindexing elasticsearch
Stopping all services
Starting all services
Taking backup of required files for re-joining cluster at [/opt/cyops-auth/.re-join-cluster.sql]
-----
Note:
Once you are done with testing, make sure to perform 'restore' operation
-----
Successfully performed firedrill
```




Note that the `celerybeatd` service is disabled on this node during the fire drill process. This is done intentionally since the `celerybeatd` is responsible for running schedules and playbooks, and any configured schedules and playbooks should not run when the node is in fire drill mode.

Confirm that you see the Successfully performed firedrill message.

6. Enter the following command to check the status and role of the fire drill node:

```
csadm ha list-nodes
```

nodeId	nodeName	status	role	comment
* 520e5f3badcd9647b829c8905fec465b	primary.internal.lab	active	primary	Firedrill in progress. The operation was run at Feb-09-2023 18:13:46PM UTC

To restore the cluster

1. Continuing on the FortiSOAR CLI, enter the following command:

```
csadm ha restore
```

```
[root@fortisoar ~]# csadm ha restore
Starting restore operation
```

```
Restoring postgresql configuration files
Modifying settings in /var/lib/pgsql/14/data/postgresql.conf
Adding primary server details in /var/lib/pgsql/14/data/postgresql.conf
Creating a standby signal file at [/var/lib/pgsql/14/data/standby.signal]
Checking whether postgresql server is running
Postgresql server is not running
Starting postgresql server
Successfully started postgresql server
Synchronizing license
Synchronizing jwt keys with primary server
Synchronizing parameters prod keys with primary server
Clearing API cache
Reset message queues
Stopping all services
Disabling all services except HA, auth and nginx, for passive server
Enabling HA service
Enabling nginx service
Enabling auth service
Enabling cyops-postman service
Enabling postgresql service
Starting all services
Setting up API files as per the new model definition
Removing backup directory
Successfully configured as passive server
Removing firedrill backup directory
Successfully restored
```

2. Enter the following command to confirm that the FortiSOAR VM has been reinstated as a passive secondary node in the cluster:

```
csadm ha list-nodes
```

nodeId	nodeName	status	role	comment
927ee24c7eabd15e684c996bd04ea968	secondary.internal.lab	active	primary	Performed Takeover at Feb-09-2023 17:27:14PM UTC
* 520e5f3badcd9647b829c8905fec465b	primary.internal.lab	passive	secondary	Firedrill is complete. Joined cluster with secondary.internal.lab

To configure a takeover on the primary

1. Continuing on the FortiSOAR primary's CLI, enter the following command to initiate another takeover:

```
csadm ha takeover
```
2. At the prompt to confirm the takeover operation, enter `yes`.
3. At the prompt to confirm the invoking of the join-cluster operation, enter `yes`.

4. Type `1` (one), and then press `Enter`.
5. The process should take 2–5 minutes, and then you should see the message `Successfully configured as passive server`.

To remove the secondary node from the cluster

1. Open an SSH session to the FortiSOAR-Secondary VM and run the following command:

```
csadm ha leave-cluster
```

2. Type `yes`, and then press `Enter`.

```
[root@secondary cyops-auth]# csadm ha leave-cluster
Do you want to leave the cluster? [yes/no]:yes
Starting service postgresql-14
Successfully started service postgresql-14
Starting service elasticsearch
Successfully started service elasticsearch
Promoting node as primary
Marking checkpoint
Checking if physical replication slot slot_phy_fsr_927ee24c7eabd15e684c996bd04ea968 exists on primary.internal.lab
Deleting physical replication slot slot_phy_fsr_927ee24c7eabd15e684c996bd04ea968 on primary.internal.lab
Resetting the password for 'cyberpgsql' user
Resetting the password for elasticsearch 'elastic' user
Taking backup of required files for re-joining cluster at [/opt/cyops-auth/.re-join-cluster.sql]
Enabling all services
Stopping all services
Starting all services
Clearing API cache
Setting up API files as per the new model definition
Removing default database configuration file
Successfully performed leave-cluster
```

3. Enter the following command to confirm that the FortiSOAR secondary VM has left the cluster and currently has the role of primary:

```
csadm ha list-nodes
```

```
[root@secondary cyops-auth]# csadm ha list-nodes
nodeId          nodeName          status    role    comment
-----
* 927ee24c7eabd15e684c996bd04ea968 secondary.internal.lab active    primary Left cluster
```

4. Close the FortiSOAR primary and FortiSOAR secondary SSH sessions.

To check email notifications for HA events

5. On the Local-Host VM, open Firefox, and then on the FortiMail webmail GUI, log in with the username `csadmin` and password `Fortinet1!`.
6. View the notification email with the subject **Cluster Health Warning**.

```
Subject: FortiSOAR Notification - Cluster Health Warning
From: soar@internal.lab
To: csadmin@internal.lab
Date: Jan 18, 2023 11:28:03 PM

Firedrill is in progress on following node(s):
Node:primary.internal.lab
Current lag with primary node: 366 KB

Please troubleshoot the same to ensure continued high availability. For more information on the error, log on to your FortiSOAR instance.
```

7. Log out of the FortiMail webmail GUI.

Exercise 4: Externalizing the FortiSOAR Database

In this exercise, you will externalize the FortiSOAR database to an external PostgreSQL server.

Database Externalization

You will create a database externalization configuration file, set its parameters, and begin the externalization process.

To create the database externalization configuration file

1. Open an SSH connection to the FortiSOAR-Primary VM.
2. Log in with the username `csadmin` and password `changeme`.
3. Enter the following command to log in as the root user:

```
sudo -i
```
4. To check the current database status, run the following command:

```
csadm db --check-connection
```



You will see a warning regarding a missing externalization configuration file. The easiest way to create it is to clone the existing `db_config.yml` file.

5. Change the working directory with the following command:

```
cd /opt/cyops/configs/database
```
6. Make a copy of the existing `db_config.yml` file with the following command:

```
cp db_config.yml db_external_config.yml
```



Ensure you are in the `/opt/cyops/configs/database` directory before issuing the `cp` command above.

7. Open the `db_external_config.yml` file for editing by using the following command:

```
vi db_external_config.yml
```
8. Press `i` to enter insert mode.
9. Change the following parameters in the file:

Field	Value
<code>pg_external</code>	<code>true</code>
<code>pg_host</code>	<code>10.0.1.170</code>
<code>pg_password</code>	<code>FortinetSQL</code>

Field	Value
pg_archival_external	true
pg_archival_host	10.0.1.170
pg_archival_password	FortinetSQL

```
postgres:
pg_external: true
pg_host: 10.0.1.170
pg_password: FortinetSQL
pg_port: 5432
pg_ssl_cert_path: ''
pg_use_ssl: false
pg_user: cyberpgsql
postgres_archival:
pg_archival_external: true
pg_archival_host: 10.0.1.170
pg_archival_password: FortinetSQL
pg_archival_port: 5432
pg_archival_ssl_cert_path: ''
pg_archival_use_ssl: false
pg_archival_user: cyberpgsql
```

- After updating those fields, press `Esc` to exit insert mode, and enter `:wq` to save the file and quit the editor.

To initiate the externalization process and verify the connection

- To initiate the externalization process, run the following command:

```
csadm db --externalize
```



The PostgreSQL server has been set up for you. It is deployed on a Linux VM and its configuration is not covered in this lab exercise. The focus of the exercise is on configuring FortiSOAR to connect to the external server.

- When prompted to specify a path to save your database backup file, press `Enter` to accept the default path.
 - Wait for the externalization process to finish. The process should take approximately 10 minutes.
 - Run the following command to check on the database connection again:
- ```
csadm db --check-connection
```
- Review the output and confirm the connection is established between FortiSOAR and the PostgreSQL server at 10.0.1.170.

```
[root@fortisoar database]# csadm db --check-connection
List of databases

```

| Name          | Owner      | Encoding | Collate     | Ctype       | Access privileges     |
|---------------|------------|----------|-------------|-------------|-----------------------|
| connectors    | cyberpgsql | UTF8     | en_US.UTF-8 | en_US.UTF-8 |                       |
| das           | cyberpgsql | UTF8     | en_US.UTF-8 | en_US.UTF-8 |                       |
| data_archival | cyberpgsql | UTF8     | en_US.UTF-8 | en_US.UTF-8 |                       |
| gateway       | cyberpgsql | UTF8     | en_US.UTF-8 | en_US.UTF-8 |                       |
| notifier      | cyberpgsql | UTF8     | en_US.UTF-8 | en_US.UTF-8 |                       |
| postgres      | postgres   | UTF8     | en_US.UTF-8 | en_US.UTF-8 |                       |
| postman       | cyberpgsql | UTF8     | en_US.UTF-8 | en_US.UTF-8 |                       |
| sealab        | cyberpgsql | UTF8     | en_US.UTF-8 | en_US.UTF-8 |                       |
| template0     | postgres   | UTF8     | en_US.UTF-8 | en_US.UTF-8 | =c/postgres           |
| template1     | postgres   | UTF8     | en_US.UTF-8 | en_US.UTF-8 | =c/postgres           |
| venom         | cyberpgsql | UTF8     | en_US.UTF-8 | en_US.UTF-8 | postgres=Cto/postgres |
| (11 rows)     |            |          |             |             |                       |

The connection established between 10.0.1.170 and Cyops instance.

- Run the following command to view the database logs (use the arrow keys to navigate up or down):

```
vi /var/log/cyops/csadm/db.log
```

7. Log out of the FortiSOAR SSH session.

## Lab 5: Machine Learning and War Rooms

In this lab, you will configure machine learning on FortiSOAR to train the recommendation engine to deliver better search results. You will also configure and work in a war room.

### Objectives

- Train the FortiSOAR recommendation engine using machine learning
- Configure a war room

### Time to Complete

Estimated: 45 minutes

## Exercise 1: Configuring Machine Learning

In this exercise, you will configure machine learning on FortiSOAR, which is based on training the machine learning engine using the data on your FortiSOAR VM.

### Configure Machine Learning

You will configure the recommendation engine on FortiSOAR to use machine learning.

#### To configure the recommendation engine

1. Log in to the FortiSOAR-Primary GUI with the username `csadmin` and password `Fortinet1!`.  
You may need to refresh the browser page after the last lab exercise.
2. Click **Settings**.
3. Click **Recommendation Engine**.
4. Confirm that **Status** is set to **Enabled**.
5. In the **Recommendation Strategy** section, click **Machine Learning Based Clustering**.
6. In the **Selected Recommendation Connector** field, confirm that **FortiSOAR ML Engine** is selected.
7. In the **Select Configuration** field, confirm that **Add new configuration** is selected.
8. In the **Configuration Name** field, type `ML-Dataset-1`.
9. Select the **Mark As Default Configuration** checkbox.
10. Confirm that the **Visibility** field is set to **Public**.

**Recommendation Engine**

FortiSOAR Recommendation Engine empowers the product intelligence framework to suggest ranked threat similarities and field predictions, thereby helping to orchestrate better, informed and holistic investigations. You can choose a suitable recommendation strategy and tune it to match your requirements.

Status **Enabled**

**Recommendation Strategy**  
Select a preferred recommendation strategy.

☐ Elasticsearch Based Text Classification  
*Based on the analysis of the similar record search, powered by Elasticsearch's highly efficient algorithms and on analyzing the search results.*

☒ **Machine Learning Based Clustering**  
*Based on training the engine on your existing FortiSOAR instance data, using the traditional machine learning supervised classification algorithms, such as K-Nearest Neighbour.*

**Selected Recommendation Connector**  
FortiSOAR ML Engine

**Select Configuration**  
+ Add new configuration

⚙️ CONFIGURATION: IN PROGRESS

**Configuration Name**  
ML-Dataset-1

☒ Mark As Default Configuration

Visibility **Public** Private

Module to train for:

11. In the **Module to train for** field, select **Alerts**.
12. In the **Feature Set** field, select **Escalated**.
13. In the **Feature Set** field, hold the **Ctrl** key and select **Type**.
14. In the **Verdict** field, select **Assigned To**.
15. In the **Verdict** field, hold the **Ctrl** key and select **Severity**.
16. In the **Verdict** field, hold the **Ctrl** key and select **Type**.
17. In the **Date Range** field, select **Last 5 years**.
18. In the **Training Set Size** field, type 1000.

The **Training Set Size** specifies the number of records that make up the training set. However, the value that you select in the **Date Range** field overrides this parameter.

19. In the **Algorithm** field, select **K-Nearest Neighbors**.
20. Confirm that the **Listener Port** field is set to **10443**.

21. Click **Save & Train Model**.
22. Scroll to the top of the page, and then confirm that the **CONFIGURATION** status is **COMPLETED** and the **HEALTH CHECK** status is **AVAILABLE**.

You may need to wait a few minutes if the training is showing **In Progress**.



## To show similar records

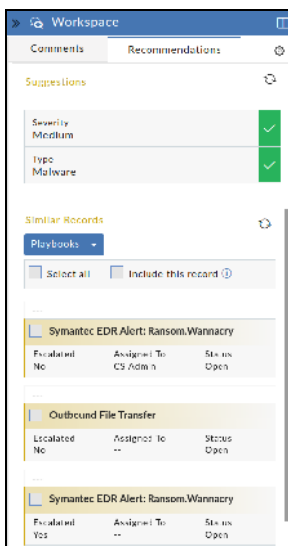
1. Continuing on the FortiSOAR-Primary GUI, click **Incident Response**.
2. Click **Alerts**.
3. Click to edit the **Symantec EDR Incident** alert.

The screenshot shows the 'Alert-14 Symantec EDR Incident' page in the FortiSOAR GUI. The alert is titled 'Alert-14 Symantec EDR Incident' and is categorized as 'epmp incident-2018-04-17/incident'. It has a 'Medium' severity level. The 'Last Modified' timestamp is '24/12/2021 01:05 PM by Playbook'. Below the header, there are tabs for 'Alert Details', 'Playbooks', and 'Audit Logs'. The 'Alert Details' tab is active, showing 'SLA Details'. The 'SLA Details' section includes fields for 'Ack Due Date' (NA), 'Ack Date' (NA), 'Ack SLA' (Awaiting Action), 'Response Due Date' (-NA-), 'Response Date' (-NA-), and 'Response SLA' (Awaiting Action).

4. Click the **Workspace** icon (🏠).
5. In the **Workspace** panel, click the **Recommendations** tab.
6. Click the **Settings** icon.
7. On the **Similar Records** tab, enable **Show Similar Records Suggestions**.
8. In the **Choose Playbook** field, select **Escalate**.
9. In the **Similarity Record Layout** section, select the **Name**, **Escalated**, **Assigned To**, and **Status** fields.
10. In the **Card Left Border Color Based On** field, select **Severity**.

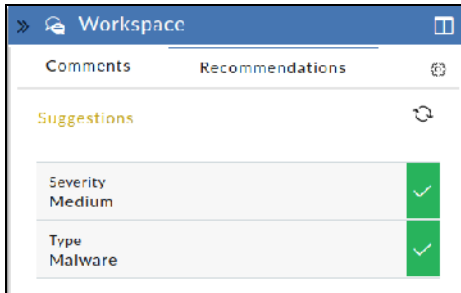
The screenshot shows the 'Recommendations Settings' dialog box. It has two tabs: 'Similar Records' and 'Suggestions'. The 'Similar Records' tab is active. The 'Show Similar Records Suggestions' toggle is turned 'ON'. Below this, there is a 'Select Playbooks' section with a 'Choose Playbook' dropdown menu. The 'Escalate' option is selected. The 'Similarity Record Layout' section has a 'Choose fields to show on the similarity suggestion cards' dropdown menu. The 'Name', 'Escalated', 'Assigned To', and 'Status' fields are selected. The 'Card Left Border Color Based On' dropdown menu is set to 'Severity'.

11. Click the **Suggestions** tab.
  12. Enable **Show Suggestions**.
  13. Click **Save**.
- You will see **Suggestions** and **Similar Records**.



After you have trained your dataset, FortiSOAR starts to analyze your dataset. Based on this analysis, FortiSOAR displays records that are similar to the record you are working on, and predicts the values of field records that you added to the **Verdict** field. Since you trained the dataset to predict the **Assigned To**, **Severity**, and **Type** fields based on the **Type** field, FortiSOAR provides suggestions for those fields, as shown in the image above.

14. Click the green checkboxes beside **Severity Medium** and **Type Malware** to agree with the recommendations.



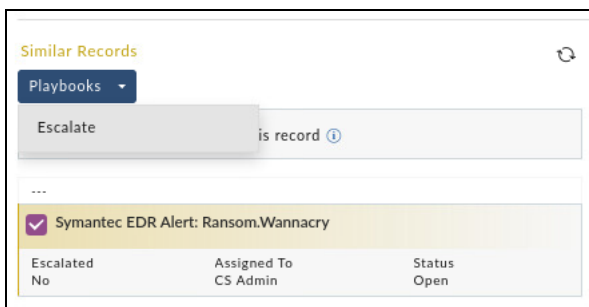
### To escalate alerts from the recommended list

1. Continuing on the FortiSOAR-Primary GUI, in the **Similar Records** section, select the **Symantec EDR Alert: Ransom.Wannacry** record that has the **Escalated** status of **No**.



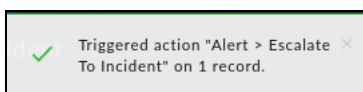
The two ransomware records are shown because their **Type** is **Malware**, and FortiSOAR used the **Type** field to display them under the **Similar Records** category.

2. Click the **Playbooks** field, and then select **Escalate**.



3. In the **Incident Name** field, type Wannacry Malware Investigation.
4. In the **Severity** field, select **High**.
5. In the **Incident Lead** field, select **John Smith**.

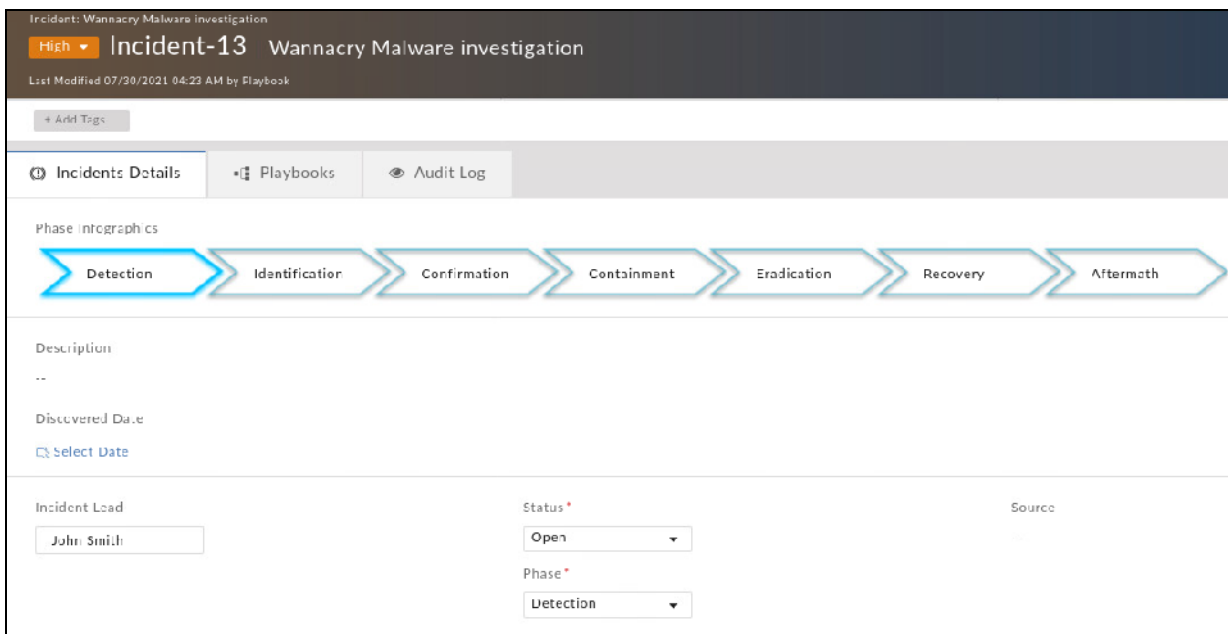
6. Click **Escalate**.
7. Wait for confirmation that the alert has been escalated to an incident.



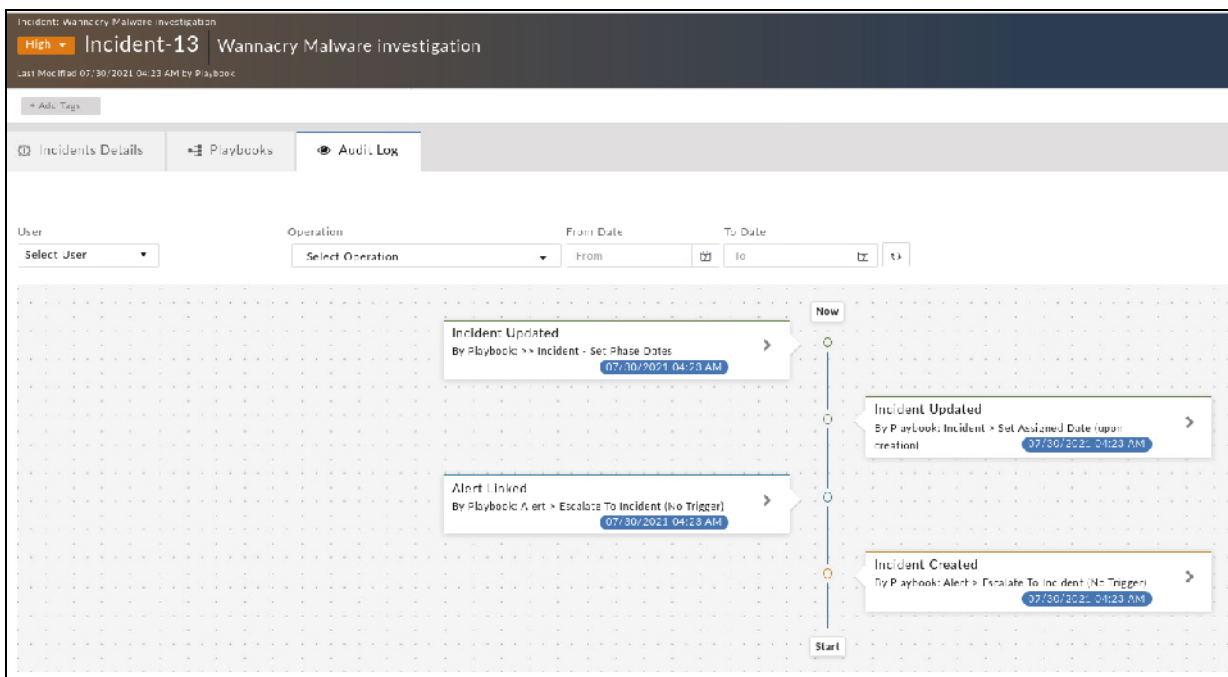
8. Close the alert.
9. Click **Incident Response**.
10. Click **Incidents**.
11. Click the **Wannacry Malware investigation** incident that was just created.

| ID | Severity | Name                                  | Source | Type | Phase     | Status   | Incident Lead       | Created On      | Ack SLA | Response SLA |
|----|----------|---------------------------------------|--------|------|-----------|----------|---------------------|-----------------|---------|--------------|
| 13 | High     | Wannacry Malware investigation        |        |      | Detection | Open     | John Smith          | 07/20/2021 0... |         |              |
| 12 | Medium   | Malware incident                      |        |      | Detection | Open     | John Smith          | 07/20/2021 0... |         |              |
| 10 | Low      | CUSTOM Windows Scheduled Task Created |        |      | Recovery  | Awaiting | Fortinet, traine... | 04/12/2021 1... |         |              |

12. Confirm that the incident was assigned to John Smith, and the severity was changed to **High**.



13. Click the **Audit Log** tab.  
You can see a timeline of the escalated incident.



14. Log out of the FortiSOAR GUI.

## Exercise 2: Configuring a War Room

In this exercise, you will provision a war room and define the policies that will achieve the functionality required to effectively run a war room. War rooms enable SOC teams to get into a collaborative space to mitigate a critical cyber threat scenario or campaign. FortiSOAR makes it easy for analysts to quickly provision a war room and ensures that the task force is well-equipped to coordinate during critical situations.

### Provision a War Room

You will create a war room, designate a response team, assign tasks, investigate the incident, and provide a report based on the investigation.

#### To assign the appropriate permissions

1. Log in to the FortiSOAR-Primary GUI with the username `csadmin` and password `Fortinet1!`.
2. Click **Settings**.
3. Click **Roles**.
4. Click the **Security Administrator** role to edit it.
5. In the **Set Role Permissions** section, select **+** beside the **Announcements** module to enable CRUD permissions.

| Set Role Permissions |                                                                         |                                                   |                                                   |                                                   |                                                   |                                       |
|----------------------|-------------------------------------------------------------------------|---------------------------------------------------|---------------------------------------------------|---------------------------------------------------|---------------------------------------------------|---------------------------------------|
| Module               |                                                                         | Create                                            | Read                                              | Update                                            | Delete                                            | Fields Permissions                    |
|                      | <input type="checkbox"/> <input type="checkbox"/>                       | <input type="checkbox"/> <input type="checkbox"/> | <input type="checkbox"/> <input type="checkbox"/> | <input type="checkbox"/> <input type="checkbox"/> | <input type="checkbox"/> <input type="checkbox"/> |                                       |
| Agents ⓘ             | <input type="checkbox"/> <input type="checkbox"/>                       | <input type="checkbox"/>                          | <input type="checkbox"/>                          | <input type="checkbox"/>                          | <input type="checkbox"/>                          | <a href="#">Set Field Permissions</a> |
| Alerts ⓘ             | <input type="checkbox"/> <input type="checkbox"/>                       | <input type="checkbox"/>                          | <input type="checkbox"/>                          | <input type="checkbox"/>                          | <input type="checkbox"/>                          | <a href="#">Set Field Permissions</a> |
| Announcements ⓘ      | <input checked="" type="checkbox"/> <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/>               | <input checked="" type="checkbox"/>               | <input checked="" type="checkbox"/>               | <input checked="" type="checkbox"/>               | <a href="#">Set Field Permissions</a> |
| Appliances ⓘ         | <input type="checkbox"/> <input type="checkbox"/>                       | <input type="checkbox"/>                          | <input type="checkbox"/>                          | <input type="checkbox"/>                          | <input type="checkbox"/>                          | <a href="#">Set Field Permissions</a> |
| Application ⓘ        | <input type="checkbox"/> <input type="checkbox"/>                       | <input checked="" type="checkbox"/>               | <input checked="" type="checkbox"/>               | <input checked="" type="checkbox"/>               | <input checked="" type="checkbox"/>               |                                       |

6. Repeat the above step and enable CRUD permissions for the following modules: **Comment**, **Connectors**, **Dashboard**, **Incidents**, **People**, **Tasks**, **War Rooms**, and **Widgets**.



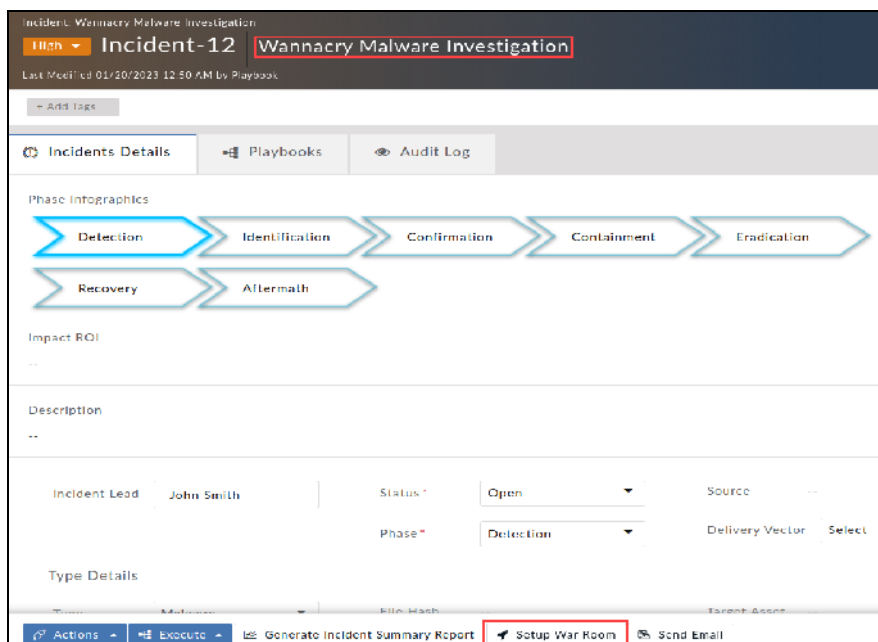
Ensure you have all the required module permissions listed above in steps 5-6 added to the Security Administrator role before continuing this exercise.

7. Click **Save**.
8. Click **Content Hub** on the navigation bar.
9. Click on the **Manage** tab.
10. Search for **Access Control**. Double-click the widget, review its description, and confirm the widget has been installed.

11. Close the widget.
12. Search for **Task Management**. Double-click the widget, review its description, and confirm the widget has been installed.
13. Close the widget.

### To set up a war room

1. Continuing on the FortiSOAR-Primary GUI, click **Incident Response**.
2. Click **Incidents**.
3. Click to edit the **Wannacry Malware Investigation** incident that you escalated from an alert in the last exercise.
4. Click **Set up War Room**.



5. In the new **Set up War Room** window, confirm that **Status** is set to **Draft**.
6. In the **War Room Title** field, type Wannacry Malware Investigation War Room.
7. In the **War Room Description** field, type the following:  
Wannacry malware detected. Please investigate as per the following guidelines:
  1. Isolate the infected device.
  2. Run malware scanner on infected device.
  3. Confirm if it is a real threat or false positive.
8. Confirm that the **Include all analysts with this incident as War Room responders** checkbox is selected.

Setup War Room

Status \*

Draft

War Room Title \*

Wannacry Malware Investigation War Room

War Room Description

Write Preview

Wannacry malware detected. Please investigate as per the following guidelines:

1. Isolate the infected device.
2. Run malware scanner on infected device.
3. Confirm if it is a real threat or false positive.

Include all analysts involved with this incident as War Room responders

☒

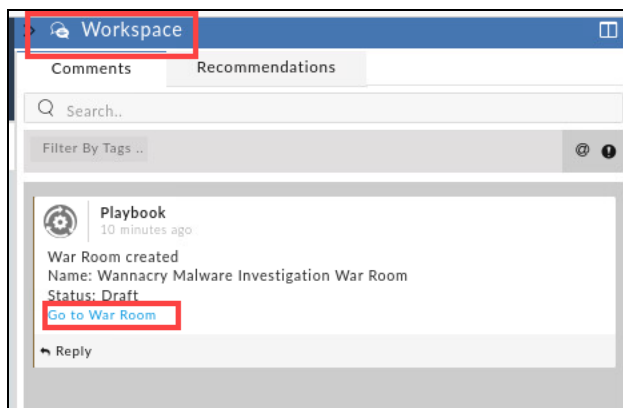
Create Cancel

9. Click **Create**.

You should see the **Triggered action "Set up War Room from Incidents" on 1 record** message.

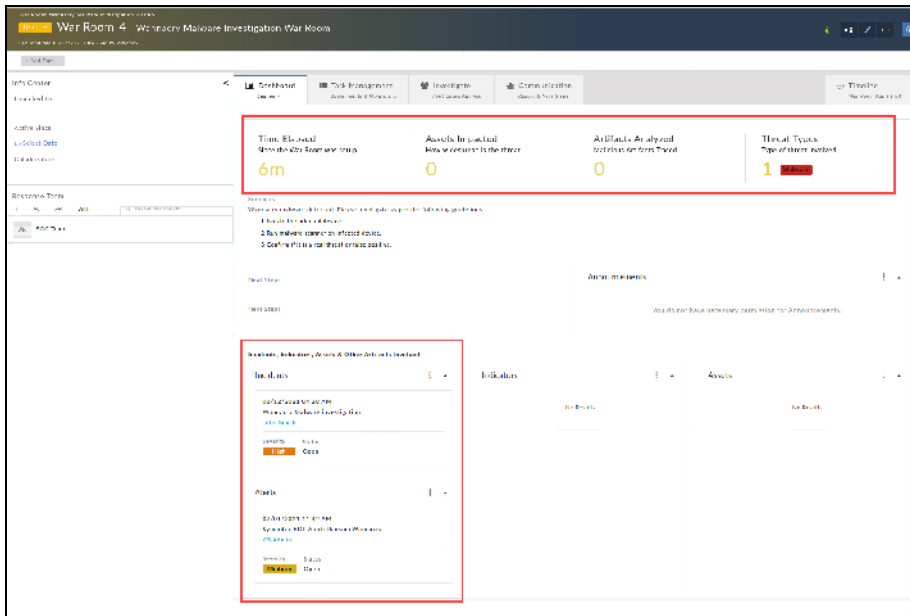
## To use the war room

1. Continuing on the **Incident** page, click the **Workspace** icon to expand it.
2. Click **Go to War Room**.

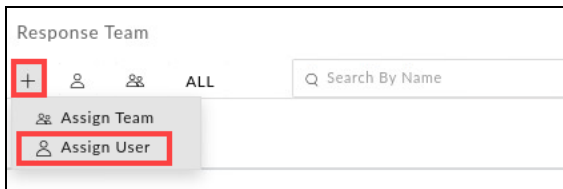


The **War Room** page opens. You may need to refresh your browser.

The **Dashboard** tab shows important information, such as **Assets Impacted**, **Artifacts Analyzed**, and **Threat Types**. It also shows all the incidents, indicators, and alerts associated with this incident.

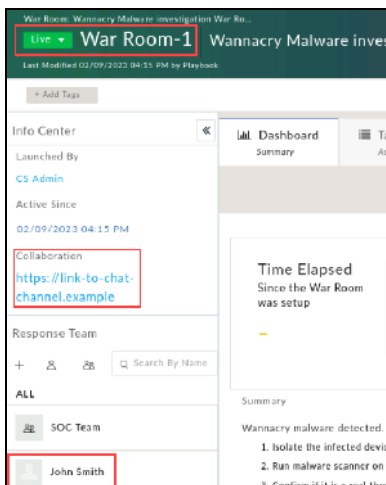


3. In the **Response Team** section, click **+**.
4. In the **+** field, select **Assign User**.



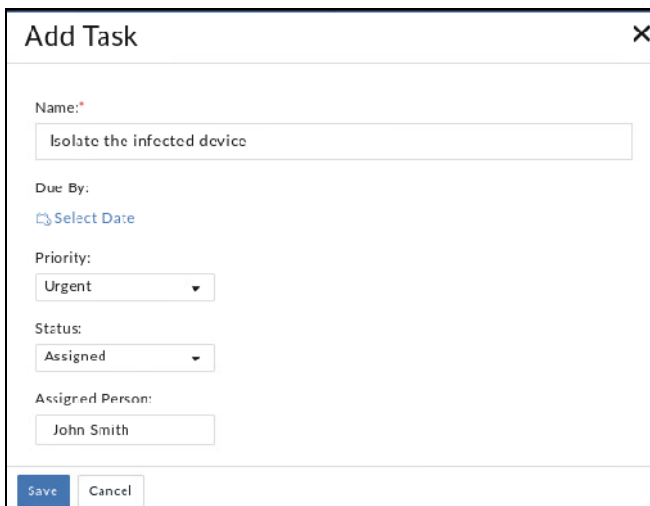
5. In the **Individual Owners** field, select **John Smith**.
6. Click **Save**.
7. Click **Go Live**.
8. In the **External Collaboration Link**, type `https://link-to-chat-channel.example`.
9. Click **Go Live** again to confirm.

Notice that the status of the **War Room** changes to **Live**.

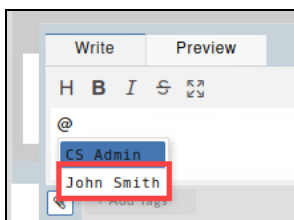




10. Click **Task Management**.
11. In the **Task List** field, click **+**.
12. In the **Name** field, type `Isolate the infected device`.
13. In the **Priority** field, select **Urgent**.
14. In the **Status** field, select **Assigned**.
15. In the **Assigned Person** field, select **John Smith**.



16. Click **Save**.  
Notice that the task has been added under the **Assigned** queue.
17. Click the **Workspace** icon to expand it.
18. Click **New Comment**.
19. Type @, and then select **John Smith** in the @ field.



20. Type `Please perform the assigned task`.
21. Press `Enter`.
22. Click the **Communication** tab.
23. Click **Next Steps**.
24. Type `Review firewall policies`.
25. In the **Announcement** section, click **Add**.
26. In the **Criticality** field, select **Generic**.
27. In the **Title** field, type `Review Firewall Policies`.
28. In the **Description** field, type `We are going to review firewall policies, which might affect some traffic, and we will be sending updates during our investigation.`
29. Select the **Notify Responders On Email** checkbox.

**Add New Announcement**

Announcement Type:  
Information

Criticality:  
Generic

Title:  
Review Firewall Policies

Description:  
We are going to review firewall policies, which might affect some traffic, and we will be sending updates during our investigation

Notify Responders On Email:  
☒

Create Cancel

30. Click **Create**.

**Communication**  
Report & Next Steps

**Reported Status Fields**

Summary

Wannacry malware detected. Please investigate as per the following guidelines:

1. Isolate the infected device.
2. Run malware scanner on infected device.
3. Confirm it is real threat or a false positive.

**Current Status**  
--

**Next Steps**  
Review Firewall policies.

**Announcements**

Announcements: 1 | Communications: 0

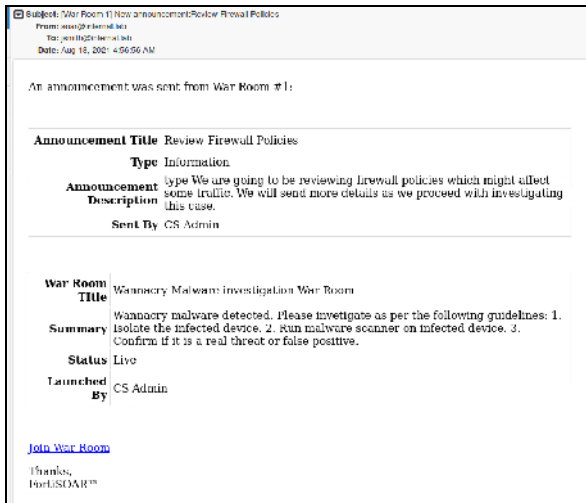
| 1 Items                  | Search      | Add                      | eLink             |
|--------------------------|-------------|--------------------------|-------------------|
| <input type="checkbox"/> | Criticality | Title                    | Announcement Type |
| <input type="checkbox"/> | Generic     | Review Firewall Policies | Information       |
|                          |             |                          | 1                 |

31. Close the war room window.

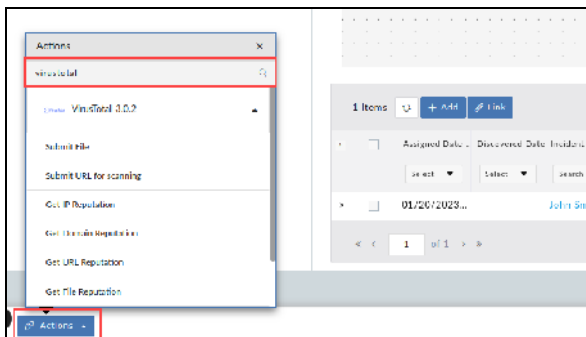
32. Log out of the FortiSOAR GUI.

### To perform tasks in a war room

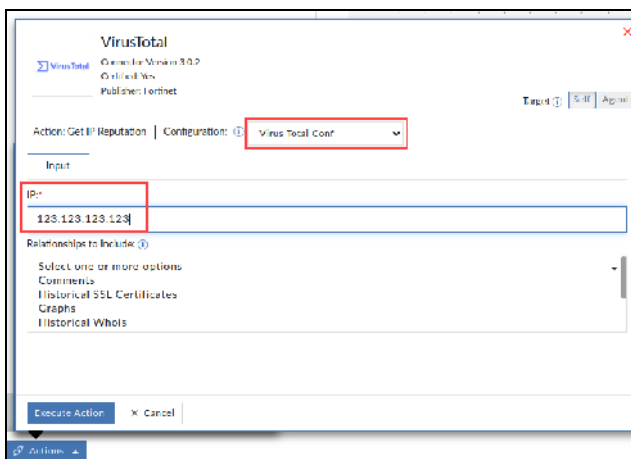
1. Log in to the FortiMail webmail GUI with the username `jsmith` and password `Fortinet1!`.
2. Open the emails that notify John Smith about being added to a war room, a task being assigned, and a new announcement on the war room being broadcast.



3. Return to the FortiSOAR GUI, and then log in with the username `jsmith` and password `Fortinet1!`.
4. Click **Incident Response**.
5. Click **War Rooms**.
6. Click **Wannacry Malware Investigation War Room**.
7. Click the **Task Management** tab.
8. Click the **Isolate the infected device** card to edit it.
9. Set the **Status** to **Accepted**.
10. Click **Save**.
11. Click the **Workspace** icon to expand it.
12. Click **New Comment**.
13. Type `I'm working on it`.
14. Press `Enter`.
15. Click **Investigate**.
16. Click **Actions**.
17. In the **Search Actions** field, type `VirusTotal`.



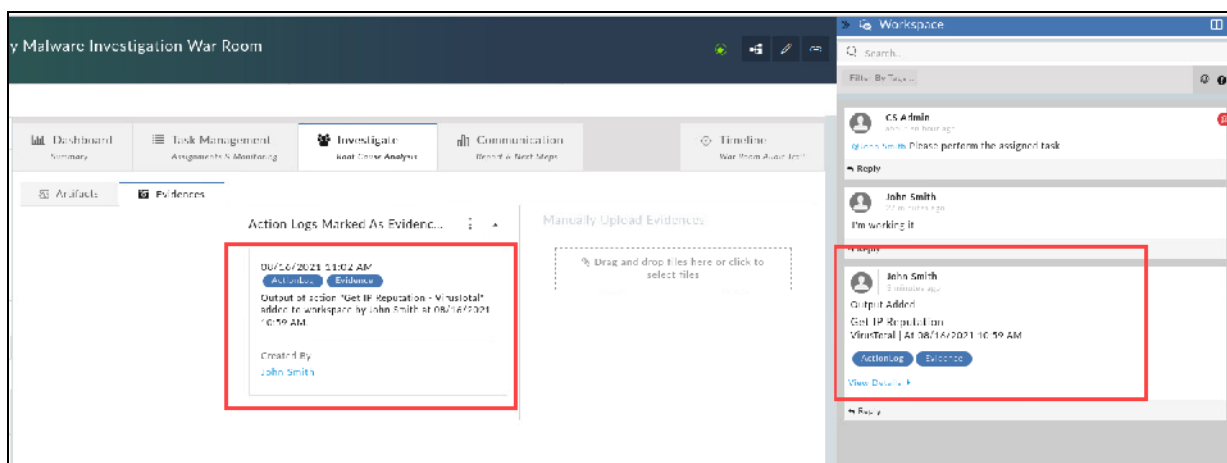
18. Click the **VirusTotal 3.0.2** field, and then select **Get IP Reputation**.
19. Confirm that **Virus Total Conf** is selected in the **Configuration** field.
20. In the **IP** field, type `123.123.123.123`.



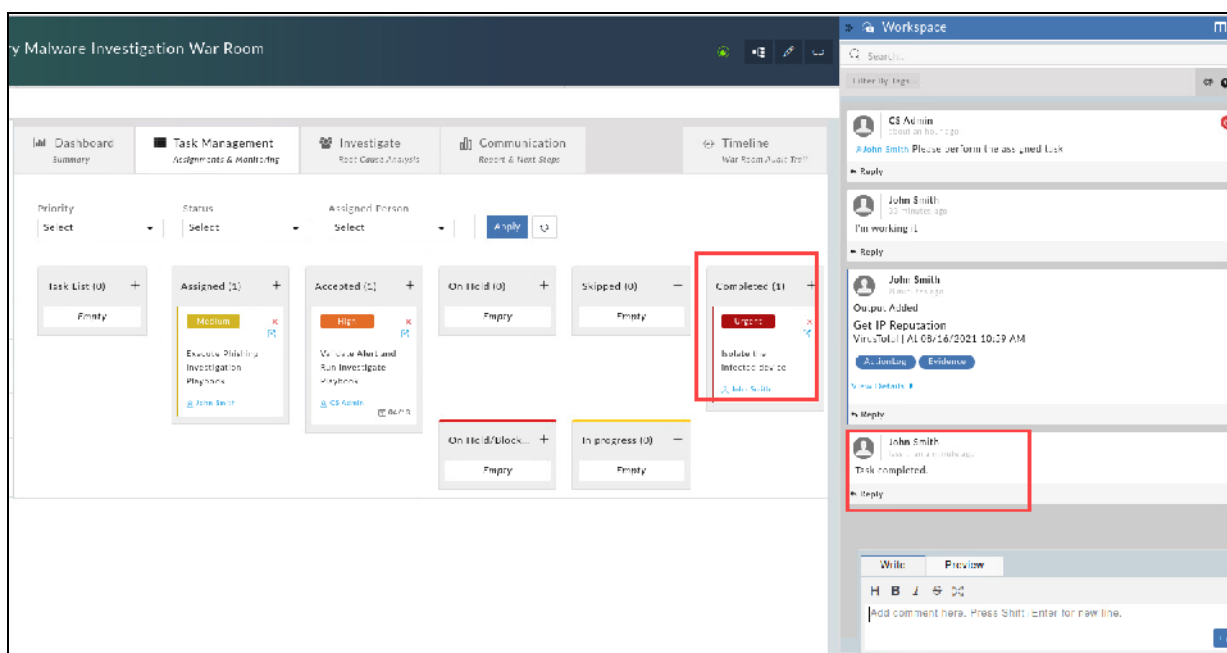
21. Click **Execute Action**.
22. Click **Add Tags**.
23. Type **Evidence**, and then press **Enter** to add this word as a tag.
24. Select the **Key** checkbox.



25. Click **Save selected output keys**.  
Notice that the output is added to the workspace comments section.
26. Click the **Evidences** tab.  
The output has also been added as evidence.



27. Click the **Task Management** tab.
28. Under the **Accepted** field, click the **Isolate the infected device** card.
29. Change the **Status** to **Completed**.
30. Click **Save**.
31. In the **Workspace** comment section, click **New Comment**.
32. Type **Task completed**.
33. Press **Enter**.



34. Close the war room window.
35. Log out of the FortiSOAR GUI.



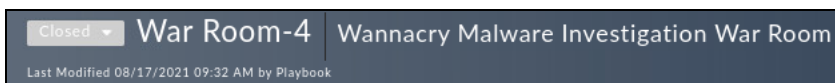
Note that you are not going to be investigating this incident as in real life because of limitations of the lab environment, but you are learning the framework of how a war room works.

### To close the war room

1. Continuing on the FortiSOAR-Primary GUI, log in with the username `csadmin` and password `Fortinet1!`.
2. Click **Incident Response**.
3. Click **War Rooms**.
4. Click **Wannacry Malware Investigation War Room**.
5. Click the **Workspace** icon to expand it.

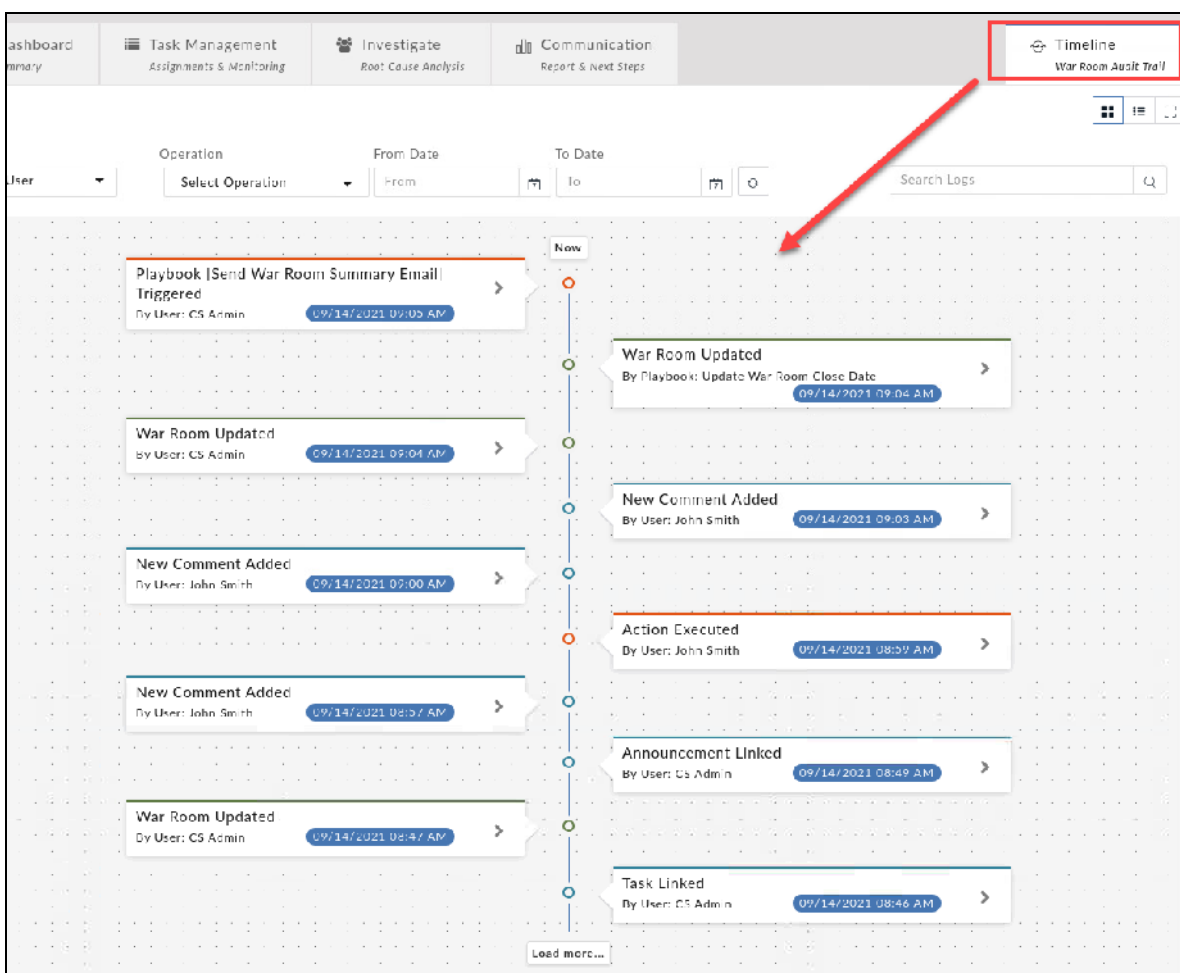
Note that all communication between the `csadmin` and `jsmith` users is seen in the workspace comments section by all teams and members who belong to this war room.

6. Click the **Live** field, and then select **Closed**.



7. Click **Send Email Update**.
8. Click **Send Email**.
9. Click the **Timeline** tab.

This provides a timeline of all the actions taken in this war room from when it is opened until it is closed.



10. Log in to the FortiMail webmail GUI using the username `jsmith` and password `Fortinet1!`,
11. Check if the **WarRoomSummaryReport** was received as an attachment over email.  
You may need to wait a few minutes for the email to arrive.
12. Open the PDF and review its contents.
13. Log out of the FortiSOAR GUI.
14. Log out of the FortiMail GUI.

## Lab 6: FortiSOAR Services Monitoring

In this lab, you will configure a monitoring dashboard on FortiSOAR, and set up monitoring of system resources. You will also view the status of FortiSOAR services, and collect log files through the CLI and the GUI.

### Objectives

- Configure monitoring dashboards on FortiSOAR
- Configure monitoring of system resources
- Monitor FortiSOAR services in the CLI
- Collect FortiSOAR logs through the CLI and the GUI

### Time to Complete

Estimated: 50 minutes



## Exercise 1: Monitoring Dashboards

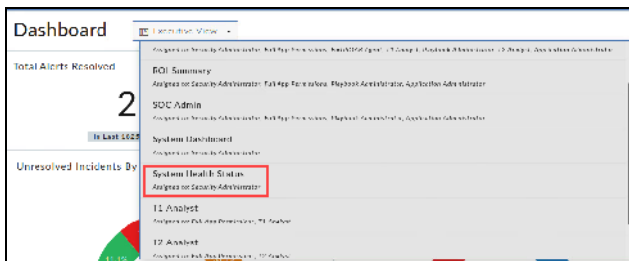
In this exercise, you will configure monitoring dashboards for system resources. FortiSOAR includes a default system monitoring dashboard, the **System Health Status** dashboard, which allows you to monitor various FortiSOAR system resources, such as CPU, disk space and memory utilization, and the status of various FortiSOAR services.

### Configure the System Monitoring Dashboard

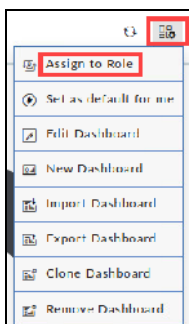
You will configure a new role to assign a dashboard to. You will also configure monitoring dashboards for various system resources.

#### To configure roles and monitoring thresholds

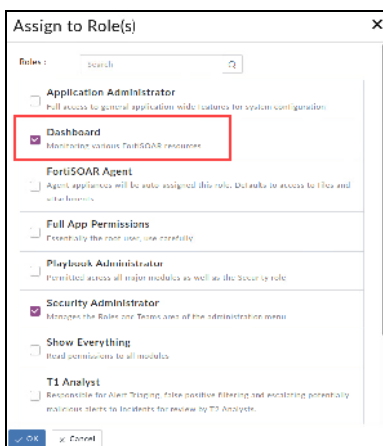
1. Log in to the FortiSOAR GUI with the username `csadmin` and password `Fortinet1!`.
2. Click **Settings**.
3. Click **Roles**.
4. Click **Add** to add a new role.
5. In the **Role Name** field, type `Dashboard`.
6. In the **Description** field, type `Monitoring various FortiSOAR resources`.
7. In the **Set Role Permissions** section, in the **Dashboard** module, select **Create** and **Read**.
8. In the **Set Role Permissions** field, in the **Security** module, select **Create** and **Read**.
9. Click **Save**.
10. Click **Dashboard**.
11. In the **Dashboard** drop-down menu, select **System Health Status**.



12. Click the **Dashboard settings** icon.
13. Click **Assign to Role**.



14. Select **Dashboard**.



15. Click **OK**.

16. Click the **Dashboard Settings** icon.

17. Click **Edit Dashboard**.

18. Click **Configure Inputs**.

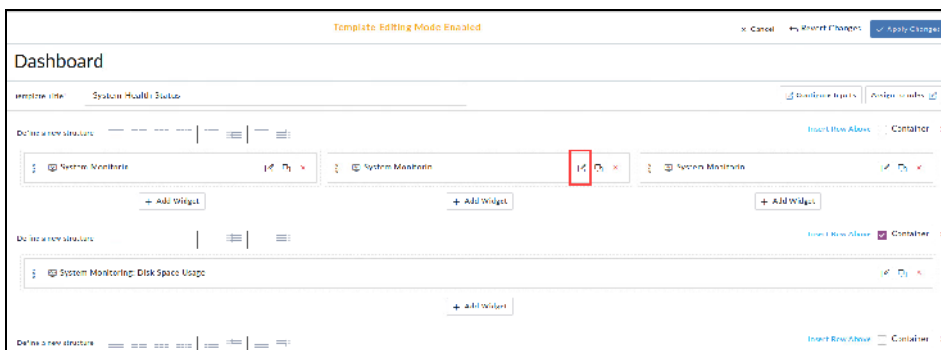
19. Select **Enable Auto-refresh**.

20. Change the auto refresh to **Every 5 Minutes**.



21. Click **Save**.

22. Click the **Edit Widget** button in the middle column to edit the **Virtual Memory usage** widget.



23. In the **Threshold Percentage** field, type 5.

The screenshot shows the 'Edit System Monitoring Metrics' dialog box. It has a 'Title' field with the value 'Virtual Memory Usage'. Below it is a 'Choose Type Of System Monitoring' dropdown menu with 'Memory Utilization' selected. Below that is a 'Threshold Percentage' field with the value '5'. At the bottom, there are 'Save' and 'Close' buttons.

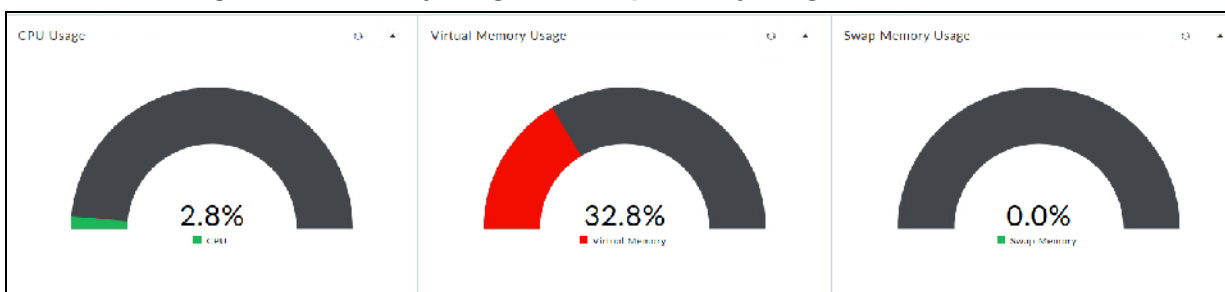
24. Click **Save**.

25. Click **Apply Changes**.

### To create a dashboard user

1. Continuing on the FortiSOAR GUI, click **Settings**.
2. Click **Users**, and then click **Add**.
3. In the **Name** field, type `Jane Doe`.
4. In the **Type** drop-down list, select **Admin**.
5. In the **Email** field, type `jd@internal.lab`.
6. In the **Teams** section, select **SOC Team**.
7. In the **Roles** section, select **Dashboard**.
8. In the **Authentication** section, ensure **User Type** is set to **Dashboard User**.
9. In the **Username** field, type `jd`.

10. Click **Save**.
11. From the **Users** interface, double-click the user **jdoe** to edit it again.
12. In the **Authentication** section, click **Reset Password**.
13. Set the password to `Fortinet1!`.
14. Uncheck the **Send Email to User** box.
15. Click **Submit**.
16. Log out of the FortiSOAR GUI.
17. Log in to the FortiSOAR GUI with the username `jdoe` and password `Fortinet1!`.
18. Click **Dashboard**.
19. In the **Dashboard** drop-down menu, select **System Health Status**.
20. Note the **CPU Usage**, **Virtual Memory Usage**, and **Swap Memory Usage** on FortiSOAR.



### Stop and think!

Why is the **Virtual Swap Memory Usage** graph in red?

Recall the threshold percentage set earlier in this lab exercise.

Note that the **Disk Space Usage** widget includes the path to the directories in FortiSOAR that the disk usage is monitoring.



The dashboard also shows the **Service Status** widget, which monitors the status of all the services running on FortiSOAR.

| Service Status           |   |
|--------------------------|---|
| rabbitmq-server          | ✓ |
| elasticsearch            | ✓ |
| postgresql-14            | ✓ |
| nginx                    | ✓ |
| cyops-auth               | ✓ |
| php-fpm                  | ✓ |
| uwsgi                    | ✓ |
| celeryd                  | ✓ |
| celerybeatd              | ✓ |
| cyops-tomcat             | ✓ |
| cyops-search             | ✓ |
| cyops-postman            | ✓ |
| cyops-integrations-agent | ✓ |
| fsr-workflow             | ✓ |
| fsr-api-consumer         | ✓ |
| cyops-ha                 | ✓ |

### Stop and think!

Why is the **Connector Health Status** widget not loading?

Recall the permissions given to the role **Dashboard** earlier in this lab exercise. Which permissions were missing?



- Log out of the FortiSOAR GUI.
- Log back in with the username `csadmin` and password `Fortinet1!`.
- Click **Settings**.
- Click **Roles**.
- Click **Dashboard**.
- In the **Set Role Permissions** section, enable **Create** and **Read** permissions for the **Connectors** module.
- Click **Save**.
- Log out of the FortiSOAR GUI.
- Log back in with the username `jdoe` and password `Fortinet1!`.
- In the **Dashboard** drop-down menu, select **System Health Status**.

The **Connector Health** widget is loading now.

| Connector Health Status    |  |                     |
|----------------------------|--|---------------------|
|                            |  |                     |
| FortiSOAR SOC Simulator    |  |                     |
| 1 Configuration Monitored  |  |                     |
| Configuration Name         |  | Health Check Status |
| lab                        |  | Available           |
| Symantec ATP               |  |                     |
| 1 Configuration Monitored  |  |                     |
| Configuration Name         |  | Health Check Status |
| lab                        |  | Disconnected        |
| SMTP                       |  |                     |
| 2 Configurations Monitored |  |                     |
|                            |  | All Available       |

31. Log out of the FortiSOAR GUI.

## Exercise 2: Configuring System Monitoring

In this exercise, you will configure system monitoring to send out email notifications when resources reach or exceed thresholds. You will also learn how to view and manage services. It is important to configure effective monitoring for your FortiSOAR environment and know how to view processes and their statuses if troubleshooting is required.

### Configure System Monitoring

You will configure FortiSOAR to send email notifications if the memory of the instance exceeds a certain threshold. You will also learn how to view and manage service statuses using the CLI.

#### To configure system monitoring

1. Log in to the FortiSOAR GUI with the username `csadmin` and password `Fortinet1!`.
2. Click **Settings**.
3. Select the **Enable Notification** checkbox.
4. In the **Email** field, type `csadmin@internal.lab`.
5. In the **Monitoring Interval (Minutes)** field, type `2`.
6. In the **Memory Utilization (%)** field, type `3`.
7. In the **Disk Utilization (%)** field, type `3`.
8. Click **Save**.

The screenshot shows the 'System Configuration' page in FortiSOAR. The 'System & Cluster Health Monitoring' section is active. The 'Enable Notification' checkbox is checked. The 'Email' field is set to 'csadmin@internal.lab'. The 'Monitoring Interval (Minutes)' is set to '2'. The 'System Health Thresholds' section shows the following values: 'Memory Utilization (%)' is 3, 'CPU Utilization (%)' is 80, 'Disk Utilization (%)' is 3, 'Swap Memory Utilization (%)' is 80, 'Workflow Queue Threshold' is 100, and 'WAL Files Size (GB)' is 20.

9. Log in to the FortiMail GUI (10.0.1.200) with the username `csadmin` and password `Fortinet1!`.
10. Click **Inbox**.
11. Click on any email with the subject **FortiSOAR Notification - Node Health Needs Immediate Attention**.
12. Review the email notification.


**Subject:** FortiSOAR Notification - Node Health: Needs Immediate Action

**From:** ccsd@fortissoc.com  
**To:** ccsd@fortissoc.com  
**Date:** Jan 24, 2023 12:39:54 PM

**Disk usage exceeds the configured threshold (95%).**

Details are below:

| mountpoint        | device                     | total   | used   | avail  | percent |
|-------------------|----------------------------|---------|--------|--------|---------|
| /                 | /dev/mapper/vgusr-root     | 8.0G    | 4.1G   | 1.9G   | 68      |
| /boot             | /dev/sda1                  | 1.5G    | 226.4M | 1.3G   | 15      |
| /opt              | /dev/mapper/vgapp-csapps   | 9.5G    | 2.9G   | 6.6G   | 30.1    |
| /var              | /dev/mapper/vgusr-var      | 4.0G    | 849.9M | 3.2G   | 20.8    |
| /var/lib/rabbitmq | /dev/mapper/vgusr-rabbitmq | 7.5G    | 469.0M | 7.0G   | 6.1     |
| /var/log          | /dev/mapper/vgusr-log      | 5.0G    | 337.9M | 1.7G   | 6.6     |
| /var/log/audit    | /dev/mapper/vgusr-audit    | 1014.0M | 72.9M  | 941.1M | 7.2     |

**Memory threshold at 93% is reached.**

Details are below:

| total | used  | avail | percent |
|-------|-------|-------|---------|
| 31.2G | 16.5G | 20.1G | 35.6    |

Details of the top 3 memory consuming process are below:

| user     | uid | pid  | rss    | vsz   | memory_percent | cpu_percent | cmdline                                                                                                                                                  |
|----------|-----|------|--------|-------|----------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| elastic  | 989 | 2627 | 4.6G   | 10.0G | 14.6           | 0.5         | /usr/bin/java -Xshare:auto -Des.networkaddress.cache.on=60 -Des.networkaddress.cache.negative.ttl=10 -XX:+AlwaysPreTouch -Xss1m -Djava.awt.headless=true |
| tomcat   | 988 | 1707 | 691.8M | 10.0G | 2.1            | 0.3         | /usr/bin/java -Djava.util.logging.config.file=/opt/cnps/loraca/conf/logging.properties -Djava.util.logging.manager=org.apache.juli.ClassLoaderLogManager |
| rabbitmq | 992 | 1618 | 213.4M | 2.9G  | 0.6            | 0.8         | /usr/lib64/erlang/erts-12.3/bin/beam smp -W w -Mbs ageffcbf -Mbs ageffcbf -Mbsmbcs 512 -Mbsmbcs 512 -MMcsms 30 -t 1048376 -t 5000000 -stbt db-zdbbl      |

Identify the components reporting a failure and refer to the documentation [Help > Administration Guide > Monitoring FortiSOAR section](#) for more help on



The notification includes useful information to assist in troubleshooting. For example, for disk usage it will show the mountpoint, device, and a numerical overview of resources. For memory usage it will show a numerical overview as well, in addition to some of the top memory-consuming processes.

You can also use the monitoring dashboard from the previous lab exercise to help identify a baseline for resource usage during peak hours of your environment, and then tune the thresholds accordingly.

## To check the service status

1. Open an SSH session to the FortiSOAR-Primary VM at IP 10.0.1.160.
2. Log in with the username `csadmin` and password `changeme`.
3. Enter the following command to log in as the root user:  

```
sudo -i
```
4. Enter the following command to check the status of all the FortiSOAR services:

```
csadm services --status
```

```
root@fortisoar:~# csadm services --status
rabbitmq-server.....[Running]
elasticsearch.....[Running]
nginx.....[Running]
cyops-auth.....[Running]
php-lpm.....[Running]
uwsgi.....[Running]
celeryd.....[Running]
celerybeatd.....[Running]
cyops-tomcat.....[Running]
cyops-search.....[Running]
cyops-postman.....[Running]
cyops-integrations-agent.....[Running]
fsr-workflow.....[Running]
fsr-api-consumer.....[Running]
cyops-ha.....[Running]
```

5. Enter the following command to view the status of an individual FortiSOAR service:  
`systemctl status nginx`



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## Exercise 3: Collecting Logs

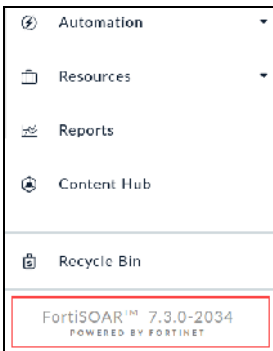
In this exercise, you will collect FortiSOAR logs using both the GUI and the CLI. You will also examine the folders and files within the logs. Collecting application logs is an important step in troubleshooting FortiSOAR issues. You can directly download logs from the GUI and review them for further investigation. It is useful to also know how to collect logs by using the CLI in cases where the GUI is not accessible.

### Collect FortiSOAR Logs

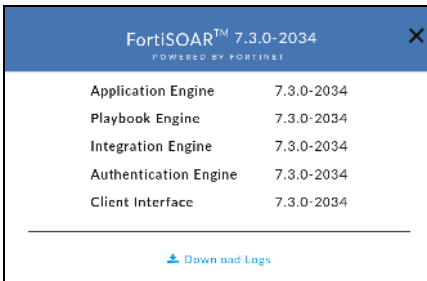
You will collect FortiSOAR logs using both the GUI and the CLI and then review them.

#### To collect logs using the GUI

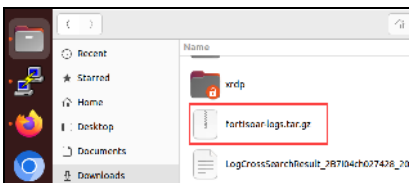
1. Log in to the FortiSOAR-Primary GUI with the username `csadmin` and password `Fortinet1!`.
2. On the bottom of the navigation bar, click the **FortiSOAR** box.



3. Click **Download Logs**.



4. Click **No, download without password** when prompted at the dialog window.
5. Wait a few minutes, then check the Downloads folder.



6. Right-click the downloaded file, and then click **Extract to....**



Note that your file name may be different depending on how many times you downloaded logs from FortiSOAR. Sort by date to ensure you are extracting the one you downloaded most recently.

7. In the **Select Extract Destination** window, click the **Create Folder** icon.



8. In the **Folder Name** field, type `fortisoar-logs-gui`.
9. Click **Select**.
10. Using the File Explorer, navigate to the new folder and examine the folder structure.
11. Navigate to `opt` and examine its subdirectories.
12. Navigate to `var` and examine its subdirectories.



You should notice that almost all logs are located under `var` with the exception of the `tomcat` service, which is nested under `opt`.

13. Navigate to the `var/log/` directory and double-click the `messages` file.
14. Review the contents of the file.



You can see a log of system messages, including statuses for the processes on the system. For example, you can search for "nginx.service" and locate the logs for the service you stopped and restarted in the last lab.

15. Close the text editor.
16. Log out of the FortiSOAR GUI.

### To collect logs using the CLI

1. Attempt to log in to the FortiSOAR-Primary GUI with the incorrect username `failedadminlogin` and password `Test`.  
You will generate a failed login attempt, which you will observe later in this exercise.
2. Open an SSH connection to the FortiSOAR VM at IP `10.0.1.160`.
3. Log in with the username `csadmin` and password `changeme`.
4. Enter the following command to log in as the root user:  

```
sudo -i
```
5. Enter the password `changeme`, if prompted.
6. Run the following command to collect log files inside the `/tmp/` folder.  

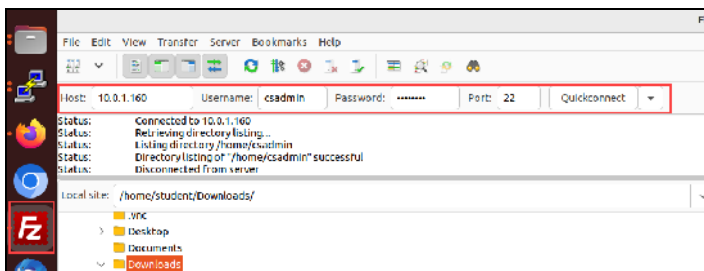
```
csadm log --collect /tmp/
```
7. After the collection is complete, make a note of the file name.

```
[root@primary tmp]# csadm log --collect /tmp/
Thu Jan 12 18:51:09 UTC 2023 Total logs size: 72M
Thu Jan 12 18:51:09 UTC 2023 Available size: 2.0G
Thu Jan 12 18:51:09 UTC 2023 Compressed log file size: 11M
Thu Jan 12 18:51:09 UTC 2023 Required size: 82M
Thu Jan 12 18:51:09 UTC 2023 Threshold size: 1.6G
Thu Jan 12 18:51:09 UTC 2023 Collecting logs
Thu Jan 12 18:51:11 UTC 2023 Logs collected at: /tmp/fortisoar-logs.tar.gz
```

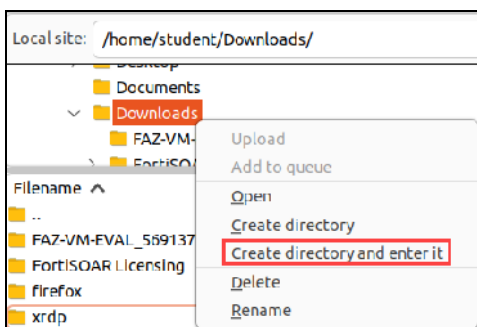
8. To change permissions on the file so you can transfer it to the Local-Host VM, run the following command:

```
chmod 777 /tmp/fortisoar.logs.tar.gz
```

9. On the Local-Host VM, open FileZilla, enter IP 10.0.1.160, username csadmin, password changeme, port 22, and then click **Quick Connect**.



10. In the **Local Site** section, right-click any folder, and then click **Create directory and enter it**.

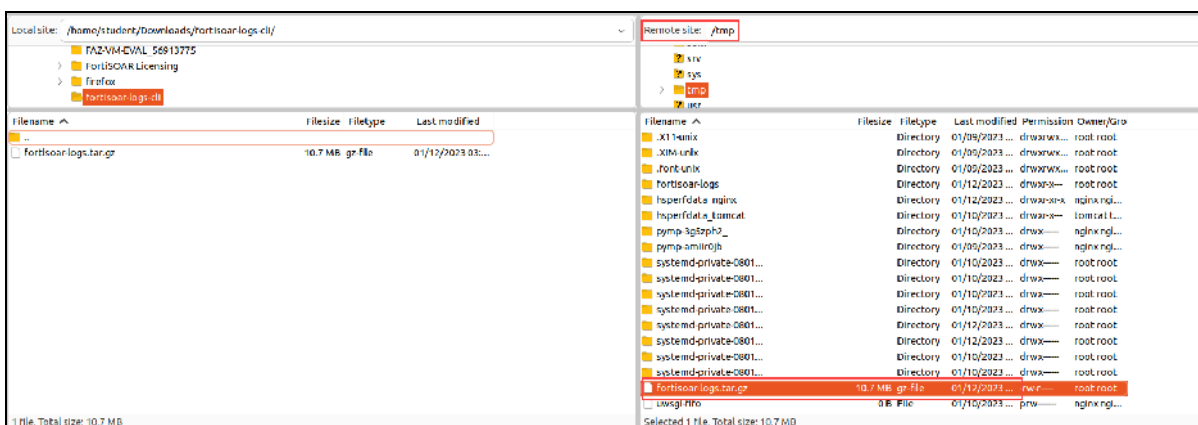


11. Enter the following path:

```
/home/student/Downloads/fortisoar-logs-cli
```

12. In the **Remote Site** section, navigate to the folder /tmp/, and then double-click the fortisoar-logs.tar.gz file.

This will initiate a file transfer onto the Local-Host VM.



13. Open a terminal on Local-Host VM.
14. Change the working directory by running the following command:  
`cd /home/student/Downloads/fortisoar-logs-cli`
15. Run the following command to extract the contents of the file:  
`tar -xvf fortisoar-logs.tar.gz`
16. Run the following command to view the contents of the `das` file:  
`vi fortisoar-logs/var/log/cyops/cyops-auth/das.log`
17. Press `shift` and `g` together to reach the end of the file.  
You should see a log entry for a failed login attempt.

```
2023-02-09,23:09:03 ERROR gen: wrapper(): 216: User failedadminlogin does not exist in the system
```

18. Press `Esc` and type `:q` to return to the CLI.
19. Close the FortiSOAR CLI session.

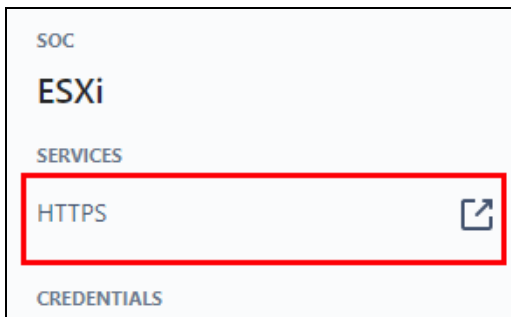
## Appendix

### Revert FortiSOAR

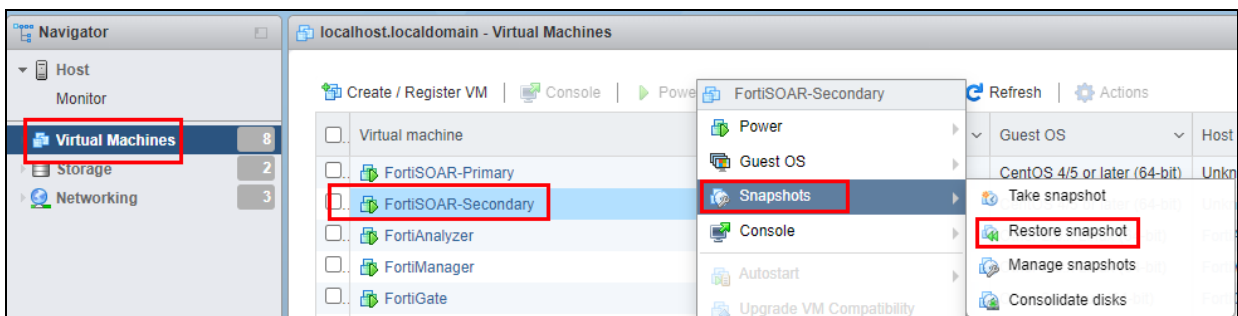
The following procedure is included in the lab guide in cases where students may run into a duplicate license issue with the FortiSOAR cluster. Do *not* perform these steps if you do not have a duplicate license issue. The revert operation will *undo* all of your Lab 4 configuration. Do *not modify or restore any other VMs*.

#### To revert FortiSOAR

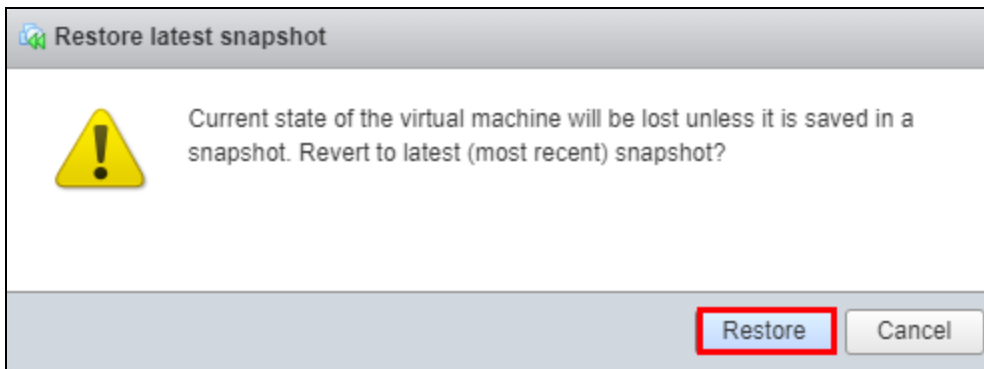
1. From the Fortinet Training Institute lab landing page, connect to the ESXi VM using HTTPS.



2. Login with the username `student` and password `Fortinet1!`.
3. Click **Virtual Machines**.
4. Right-click the **FortiSOAR-Secondary** VM, and select **Snapshots > Restore snapshot**.

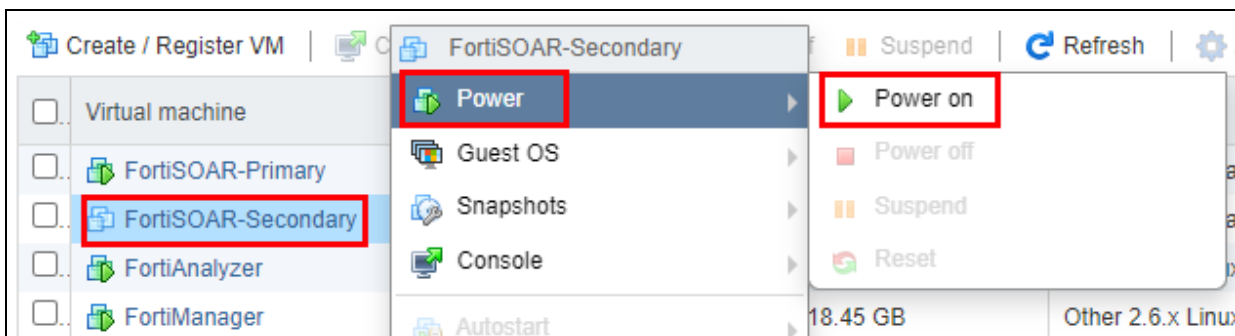


5. Click **Restore**.

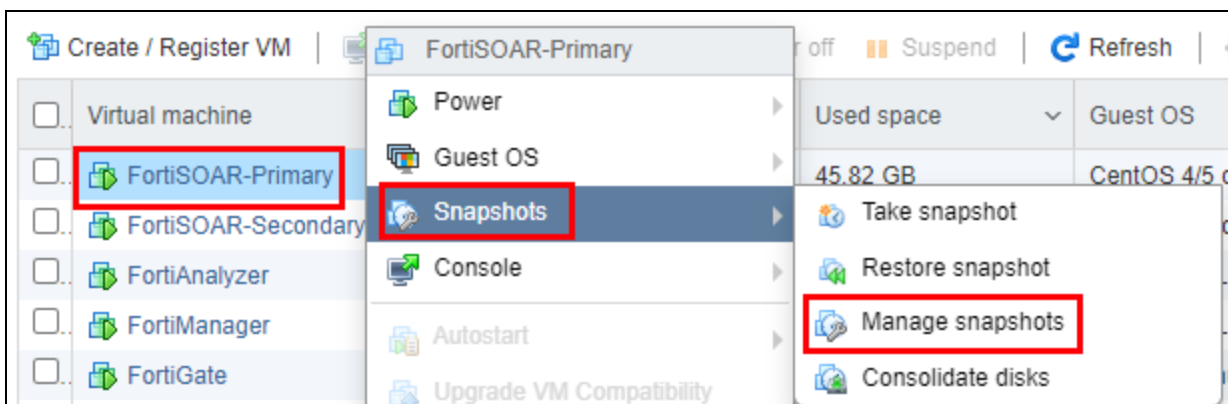


The VM powers off

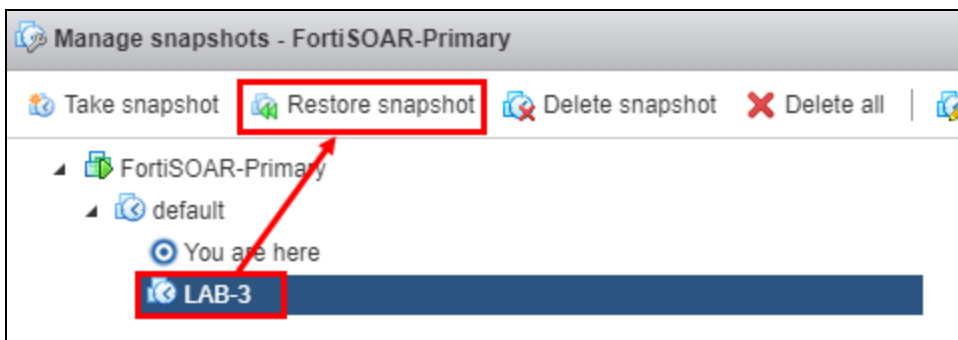
6. Right click the **FortiSOAR-Secondary** VM, and select **Power > Power On**.



7. Right-click the **FortiSOAR-Primary** VM, and select **Snapshots > Manage snapshots**.

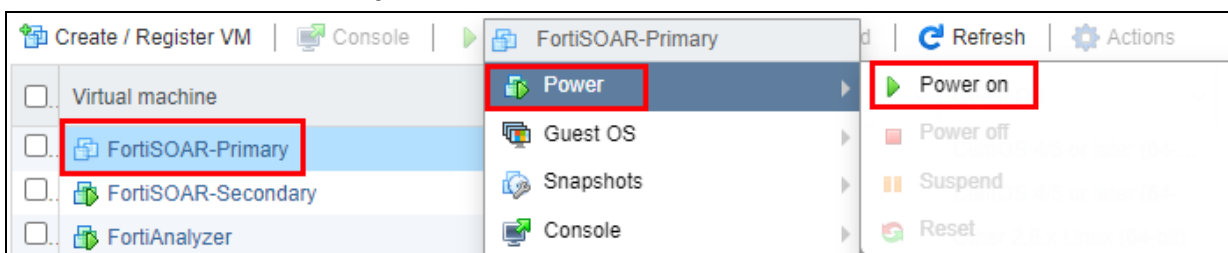


8. Select **LAB-3**, and click **Restore snapshot**.



The VM powers off.

9. Click **Close**.
10. Power on the **FortiSOAR-Primary** VM.



11. Log out of the ESXi session.



The revert procedure restores both FortiSOAR VM to their state after Lab 3. You must do [Configure an HA Cluster on page 65](#) again to proceed.



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